



ECONOMICS DECISION IN BUSINESS

Self-Instructional Manual

The Self-Instructional Manual (SIM) for Economics Decision in Business (MBA6664) is designed to equip MBA students with the core tools of economic theory, enabling sound decision-making across microeconomic, macroeconomic, and strategic contexts.

About This Manual

This manual comprises seven learning units spanning microeconomics, macroeconomics, market structures, development economics, international trade, game theory, and the digital economy. Each unit integrates Sarawak and Malaysian case studies to ground theory in local business realities, with self-check questions, discussion prompts, and summaries to reinforce independent learning.



MBA6664 | Major (Core) | Credit 4 | 160 SLT | Semester 2

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INSPIRING MINDS, SHAPING FUTURE

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Preface

Course Information

How to Use This Manual

Topic 1: Microeconomics

Topic 2: Macroeconomics

Topic 3: Market Structures

Topic 4: Macroeconomic Variables and Economic Development

Topic 5: Current Issues in World Economics — Trade and International Finance

Topic 6: Game Theory in Decision Making

Topic 7: Digital and Knowledge Economy

PREFACE

ECONOMICS DECISION IN BUSINESS

(MBA6664)

Welcome to MBA6664 Economics Decision in Business. This module equips you with the conceptual frameworks, analytical tools, and applied reasoning skills required to make sound economic decisions in real business and policy environments. Economics is not an abstract academic exercise — it is the intellectual foundation of every significant managerial decision: how to price a product, how to respond to a competitor, how to evaluate an investment, how to interpret government policy, and how to allocate scarce resources across competing priorities.

Sarawak's Economic Context: Sarawak occupies a unique and fascinating position in Malaysia's economic landscape. As a state whose prosperity has historically been built on commodity exports — crude oil, liquefied natural gas, timber, and palm oil — Sarawak is now undergoing a deliberate structural transformation toward a knowledge-based, digitally enabled economy. The Sarawak Digital Economy Blueprint 2030, the expansion of Sarawak Energy Berhad's (SEB) hydroelectric generation capacity, the establishment of PETROS (Petroleum Sarawak Bhd) in 2018, and massive infrastructure investments including the Pan Borneo Highway and Baleh Dam are reshaping the state's economic structure in real time. These transformations generate the full range of economic phenomena that this module examines: resource allocation decisions under scarcity, market structure and pricing power, macroeconomic stabilisation, inflation, unemployment, fiscal policy, and long-run growth. Students at i-CATS University College are uniquely positioned to observe and analyse these processes from within the very economy they are studying.

Why Economics Matters for Managers: Three managerial capabilities depend directly on economic understanding. First, resource allocation — every organisation operates under resource constraints, and economics provides the analytical framework for allocating scarce resources to their highest-value uses. Second, pricing decisions — whether you work in a competitive market, a monopoly, or an oligopoly, understanding demand elasticity, cost structures, and market equilibrium is essential for setting prices that maximise revenue and meet strategic objectives. Third, policy impact assessment — government policies (taxation, subsidies, monetary policy, trade regulation) directly affect business costs, consumer demand,

and competitive dynamics. Managers who understand economic models can anticipate policy impacts and adapt their strategies proactively.

Examples and case studies in this SIM prioritise Sarawak first, then East Malaysia (Sabah and Sarawak), then Malaysia, and only then regional or global cases where necessary. This ensures economic models are applied primarily to the real policy and business environments you operate in. You will encounter examples drawn from Sarawak Energy Berhad, PETROS, Sarawak pepper farming cooperatives, the CelcomDigi merger, Bank Negara Malaysia's interest rate policy, and the Sarawak State Budget — all of which provide rich, first-hand applications of the economic theory covered in each Learning Unit.

How This SIM is Structured: LU1 introduces microeconomics — the economics of individual markets, firms, and consumers, covering supply and demand, cost of production, and market structures. LU2 introduces macroeconomics — the economics of national income, monetary systems, unemployment, aggregate demand and supply, and fiscal and monetary policy. LU3 deepens the market structures analysis from LU1, examining perfect competition, monopoly, oligopoly, monopolistic competition, factor markets, and income inequality. LU4 advances macroeconomic analysis into development economics, examining GDP limitations, inflation measurement, the Solow growth model, financial systems, and debt sustainability. LU5 through LU7, covered in a separate volume, address international trade, business strategy applications of economics, and economic policy evaluation.

A Note on Artificial Intelligence and Academic Integrity: AI tools such as ChatGPT, Copilot, and Gemini can assist with research and concept clarification. However, submitting AI-generated analysis as your own work without disclosure constitutes academic misconduct under i-CATS University College's Academic Integrity Policy. All three assessments in this course — CA1 Assignment, CA2 Case Study, and the Final Assignment — require original analytical reasoning applied to specific Malaysian and Sarawak economic scenarios. Economics is learned by doing: working through problems, constructing diagrams, and applying models to real data. Use AI as a thinking partner, not a substitute for your own intellectual engagement.

We wish you an analytically rigorous and practically rewarding engagement with MBA6664. The economic reasoning you develop in this module will inform your managerial decisions for your entire career.

COURSE INFORMATION

Field	Details
Course Code	MBA6664
Course Title	Economics Decision in Business
Credit Hours	4
Student Learning Time (SLT)	160 hours
Classification	Major (Core)
Semester	2 (Year 1, Semester 2)
Pre-requisite	None
Module Writer	Dr. Colin Wong Koh King; Prof. Dr. Dato Shazali Abu Mansor; Prof. Dr. Frank Kiong
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Institution	i-CATS University College, Kuching, Sarawak
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Course Synopsis

This course uses the tools of a core economic theory that will enhance a student's ability to apply economics models to arrive at appropriate decision-making, how markets allocate resources, how the structure of markets affects choices and social welfare, and the ways that government intervention can improve or impair the functioning of markets. The student will be given the opportunity to apply these models to describe real world current events and to study various policy problems in-depth.

Course Learning Outcomes (CLOs)

CLO	Statement	Bloom's Level	PLO
CLO1	Analyze microeconomics and macroeconomics model for economics decisions making.	C4	PLO1
CLO2	Recommend these models to real-world situation in a correct and proficient manner.	C5	PLO2
CLO3	Apply appropriate models in business decision making.	A5	PLO5

CLO–PLO Mapping and Teaching–Assessment Alignment

CLO	PLO	Teaching Method	Assessment Method
CLO1	PLO1	Lecture	Assignment
CLO2	PLO2	Tutorial	Assignment
CLO3	PLO5	Tutorial	Case Study

Assessment Structure

Assessment	Component	Weightage	CLOs Assessed	SLT (hrs)
CA1	Assignment	35%	CLO1 + CLO2	30
CA2	Case Study	35%	CLO2 + CLO3	30
Final Assessment	Assignment	30%	CLO1 + CLO2 + CLO3	30

LU Independent Study	(LU1–LU7)	—	—	70
	GRAND TOTAL	100%		160 hrs

MQF Alignment

CLO	MQF Domain	MQF Descriptor
CLO1	Knowledge — C1 + C2	MQF Level 7 — Analyse microeconomic and macroeconomic models
CLO2	Communication — C3C	MQF Level 7 — Recommend models proficiently to real-world situations
CLO3	Application — A5 (high-order)	MQF Level 7 — Apply economic models to business decision-making

HOW TO USE THIS MANUAL

This Self-Instructional Material (SIM) is your primary learning resource for MBA6664 Economics Decision in Business. It has been written for Open and Distance Learning (ODL), which means you are expected to study the content independently, at your own schedule, and come prepared for tutorials with questions and reflections already formed. Economics requires active engagement: reading alone is insufficient. You must work through calculations, reconstruct diagrams, and apply models to real examples to develop genuine analytical competence.

Feature	Description	How to Use It
Self-Reflection 	A prompt inviting you to connect the economic theory to your	Take 10–15 minutes to write a brief personal response before your next

	personal, professional, or Sarawak context.	tutorial. These are not graded but build the applied reasoning skills needed for CA1 and CA2.
Managerial Insight 	A focused box highlighting a real-world Sarawak or Malaysian application of the economic concept, often with numeric data.	Read carefully — these examples are frequently the basis for discussion questions and assignment scenarios. The Sarawak Energy and PETROS examples are especially important.
Summary Box	A structured summary of the key economic concepts covered in the Learning Unit.	Review this before assessments. If any concept is unfamiliar, return to the relevant subtopic. Use the summary to check your diagram recall.
MCQs	Multiple Choice Questions testing conceptual understanding and applied calculation skills.	Attempt independently before checking the Answer Key. For quantitative questions, show your working. Identify gaps and re-read the relevant sections.
Discussion Questions	Higher-order analytical questions aligned to the CLOs, suitable for tutorial discussion.	Prepare a structured response (250–400 words) before each tutorial. These mirror the style of CA1 and CA2 questions.
Diagram Note 	A text description of an economic diagram — axes, curves, shifts, and equilibrium points — since actual diagrams cannot be embedded in this ODL format.	When you encounter a Diagram Note, draw the diagram yourself on paper. Try to reproduce it from memory before reading the description — this is the most effective method for learning economic models.
Online Resources	Curated links to free or freemium videos, tools, and data	Access these at your own pace. Khan Academy and MRU (Marginal

	sources that supplement the unit content.	Revolution University) videos are especially useful for visual learners reviewing diagrams and calculations.
Assignment Box	A formative exercise at the end of selected LUs to prepare you for summative assessments.	Not graded — but completing these exercises significantly improves your CA1 and CA2 submissions. The elasticity calculations and GDP exercises in LU1 and LU2 are especially important.

Diagram Study Tip: Economic models are primarily communicated through diagrams — the supply-demand diagram, the production possibility frontier, the cost curve, and the AD-AS model. When you encounter a diagram description in this SIM, reproduce it on paper from the text description. Label all axes, all curves, and all equilibrium points. Then close the book and reproduce it from memory. If you can draw and explain every diagram in this SIM from memory, you are well-prepared for all three assessments.

Students are strongly encouraged to apply all concepts and models to Sarawak and Malaysian economic contexts wherever possible. The Sarawak Economic Planning Unit (EPU Sarawak), the Department of Statistics Malaysia (DOSM), Bank Negara Malaysia (BNM), and the Sarawak State Budget documents are all freely available online and provide real data for applying the theoretical models in this SIM.

14-WEEK COURSE SCHEDULE

The following schedule maps the Learning Units across the 14-week semester, including assessment milestones and indicative Student Learning Time (SLT) allocations. Total SLT for this course is 160 hours: 70 hours across the 7 LUs (independent study), plus 90 hours for the three assessments (CA1 Assignment: 30 hrs, CA2 Case Study: 30 hrs, Final Assignment: 30 hrs).

Week	Learning Unit	Topic	Ind. Study (hrs)	Assessment Milestones
1–2	LU1	Microeconomics: Supply, Demand, Costs, and Market Structures	8	—
3–4	LU2	Macroeconomics: GDP, Monetary System, Unemployment, AD-AS, Policy	9	—
5–6	LU3	Market Structures: Perfect Competition, Monopoly, Oligopoly, Factor Markets	9	CA1 (Assignment) introduced — Week 5
7–8	LU4	Macroeconomic Variables and Economic Development	10 (+1 online)	CA1 due — Week 8
9–10	LU5	International Trade and Open Economy Macroeconomics	10 (+1 online)	CA2 (Case Study) introduced — Week 9
11–12	LU6	Game Theory and Strategic Business Decisions	10 (+1 online)	CA2 due — Week 12
13–14	LU7	Economic Policy Evaluation and Business Strategy	10 (+1 online)	Final Assignment introduced — Week 13; due post-Week 14
Post Week 14	Final Assignment	Self-directed study and submission	90 (CA1+CA2+Final assessment SLT)	Final Assignment due — per examination timetable

	TOTAL		70 (LU content) + 90 (Assessments) = 160 hrs	
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Note: Tutorial hours and scheduled contact sessions are organised by your programme coordinator and are separate from the SLT hours listed above. Please refer to your official timetable for tutorial session details. The online hours noted for LU4–LU7 refer to required engagement with online data sources (BNM, DOSM, World Bank) as part of the independent study for those units.



TOPIC

1

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MICROECONOMICS

Lecturers: Dr. Colin Wong Koh King; Prof. Dr. Dato Shazali Abu Mansor; Prof. Dr. Frank Kiong

Topic 1: Microeconomics

Learning Outcomes

By the end of this topic you should be able to:

- **Define economics and explain the core concepts of scarcity, opportunity cost, and trade-offs**
- **Analyse supply and demand to explain price and quantity determination**

- Calculate and interpret price elasticity of demand and supply
- Explain production costs and apply cost concepts to firm decision-making
- Describe different market structures and their implications for pricing and output



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LO Learning Objectives

By the end of this topic you should be able to:

- **Define core economic concepts** including scarcity, opportunity cost, and the production possibility frontier.
- **Analyse supply and demand dynamics** and predict how price and quantity respond to market changes.
- **Evaluate different cost structures and production functions** to guide firm-level decision-making.
- **Apply microeconomic models** to assess firm behaviour and market efficiency in Malaysian and Sarawak contexts.

KC Key Concepts and Definitions

Concept	Definition	Example / Notes
Scarcity	The fundamental economic problem: limited resources relative to unlimited wants.	<i>Every household, firm, and government faces trade-offs due to scarcity.</i>
Opportunity Cost	The value of the next-best alternative forgone when a decision is made.	<i>Building a factory vs. investing in R&D — both cannot be fully funded.</i>
Supply and Demand	The forces that determine price and quantity in a market.	<i>A rise in palm oil prices increases supply from Sarawak plantations.</i>
Price Elasticity	Measures the responsiveness of quantity demanded or supplied to a price change.	<i>Inelastic demand for petrol; elastic demand for luxury goods.</i>

Cost of Production	The total expenses incurred by a firm to produce goods or services.	<i>Fixed costs (rent), variable costs (labour), and total cost (both).</i>
Market Equilibrium	The point where quantity demanded equals quantity supplied.	<i>Equilibrium price of Sarawak timber in the export market.</i>

1.1 Introduction to Economics

Economics is the social science that studies how individuals, firms, and societies allocate scarce resources among competing ends. The word "scarce" is the foundation of the entire discipline: if everything we wanted were available in unlimited quantities, there would be no need to make choices, and therefore no economics. In reality, all resources — time, capital, land, labour, entrepreneurial talent — are finite. Choices must be made, and every choice involves a cost.

Scarcity and Opportunity Cost

Scarcity forces every economic actor — a household, a firm, a government — to make trade-offs. When a government allocates RM 2 billion to hydropower infrastructure, it cannot simultaneously allocate that same RM 2 billion to digital economy development. When a student spends four hours studying economics, they cannot simultaneously spend those four hours working a part-time job. The concept of opportunity cost captures this reality: the opportunity cost of any decision is the value of the next best alternative that must be foregone. Opportunity cost is not merely monetary — it includes the value of time, leisure, and any other resource that could have been deployed differently. Economists argue that rational decision-makers should always think in terms of opportunity cost, because the true cost of any choice is what you give up to make it.

The Production Possibility Frontier (PPF)

The Production Possibility Frontier (PPF) is one of the most fundamental tools in economics. It represents all possible combinations of two goods (or broad categories of goods) that an economy can produce if it fully and efficiently utilises all available resources. Points on the

PPF are efficient — the economy is producing as much as possible. Points inside the PPF indicate inefficiency — resources are either idle or being deployed sub-optimally (perhaps due to unemployment, poor management, or institutional failures). Points outside the PPF are currently unattainable but may become attainable over time through economic growth — achieved through capital accumulation, technological progress, or improved resource utilisation.

The PPF is concave (bowed outward) because resources are not perfectly adaptable — as an economy shifts production increasingly toward one good, it must divert resources that are progressively less suited to producing that good, causing increasing opportunity costs. A straight-line PPF would imply constant opportunity costs and perfect substitutability of resources, which is rarely realistic.

Diagram Note — Figure 1.1: Production Possibility Frontier

Figure 1.1: Production Possibility Frontier — The PPF shows all combinations of two goods (e.g., energy infrastructure investment and digital economy investment) that an economy can produce with its available resources. The horizontal axis (Q-axis) measures investment in digital economy programmes (RM billions); the vertical axis (Q-axis) measures investment in energy infrastructure (RM billions). The PPF curve is concave, bowing outward from the origin. Point A, on the PPF, represents an efficient allocation — e.g., RM 1.2B digital + RM 0.8B energy. Point B, inside the PPF, represents an inefficient allocation — resources are underutilised. Point C, outside the PPF, represents a currently unattainable combination. The negative slope of the PPF at any point equals the marginal rate of transformation (MRT) — the opportunity cost of shifting one more unit of resource from energy infrastructure to digital economy investment.

Sarawak Case: The Opportunity Cost Dilemma

The PPF provides a powerful framework for understanding one of Sarawak's most important current policy debates. The Sarawak government faces a real opportunity cost dilemma: should Sarawak's limited capital be directed toward energy infrastructure — specifically the Baleh Dam hydroelectric project (RM 8.6 billion, 760 MW capacity, targeting completion in 2026) — or toward digital economy development through the SDEC's suite of programmes? Both

goals require the same scarce capital. Choosing more of one necessarily means choosing less of the other, at least in the short run. The PPF illustrates this trade-off explicitly: the government is operating on the PPF when it uses all available capital efficiently, and the slope of the PPF at the chosen point represents the opportunity cost of each marginal unit of investment shifted from one priority to the other. Over the long run, of course, strategic investments in energy infrastructure may themselves enable digital economy growth — cheap, reliable electricity (at SEB's industrial tariff of approximately RM 0.23/kWh) is itself an input into data centres and digital manufacturing that the SDEC wishes to attract. This illustrates an important point about the PPF: investing in capacity can shift the entire frontier outward over time, expanding the economy's production possibilities.

Positive vs Normative Economics

Economics is conventionally divided into two modes of inquiry. Positive economics describes and explains the world as it is — it makes factual, testable claims about cause and effect. "When the price of petrol rises, the quantity demanded falls" is a positive statement — it can be tested empirically. "The government should subsidise petrol for rural Sarawak consumers" is a normative statement — it expresses a value judgement about what the government ought to do, and reasonable people can disagree based on different values and priorities. Managers and policy analysts must be careful to distinguish between the two: positive economic analysis provides the factual foundation, while normative judgements about what should be done involve value choices that economics alone cannot resolve.

The Circular Flow Model

The circular flow model describes the fundamental economic relationships in a market economy. It shows two key actors — households and firms — interacting in two markets. In the factor markets (or input markets), households supply factors of production (labour, land, capital, entrepreneurship) to firms, and receive income in return (wages, rent, interest, profit). In the goods and services markets (output markets), firms sell goods and services to households, and receive revenue in return. The circular flow illustrates the fundamental identity of macroeconomics: total income earned by households equals total expenditure on goods and services, which equals total output produced by firms. A more complete model adds the government sector (taxation and government spending) and the external sector (exports and imports), which we will explore in LU2. In the Sarawak context, the circular flow includes a

critical additional flow: oil and gas royalty payments from PETROS and Petronas to the Sarawak state government, which the government then redistributes through the state budget — a fiscal flow that does not originate from household labour or firm output in the conventional sense.

Mankiw's Ten Principles of Economics

Mankiw's (2024) *Principles of Economics* organises the fundamental insights of the discipline into ten core principles, grouped into three categories: how people make decisions (Principles 1–4), how people interact (Principles 5–7), and how the economy works as a whole (Principles 8–10). The four decision-making principles are: (1) people face trade-offs; (2) the cost of something is what you give up to get it (opportunity cost); (3) rational people think at the margin — they compare marginal benefits with marginal costs; and (4) people respond to incentives. These four principles underpin the entire microeconomics curriculum in LU1 and LU3. The three interaction principles are: (5) trade can make everyone better off; (6) markets are usually a good way to organise economic activity; and (7) governments can sometimes improve market outcomes (when market failures exist). The three economy-wide principles are: (8) a country's standard of living depends on its ability to produce goods and services; (9) prices rise when the government prints too much money; and (10) society faces a short-run trade-off between inflation and unemployment. Principles 8–10 underpin the macroeconomics curriculum in LU2 and LU4.

Microeconomics vs Macroeconomics

Microeconomics studies individual economic units — individual consumers, households, firms, and specific markets. It examines how prices are determined, how firms decide on output and pricing, how consumers maximise utility under budget constraints, and how market structures affect economic welfare. Macroeconomics, by contrast, studies the economy as a whole — national income, aggregate price levels, total employment, economic growth, and the role of monetary and fiscal policy. While the distinction is analytically useful, the two levels are deeply interconnected: macroeconomic phenomena (such as the COVID-19 recession) originate in microeconomic decisions (firms laying off workers, households cutting consumption), and macroeconomic policy (such as BNM's interest rate cuts in 2020) works through microeconomic channels (lower borrowing costs for Sarawak SMEs stimulating

investment). LU1 addresses microeconomics; LU2 addresses macroeconomics; LU3 and LU4 deepen each.

1.2 Supply and Demand

The supply-demand model is the workhorse of microeconomic analysis. It describes how markets coordinate the independent decisions of millions of buyers and sellers through the price mechanism, and how equilibrium prices and quantities are determined. Alfred Marshall's synthesis of supply and demand in his 1890 *Principles of Economics* remains one of the most powerful analytical tools ever developed in social science. Understanding this model deeply — not just as a diagram to be reproduced, but as a framework for reasoning about real market behaviour — is the foundation of economic competence for business managers.

The Law of Demand and the Demand Curve

The law of demand states that, *ceteris paribus* (all other things being equal), when the price of a good rises, the quantity demanded falls; and when price falls, quantity demanded rises. This inverse price-quantity relationship is represented graphically by the demand curve — a downward-sloping line or curve in price-quantity space. The negative slope of the demand curve reflects two effects: the substitution effect (as price rises, consumers substitute away from the good toward cheaper alternatives) and the income effect (as price rises, the real purchasing power of consumers' income falls, leading them to buy less of the good, assuming it is a normal good).

The determinants of demand — factors that shift the entire demand curve, as opposed to causing a movement along it — include: (1) consumer income (rising income increases demand for normal goods, decreases demand for inferior goods); (2) prices of related goods (a rise in the price of a substitute good increases demand for this good; a rise in the price of a complementary good decreases demand for this good); (3) consumer tastes and preferences; (4) expectations about future prices or income; and (5) the number of buyers in the market. When any of these determinants change, the demand curve shifts — rightward for an increase in demand, leftward for a decrease.

The Law of Supply and the Supply Curve

The law of supply states that, *ceteris paribus*, when the price of a good rises, the quantity supplied increases; and when price falls, quantity supplied falls. This positive price-quantity relationship is represented by the upward-sloping supply curve. The positive slope reflects the increasing marginal cost of production — as firms produce more output, they typically incur higher per-unit costs (due to diminishing returns, higher input prices, or capacity constraints), and therefore require a higher price to justify increasing output. The determinants of supply — factors that shift the supply curve — include: (1) input prices (rising wages or raw material costs reduce supply, shifting the curve leftward); (2) technology (improvements in production technology lower costs and increase supply); (3) government policies (taxes increase costs, reducing supply; subsidies lower costs, increasing supply); (4) producer expectations; and (5) the number of sellers.

Market Equilibrium, Price Ceilings, and Price Floors

Market equilibrium occurs where the supply and demand curves intersect — at the price where quantity demanded equals quantity supplied. At the equilibrium price, the market clears: there is no unsold surplus and no unsatisfied demand. If price is above equilibrium, a surplus develops (quantity supplied exceeds quantity demanded), putting downward pressure on price. If price is below equilibrium, a shortage develops (quantity demanded exceeds quantity supplied), putting upward pressure on price. Price ceilings (maximum legal prices set below equilibrium) cause shortages — as seen in Malaysia's controlled price for RON95 petrol, which created excess demand when market prices rose above the ceiling. Price floors (minimum legal prices set above equilibrium) cause surpluses — as seen in Malaysia's minimum wage legislation, which some economists argue creates a surplus of labour (unemployment) in low-wage sectors, though this is contested by evidence that modest minimum wage increases have limited disemployment effects in practice.

Price Elasticity of Demand: Formula, Interpretation, and Application

Price elasticity of demand (PED) measures the responsiveness of quantity demanded to a change in price. The formula is: $PED = (\text{percentage change in quantity demanded}) / (\text{percentage change in price}) = (\% \Delta Q_d / \% \Delta P)$. Because demand curves are typically downward-sloping, PED is negative — a price increase causes a quantity decrease. By convention, economists often report the absolute value of PED. If $|PED| > 1$, demand is elastic — consumers are highly

responsive to price changes (quantity demanded changes by a larger percentage than the price change). If $|PED| < 1$, demand is inelastic — consumers are relatively unresponsive (quantity demanded changes by a smaller percentage than the price change). If $|PED| = 1$, demand has unit elasticity. Perfectly inelastic demand ($|PED| = 0$) means quantity demanded does not change at all when price changes — represented by a vertical demand curve. Perfectly elastic demand ($|PED| = \infty$) means consumers are infinitely price-sensitive — represented by a horizontal demand curve.

Sarawak Case: Petrol Demand Elasticity

When the Malaysian government reduced RON95 petrol subsidies in 2024, this effectively raised the price paid by consumers, including in Sarawak. Using supply-demand analysis, we can trace the economic consequences precisely. The reduction in subsidy raises the effective cost to suppliers (or equivalently, the price consumers pay at the pump). In urban areas like Kuching and Miri, where car ownership rates are high but alternatives (public transport, e-hailing, cycling) are available, the price elasticity of demand for petrol is moderately inelastic — consumers have some ability to substitute. In rural Sarawak — Kapit, Limbang, Baram — petrol is the only transport fuel and there are virtually no substitutes. Demand is therefore extremely inelastic: a 20% price increase may reduce consumption by only 4–6%, because rural residents have no alternative means of transport or heating fuel.

Numeric Example: Calculating PED

If petrol price in Sarawak rises from RM 2.05 to RM 2.50 (a change of +RM 0.45, or +21.9%), and quantity demanded falls from 10,000 litres per day to 9,400 litres per day (a change of -600 litres, or -6%), then: $PED = -6\% / 21.9\% = -0.27$. The absolute value $|PED| = 0.27$, which is well below 1 — demand is inelastic. This has an important revenue implication: when demand is inelastic, a price increase raises total revenue for the supplier, because the percentage increase in price exceeds the percentage decrease in quantity. Total revenue before the price rise: $RM\ 2.05 \times 10,000 = RM\ 20,500/\text{day}$. Total revenue after: $RM\ 2.50 \times 9,400 = RM\ 23,500/\text{day}$. The fuel supplier benefits financially from the price increase, while consumers — especially rural Sarawakians with no transport alternatives — bear a significant welfare cost.

Diagram Note — Figure 1.2: Sarawak Petrol Market

Figure 1.2: Sarawak Petrol Market — Supply-Demand Diagram. The vertical axis (P-axis) measures the price per litre in RM. The horizontal axis (Q-axis) measures litres demanded and supplied per day. The demand curve (D) slopes downward from left to right — steeply, to represent inelastic demand in rural Sarawak. The supply curve (S) slopes upward from left to right. Initial equilibrium: $P_0 = \text{RM } 2.05$, $Q_0 = 10,000$ litres/day (labelled as the intersection of D and S_0). When the government reduces the petrol subsidy, the effective cost to suppliers increases — the supply curve shifts leftward to S_1 . New equilibrium: $P_1 = \text{RM } 2.50$, $Q_1 = 9,400$ litres/day. The shaded triangular area between the two equilibrium points and the demand curve represents consumer surplus lost due to the price increase. The steep (inelastic) demand curve means this triangle is relatively tall and narrow — a large price increase for a small quantity reduction, confirming the inelastic demand conditions in rural Sarawak.

Cross-Price, Income, and Supply Elasticities

Cross-price elasticity of demand (XED) measures the responsiveness of demand for good A to a change in the price of good B. $XED = (\% \Delta Q_d \text{ of A}) / (\% \Delta P \text{ of B})$. A positive XED indicates substitutes — when the price of bus transport rises in Kuching, demand for Grab e-hailing services increases. A negative XED indicates complements — when the price of printers rises, demand for printer ink falls. Income elasticity of demand (YED) measures the responsiveness of demand to changes in consumer income. $YED = (\% \Delta Q_d) / (\% \Delta Y)$. A positive YED indicates a normal good — demand rises as income rises. A YED greater than 1 indicates a luxury good — demand rises proportionally more than income (e.g., premium restaurants, luxury cars). A negative YED indicates an inferior good — demand falls as income rises (e.g., budget rice varieties, low-quality transport). Price elasticity of supply (PES) measures the responsiveness of quantity supplied to price changes. $PES = (\% \Delta Q_s) / (\% \Delta P)$. Supply of agricultural goods (e.g., Sarawak pepper) tends to be inelastic in the short run because production requires time to adjust — farmers cannot instantly plant more pepper when prices rise.

Government Intervention: Taxes and Subsidies

A tax on a good shifts the supply curve leftward (or equivalently, raises the supply price at every quantity). This raises the equilibrium price paid by consumers, reduces the equilibrium quantity, and creates a deadweight loss — a reduction in total economic surplus (the combined value of consumer surplus and producer surplus) that represents economic welfare destroyed by the tax. A subsidy has the opposite effect: it shifts the supply curve rightward, lowering the

equilibrium price and increasing quantity. Malaysia's RON95 petrol subsidy historically shifted the effective supply curve rightward, keeping petrol prices below market equilibrium. The 2024 targeted subsidy rationalisation — eliminating the blanket subsidy and replacing it with targeted assistance for lower-income groups — represents a policy shift with significant distributional and efficiency implications. From an economics perspective, targeted subsidies are more efficient (they direct support to those who need it most) but more administratively complex and politically challenging to implement.

1.3 Cost of Production

Understanding production costs is essential for every managerial decision involving pricing, output levels, investment, and competitive strategy. The economics of cost goes considerably deeper than accounting — economic cost incorporates opportunity cost, distinguishes between the short run and the long run, and provides the analytical foundation for understanding firm behaviour across all market structures.

Explicit vs Implicit Costs and Economic Profit

Explicit costs are the direct, out-of-pocket monetary payments a firm makes — wages paid to employees, rent paid for premises, materials purchased, utilities, and so on. These are the costs that appear in a conventional accounting profit-and-loss statement. Implicit costs are the opportunity costs of the resources a firm owns and uses in production — resources for which no monetary payment is made but which have a valuable alternative use. The most significant implicit cost for small businesses is the opportunity cost of the owner's time: an entrepreneur who works in their own business foregoes the salary they could earn as a salaried employee elsewhere. If a Kuching restaurant owner works full-time in their own restaurant without drawing a salary, and could alternatively earn RM 8,000/month as a manager in another company, the RM 8,000/month is an implicit cost of running the restaurant. Economic profit = Total Revenue – Total Economic Cost (explicit + implicit costs). Accounting profit = Total Revenue – Explicit costs only. A business can earn positive accounting profit while simultaneously earning negative economic profit — meaning the owner would be better off deploying their resources elsewhere.

Fixed, Variable, and Total Costs

In the short run (a period in which at least one input is fixed), costs divide into fixed costs and variable costs. Fixed costs (FC) do not vary with output — they must be paid regardless of how much the firm produces. Examples include rent, insurance, depreciation on equipment, and interest payments on loans. Variable costs (VC) change with the level of output — raw materials, direct labour, energy consumed in production. Total cost (TC) = FC + VC. As output increases, total costs rise because variable costs increase, while fixed costs remain constant. The important cost curves for managerial analysis are the average and marginal cost curves, derived from these total costs.

Average and Marginal Costs: The Key Curves

Average Fixed Cost (AFC) = FC / Q . AFC falls continuously as output increases, because a fixed amount of fixed cost is being spread over an increasing number of units. Average Variable Cost (AVC) = VC / Q . AVC typically falls initially (as the firm becomes more efficient through division of labour and specialisation), reaches a minimum, then rises (as diminishing returns to variable inputs set in at high output levels). Average Total Cost (ATC) = $TC / Q = AFC + AVC$. The ATC curve is U-shaped — falling at first as AFC falls rapidly and as scale economies are realised, then rising as the law of diminishing returns raises AVC faster than falling AFC reduces it. Marginal Cost (MC) = $\Delta TC / \Delta Q$ — the additional cost of producing one more unit of output. MC is the most important cost concept for decision-making: a profit-maximising firm should expand output whenever marginal revenue exceeds marginal cost, and stop when $MR = MC$.

The MC–ATC Relationship

A critical relationship for managerial economics: the MC curve always intersects the ATC curve at the minimum point of ATC. This is a mathematical necessity, not a coincidence. When $MC < ATC$, producing an additional unit costs less than the average, so the average must be falling. When $MC > ATC$, producing an additional unit costs more than the average, so the average must be rising. Therefore $MC = ATC$ exactly at the minimum of ATC. The minimum of ATC is the efficient scale — the output level at which the firm minimises its cost per unit. This concept is central to understanding both pricing in competitive markets (price is driven to minimum ATC in long-run equilibrium) and the justification for natural monopoly regulation

(some industries have declining ATC across the entire relevant range of output, meaning a single firm can always produce more cheaply than multiple competing firms).

Economies and Diseconomies of Scale

In the long run, all inputs are variable — the firm can adjust its scale of operations entirely. Economies of scale occur when long-run average total cost (LRATC) falls as output increases — meaning a larger firm can produce at lower average cost than a smaller firm. Economies of scale arise from specialisation of labour and capital, spreading of fixed overhead, bulk purchasing power, and technological advantages of large-scale equipment. Diseconomies of scale occur when LRATC rises as output increases — typically due to organisational complexity, management coordination failures, and bureaucratic inefficiency at very large scales. Constant returns to scale occur when LRATC is constant as output increases. The long-run average cost curve (LRAC) is constructed as the envelope of all the short-run ATC curves for different scales of plant — the minimum short-run ATC achievable at each output level defines the LRAC.

Sarawak Case: Sarawak Energy Berhad's Cost Structure

Sarawak Energy Berhad (SEB) provides a textbook example of a firm exploiting massive economies of scale. SEB's hydroelectric generation involves enormous fixed costs (dam construction, transmission grid, substation infrastructure) but very low variable costs per unit of electricity generated (essentially the cost of water flow management and turbine maintenance). A simplified illustration: assume SEB's annual fixed infrastructure costs are RM 500 million, and its variable cost is approximately RM 0.05 per kilowatt-hour (kWh). If SEB generates 5,000 GWh (5,000,000,000 kWh) per year: $ATC = (RM\ 500,000,000 + RM\ 0.05 \times 5,000,000,000) / 5,000,000,000 = (500,000,000 + 250,000,000) / 5,000,000,000 = RM\ 750,000,000 / 5,000,000,000 = RM\ 0.15/kWh$. SEB charges industrial users approximately RM 0.23/kWh, earning an economic profit of approximately RM 0.08/kWh on the cost structure above. Critically, as SEB generates more electricity, the fixed cost is spread over more units, causing ATC to fall continuously — classic economies of scale. This explains why SEB's industrial tariff (approximately RM 0.23/kWh) is significantly lower than Peninsular Malaysia's Tenaga Nasional Berhad (TNB) industrial tariff (approximately RM 0.38/kWh): SEB operates a natural monopoly with declining long-run average costs, enabling low tariffs that attract energy-intensive industries to Sarawak.

1.4 Firm Behaviour and Market Structure

Market structure describes the competitive environment in which firms operate — the number of sellers, the degree of product differentiation, barriers to entry, and the extent of pricing power. Market structure profoundly affects firm behaviour: a firm in a perfectly competitive market has no control over price (it is a "price taker"), while a monopolist has significant pricing power constrained only by the demand curve it faces. The four canonical market structures — perfect competition, monopolistic competition, oligopoly, and monopoly — differ along these dimensions and generate different predictions about pricing, output, profits, and economic welfare.

The Profit Maximisation Rule: $MR = MC$

The profit maximisation rule applies across all market structures: a firm maximises profit by setting output at the level where marginal revenue (MR) equals marginal cost (MC). Marginal revenue is the additional revenue generated by selling one more unit. In perfect competition, each firm is a price taker: the market price is fixed for the individual firm, so $MR = P$ (the price received for every additional unit equals the market price). In monopoly and oligopoly, the firm faces a downward-sloping demand curve for its output, so $MR < P$ — to sell more units, the firm must lower the price on all units sold, making the marginal revenue from the last unit less than the price received. A firm should expand output whenever $MR > MC$ (each additional unit adds more to revenue than it adds to cost, increasing profit) and should contract output whenever $MR < MC$. At $MR = MC$, profit is maximised.

Perfect Competition: Sarawak Pepper Farming

A perfectly competitive market has many sellers, a standardised (homogeneous) product, free entry and exit, and perfect information. Individual firms are price takers. In the short run, a firm may earn economic profit or incur losses. In the long run, free entry and exit drive economic profit to zero — firms earn only normal profit (zero economic profit). Sarawak's pepper farming sector approximates perfect competition closely. Sarawak is the world's largest producer of Sarawak Black Pepper (approximately 20,000–25,000 metric tonnes per year), with thousands of small-scale Dayak and Chinese farmers in the Kuching Division and Serian

area growing the same product. No individual farmer has pricing power — they accept the market price set on the international commodity market, with the Sarawak Pepper Board providing price guidance. The pepper market is highly competitive and price fluctuations directly affect farmers' incomes.

Monopoly: PETROS and SEB

A monopoly is a market with a single seller of a product with no close substitutes and significant barriers to entry. The monopolist faces the entire market demand curve (which slopes downward) and therefore has pricing power — it can choose its price, subject to the constraint that consumers will buy less at higher prices. PETROS (Petroleum Sarawak Bhd), established in 2018, holds a monopoly position in Sarawak's onshore oil and gas sector — the state government granted PETROS the exclusive right to manage petroleum exploration and production activities in Sarawak, a role previously held by the federal government's Petronas. This represents both a political and economic monopoly: PETROS has no direct competitor for onshore Sarawak petroleum resources. Sarawak Energy Berhad (SEB) holds a de facto monopoly in electricity generation and transmission in Sarawak — a natural monopoly justified by the enormous economies of scale that make a single integrated provider cheaper than competing multiple generators and transmission networks.

Oligopoly: Sarawak Telecommunications

An oligopoly is a market dominated by a small number of large sellers, each of which must consider the reactions of rivals when making pricing and output decisions. The interdependence of oligopolistic firms is their defining characteristic — a price cut by one firm will likely trigger retaliatory price cuts by rivals, and a capacity expansion by one firm affects the profitability of all. The Sarawak telecommunications market is a clear oligopoly: following the CelcomDigi merger in 2022 (which combined Celcom and Digi into Malaysia's largest mobile operator by subscriber base), the market is dominated by three players — CelcomDigi, Maxis, and Unifi (Telekom Malaysia). These three providers control the overwhelming majority of mobile broadband, fixed broadband, and pay-TV services in Sarawak. The Malaysian Communications and Multimedia Commission (MCMC) regulates the sector, requiring CelcomDigi to maintain rural coverage including Sarawak's interior districts as a condition of the merger approval.

Monopolistic Competition: Kuching's Food and Beverage Sector

Monopolistic competition combines features of both perfect competition (many sellers, free entry and exit) and monopoly (product differentiation giving each firm some pricing power). Each firm sells a differentiated product — similar but not identical to rivals' products — and therefore faces a downward-sloping demand curve. In the long run, free entry drives economic profit to zero, as in perfect competition. Kuching's vibrant food and beverage sector is a textbook example: hundreds of cafés, restaurants, hawker stalls, and food trucks serve variations of Sarawak laksa, kolo mee, nasi ayam, and other dishes, each with their own brand identity, reputation, and loyal customer base. No single café has monopoly power — there are many alternatives — but each has a unique product (a particular recipe, ambiance, location, or social media reputation) that gives it some degree of pricing power and customer loyalty that pure commodity producers lack.

SUMMARY

Managerial Insight: SEB's Low Tariff Strategy — Economies of Scale as Competitive Policy

Sarawak Energy Berhad's hydroelectric cost structure (massive fixed costs, very low variable costs) means the more electricity it generates and sells, the lower its average total cost per kWh. By maintaining low industrial tariffs (approximately RM 0.23/kWh versus the national average of approximately RM 0.38/kWh for Peninsular Malaysia), SEB uses its cost advantage to attract energy-intensive industries to Sarawak — including aluminium smelting (Press Metal Bintulu, one of the largest aluminium producers in Asia), silicon metal production (Sarawak OSK Holdings), and data centres attracted by the SDEC's digital economy incentives. This creates a virtuous economic cycle: low-tariff energy attracts energy-intensive industries, which expand SEB's revenue base, which enables further infrastructure investment, which further reduces average costs. This is economic strategy in action — using the cost curve, economies of scale, and natural monopoly advantages to drive industrial policy and diversify Sarawak's economic base beyond commodity dependence.

Self-Reflection



1. Think about a purchase decision you made recently — perhaps choosing between two mobile plans, two meal options, or two transport modes. Identify the trade-offs involved, calculate a rough opportunity cost, and determine whether your demand was elastic or inelastic. How does this everyday decision connect to the economic concepts in LU1? If you are currently employed, reflect on your organisation's cost structure: can you identify fixed costs, variable costs, and the point at which economies of scale apply?

SUMMARY

Economics studies the allocation of scarce resources among competing ends. Scarcity forces trade-offs, and every trade-off involves an opportunity cost — the value of the next best alternative foregone.

The Production Possibility Frontier (PPF) illustrates the trade-offs an economy faces when allocating resources between competing uses. Points on the PPF are efficient; points inside are inefficient; points outside are currently unattainable. The Sarawak government's choice between energy infrastructure and digital economy investment illustrates this trade-off in a real policy context.

The supply-demand model explains how market prices and quantities are determined. The demand curve slopes downward (inverse price-quantity relationship); the supply curve slopes upward. Equilibrium is where supply equals demand.

Price elasticity of demand ($PED = \% \Delta Q_d / \% \Delta P$) measures responsiveness of quantity demanded to price changes. Inelastic demand ($|PED| < 1$) means consumers are relatively unresponsive — as demonstrated by rural Sarawak petrol demand ($|PED| \approx 0.27$).

Production costs divide into fixed costs (independent of output) and variable costs (vary with output). The key cost curves are MC, AVC, and ATC. MC intersects ATC at the minimum of ATC — the efficient scale.

Economies of scale cause long-run average cost to fall as output increases — illustrated by SEB's hydroelectric cost structure, which enables Sarawak's competitively low industrial electricity tariffs.

Market structures — perfect competition, monopolistic competition, oligopoly, and monopoly — differ in the number of sellers, product differentiation, and barriers to entry. Sarawak provides clear examples of all four: pepper farming (near-perfect competition), PETROS (monopoly), telco sector (oligopoly), and F&B sector (monopolistic competition).

Multiple Choice Questions (MCQs)

1. Opportunity cost is best defined as:
 - A) The monetary price of a good
 - B) The value of the next best alternative foregone — ANSWER
 - C) The total cost of production
 - D) The tax paid on a purchase

2. If the price of petrol rises by 20% and quantity demanded falls by 5%, the price elasticity of demand is:
 - A) -4.0
 - B) -0.25 — ANSWER
 - C) -0.05
 - D) -2.5

3. A price ceiling set BELOW the equilibrium price will cause:
 - A) A surplus
 - B) A shortage — ANSWER
 - C) No change in quantity
 - D) An increase in supply

4. Marginal cost (MC) intersects average total cost (ATC) at:
 - A) The profit-maximising point
 - B) The minimum of the ATC curve — ANSWER

- C) The maximum of the AVC curve
- D) The break-even point

5. Economies of scale occur when:

- A) Long-run average cost rises as output increases
- B) Long-run average cost falls as output increases — ANSWER
- C) Fixed costs increase with output
- D) Variable costs fall to zero

6. Which of the following best describes PETROS's market position in Sarawak's onshore gas sector?

- A) Perfect competition
- B) Monopolistic competition
- C) Monopoly — ANSWER
- D) Duopoly

7. A government subsidy on a good will:

- A) Shift the supply curve leftward
- B) Shift the demand curve rightward
- C) Shift the supply curve rightward, lowering the equilibrium price — ANSWER
- D) Have no effect on equilibrium

8. The profit-maximisation rule for a firm states that output should be set where:

- A) Total revenue equals total cost
- B) Marginal revenue equals marginal cost — ANSWER
- C) Average revenue equals average cost
- D) Price equals average total cost

Questions for Discussion

1. Sarawak's electricity tariff for industrial users is significantly lower than Peninsular Malaysia's. Using the concepts of cost of production (economies of scale, ATC, MC), supply-demand analysis, and market structure, explain why SEB can sustain such low tariffs, and analyse the economic implications of this pricing strategy for both industrial competitiveness and consumer welfare in Sarawak. (CLO1)
2. When the Malaysian government reduced RON95 petrol subsidies in 2024, the distributional impacts were uneven across different population groups. Using the supply-demand model and price elasticity concepts, analyse how rural Sarawak consumers, urban Kuching consumers, and fuel retailers were each affected differently. What does this analysis suggest about the design of targeted subsidy programmes? (CLO1)
3. Identify a specific market in Sarawak or East Malaysia and classify it according to the four market structure types (perfect competition, monopolistic competition, oligopoly, monopoly). Justify your classification using the criteria of number of sellers, product differentiation, and barriers to entry. What are the implications of this market structure for pricing power, economic profit, and consumer welfare? (CLO1)

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Formative Exercise: Elasticity Calculation

Summary

Elasticity Calculation — Sarawak's Demand for Miri Fresh Fish: The price of fresh fish at Miri public market rises from RM 12/kg to RM 15/kg. Monthly household purchases fall from 8 kg to 7 kg per household. Calculate: (a) the price elasticity of demand using the midpoint method; (b) whether demand is elastic or inelastic; (c) what this implies for fish vendors' total revenue when price rises; and (d) identify two possible determinants of demand that might cause the demand curve for fresh fish to shift rightward in Miri during the festive season.

This is a formative exercise — not graded. It prepares you for the quantitative reasoning component of CA1 and CA2.

Online Learning Resources

The following resources supplement your study of LU1. All are free or freemium.

Khan Academy — Supply and Demand (Free interactive lessons and practice)

<https://www.khanacademy.org/economics-finance-domain/microeconomics/supply-demand-equilibrium>

MRU (Marginal Revolution University) — Principles of Microeconomics (Free video series by Tyler Cowen)

<https://mru.org/courses/principles-economics-microeconomics>

Bank Negara Malaysia — Economic Data and Reports (Free, Malaysian)

<https://www.bnm.gov.my/publications>

**DOSM — Department of Statistics Malaysia —
Economic Indicators (Free, Malaysian)**

<https://www.dosm.gov.my>

END OF TOPIC

TOPIC

2



MACROECONOMICS

*Lecturers: Dr. Colin Wong Koh King; Prof. Dr. Dato
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Topic 2: Macroeconomics

Learning Outcomes

By the end of this topic you should be able to:

- **Measure national income using GDP and GNP and distinguish between the two**
- **Explain the monetary system and the role of Bank Negara Malaysia**
- **Analyse unemployment types and their implications for Sarawak**
- **Apply the AD-AS model to explain economic fluctuations**
- **Evaluate the effects of monetary and fiscal policy in the Malaysian context**



INSPIRING MINDS, SHAPING FUTURE

LO Learning Objectives

By the end of this topic you should be able to:

- **Explain the key macroeconomic indicators** including GDP, GNP, inflation, and unemployment rates.
- **Analyse the mechanisms of monetary and fiscal policy** and their effects on business environments.
- **Evaluate the role of aggregate demand and aggregate supply** in determining national economic output.
- **Apply macroeconomic models** to assess government policy decisions relevant to Malaysia and Sarawak.

KC Key Concepts and Definitions

Concept	Definition	Example / Notes
GDP	Total market value of all final goods and services produced within a country's borders.	<i>Malaysia's GDP measures total economic output including Sarawak's contribution.</i>
Inflation	A sustained increase in the general price level of goods and services.	<i>Rising cost of living in Kuching affects consumer purchasing power.</i>
Unemployment	The proportion of the labour force that is jobless and actively seeking work.	<i>Structural unemployment rises when manufacturing shifts to automation.</i>
Monetary Policy	Central bank actions to control money supply and interest rates.	<i>Bank Negara Malaysia reduces interest rates to stimulate borrowing.</i>

Fiscal Policy	Government use of taxation and public spending to influence the economy.	<i>Malaysia's national budget allocates funds to Sarawak infrastructure.</i>
Aggregate Demand	Total demand for goods and services in an economy at a given price level.	$AD = C + I + G + (X-M)$; <i>includes household consumption and exports.</i>

2.1 Measuring GNP and GDP

Gross Domestic Product (GDP) is the total market value of all final goods and services produced within a country's borders during a given time period (typically one year or one quarter). It is the most widely used measure of an economy's total output and is the primary yardstick for comparing economic performance across countries and over time. Understanding GDP — how it is measured, what it captures, and crucially, what it does not capture — is essential for any manager or policy analyst interpreting economic data.

Three Approaches to Measuring GDP

GDP can be measured using three equivalent approaches, all of which should yield the same result (in theory): (1) the expenditure approach adds up all spending on final goods and services: $GDP = C + I + G + (X - M)$, where C is private consumption, I is investment, G is government spending, X is exports, and M is imports; (2) the income approach sums all income earned in the economy: wages, rent, interest, and profit; (3) the production (value-added) approach sums the value added at each stage of production across all industries. In Malaysia, the expenditure approach is the most widely reported — DOSM publishes quarterly GDP by expenditure components. Malaysia's 2023 GDP was approximately RM 1,815 billion (approximately USD 390 billion), making it the 36th largest economy globally.

GDP vs GNP: The Distinction

Gross National Product (GNP) measures the total income earned by a country's residents, regardless of where that income is earned. $GNP = GDP + \text{Net Factor Income from Abroad (NFIA)}$, where NFIA = income earned by residents abroad minus income earned by foreigners

domestically. For a country like Malaysia — where significant oil and gas production involves foreign companies (particularly in Petronas joint ventures and Sabah offshore concessions) — the difference between GDP and GNP is meaningful: foreign companies' profits repatriated abroad are counted in Malaysia's GDP but not GNP. For Sarawak specifically, the distinction is particularly significant: Sarawak contributes approximately 12% of Malaysia's GDP but a disproportionately large share (approximately 25%) of upstream oil and gas production. Some of this oil income has historically been repatriated to Petronas and the federal government rather than remaining in Sarawak — a central grievance in Sarawak's fiscal federalism disputes, addressed in part by the establishment of PETROS in 2018.

Nominal vs Real GDP

Nominal GDP measures output at current prices — it includes the effects of both changes in output quantity and changes in price levels (inflation). Real GDP adjusts for inflation by valuing all output at the prices of a base year, isolating actual changes in production volume. If Malaysia's nominal GDP grew by 7% in a year when inflation was 4%, real GDP growth was approximately 3% — the economy produced 3% more goods and services in volume terms. The GDP deflator is one measure of economy-wide inflation: $\text{GDP Deflator} = (\text{Nominal GDP} / \text{Real GDP}) \times 100$. GDP per capita (GDP divided by population) is a rough indicator of average living standards — Malaysia's GDP per capita is approximately RM 62,000 (approximately USD 13,500) in 2023, but Sarawak's GDP per capita is somewhat lower than the national average, reflecting both lower income levels in rural areas and the concentration of oil income that benefits production statistics more than household incomes.

2.2 The Monetary System

The monetary system provides the infrastructure through which economic transactions occur. Money performs three fundamental functions: it is a medium of exchange (eliminating the inefficiencies of barter), a unit of account (providing a common measure of value), and a store of value (allowing purchasing power to be saved and transferred over time). Modern economies use fiat money — currency not backed by a commodity like gold, but by government decree (fiat) and the public's confidence in its value. Understanding the monetary system is essential for managers because monetary conditions — interest rates, credit availability, the money

supply — directly affect business borrowing costs, investment decisions, exchange rates, and economic activity.

Bank Negara Malaysia (BNM) and the Overnight Policy Rate

Bank Negara Malaysia (BNM) is Malaysia's central bank, established in 1959. BNM's primary mandate is to promote monetary and financial stability conducive to sustainable growth. BNM's key monetary policy tool is the Overnight Policy Rate (OPR) — the target interest rate at which banks lend to one another overnight. By raising the OPR, BNM makes borrowing more expensive throughout the economy, cooling credit growth, reducing investment and consumption, and moderating inflationary pressure. By lowering the OPR, BNM stimulates borrowing, investment, and economic activity. The OPR signals BNM's monetary policy stance to all financial market participants and quickly transmits through the banking system to affect commercial lending rates. BNM hiked the OPR from 1.75% to 3.00% between May 2022 and May 2023 — a 125 basis point tightening cycle in response to rising inflation — before pausing at 3.00% through 2024.

Money Supply: M1, M2, and M3

The money supply is measured at different levels of liquidity. M1 (narrow money) includes the most liquid components: currency in circulation (notes and coins held by the public) plus demand deposits (current account balances that can be withdrawn immediately). M2 (broad money) adds savings deposits and time deposits with maturities up to one year. M3 (broadest measure) adds longer-term deposits and other less-liquid financial instruments. BNM monitors the growth of M3 as an indicator of overall monetary conditions. Malaysia's dual banking system — operating conventional and Islamic (Syariah-compliant) banking in parallel — is a significant institutional feature. Islamic banks cannot charge interest (riba is prohibited) but instead use profit-and-loss sharing arrangements (mudarabah, musyarakah) and cost-plus financing (murabahah). Malaysia is one of the world's largest Islamic finance markets, and for Muslim entrepreneurs and businesses in Sarawak, Islamic banking products are an important source of business financing.

2.3 Unemployment

Unemployment is measured by the unemployment rate — the percentage of the labour force that is without work, available to work, and actively seeking employment. The labour force comprises all individuals who are either employed or actively seeking work; it excludes the economically inactive (students, homemakers, retirees, discouraged workers). A low unemployment rate is typically associated with economic health, but the unemployment rate must be interpreted carefully: it does not capture underemployment (workers in part-time or low-skill jobs who seek full-time or higher-skill employment) or hidden unemployment (discouraged workers who have given up seeking work).

Types of Unemployment: Frictional, Structural, Cyclical, and Seasonal

Frictional unemployment arises from the normal process of job search and matching — workers transitioning between jobs, recent graduates seeking their first position, and employers searching for qualified candidates. It is temporary and largely unavoidable in a dynamic economy with a functioning labour market. In Sarawak, frictional unemployment is visible in Kuching, where fresh graduates from UNIMAS, Curtin Sarawak, and i-CATS search for professional employment opportunities in the months following graduation.

Structural unemployment arises when workers' skills no longer match the requirements of available jobs — due to technological change, industry restructuring, or geographic mismatch. In Sarawak, structural unemployment is a significant challenge in communities whose livelihoods depended on the timber industry. The dramatic contraction of Sarawak's timber sector from the 1990s to the 2010s (due to depletion of forest resources, stricter environmental regulation, and reduced logging licences) left many workers — particularly in the Rejang Valley districts of Kapit and Belaga — structurally unemployed. Their traditional skills (chainsaw operation, log transport, timber grading) have limited value in the digital economy or energy sector that Sarawak is now building. Reskilling programmes, such as HRD Corp's levy-funded training schemes and SDEC's digital literacy initiatives, are the primary policy response to structural unemployment.

Cyclical unemployment is caused by downturns in the business cycle — during recessions, aggregate demand falls, firms reduce output and lay off workers. Malaysia's 2020 COVID-19 recession caused significant cyclical unemployment: the national unemployment rate peaked at 5.3% in May 2020, the highest since 1990. Sarawak's tourism-dependent workers

(particularly in hotels, tour operators, and food service in Kuching and Miri) experienced severe cyclical job losses during the pandemic lock-down periods. Seasonal unemployment occurs due to predictable, recurring seasonal demand patterns — common in agriculture (harvest vs off-season) and tourism (peak vs off-peak periods).

The Natural Rate of Unemployment and NAIRU

The natural rate of unemployment is the equilibrium unemployment rate consistent with stable inflation — it comprises frictional and structural unemployment. An economy cannot sustain unemployment below the natural rate in the long run without generating accelerating inflation. The Non-Accelerating Inflation Rate of Unemployment (NAIRU) is the more technically precise formulation of this concept. Sarawak's historical unemployment rate of 3–4% is lower than Peninsular Malaysia's, partly because Sarawak's energy, construction, and government sectors provide stable employment, and partly because the informal sector (subsistence agriculture, fishing, small-scale trading in rural areas) absorbs workers who would otherwise appear unemployed. Understanding unemployment dynamics is critical for macroeconomic policy: the correct policy response to frictional unemployment (job matching services, information) differs from the correct response to structural unemployment (reskilling, industrial policy) and cyclical unemployment (macroeconomic stimulus).

2.4 Aggregate Demand (AD)

Aggregate Demand (AD) is the total quantity of goods and services demanded in the entire economy at each price level. The AD curve slopes downward — as the aggregate price level rises, total quantity demanded falls. Three mechanisms explain this negative relationship: the wealth effect (higher prices reduce the real value of money holdings, making consumers poorer and reducing spending); the interest rate effect (higher prices lead to higher demand for money, pushing up interest rates and reducing investment); and the exchange rate effect (higher domestic prices make exports more expensive and imports cheaper, reducing net exports).

Components of Aggregate Demand

Aggregate demand equals the sum of its components: $AD = C + I + G + NX$ (net exports). Private consumption (C) is typically the largest component in most economies — in Malaysia,

private consumption accounts for approximately 54% of GDP. Investment (I) — business spending on capital goods, residential construction, and inventory — is the most volatile component and the primary channel through which monetary policy affects output. Government spending (G) is a direct component of AD that the government can manipulate through fiscal policy. Net exports ($NX = \text{Exports} - \text{Imports}$) reflect the contribution of the external sector — Malaysia is a highly open, export-oriented economy (exports represent approximately 65% of GDP), making NX an important component. Anything that increases any of these components — a rise in consumer confidence, a fall in interest rates, an increase in government spending, a depreciation of the Ringgit (making exports cheaper) — shifts the entire AD curve rightward, increasing total output and price level.

Sarawak Case: SDEC Stimulus and AD

When the Sarawak state government launched the Sarawak Digital Economy Blueprint 2030 with multi-year infrastructure investments — broadband connectivity for rural Sarawak, smart city infrastructure in Kuching, data centre development in Kota Samarahan — this represented a significant positive AD shock through the G (government spending) channel. Government spending on infrastructure directly adds to aggregate demand; it also has a multiplier effect, as construction workers spend their wages, suppliers of materials earn revenue, and this income circulates through the local economy. The Sarawak construction boom of 2022–2025 (Pan Borneo Highway Phase 2, Baleh Dam, Coastal Road) created significant employment and income in the construction sector, adding to Sarawak's aggregate demand and contributing to above-national-average economic growth in the state.

2.5 Aggregate Supply (AS)

Aggregate Supply (AS) describes the total quantity of goods and services that firms in the economy are willing and able to produce at each price level. The shape of the AS curve differs in the short run and the long run — one of the most important distinctions in macroeconomics.

Short-Run and Long-Run Aggregate Supply

In the short run, some input prices (particularly wages) are sticky — they adjust slowly to changes in the aggregate price level. This means that when the price level rises unexpectedly,

firms' revenues rise faster than their costs, making production more profitable and inducing firms to increase output. The short-run aggregate supply (SRAS) curve therefore slopes upward — higher price levels are associated with higher output. In the long run, all input prices fully adjust to the price level — workers renegotiate wages, suppliers renegotiate contracts — and the economy returns to its potential output (full employment output). The long-run aggregate supply (LRAS) curve is vertical at the potential output level, reflecting the classical principle that in the long run, real output is determined by supply-side factors (capital, labour, technology) and is independent of the price level.

AD-AS Model: Malaysia's COVID-19 Recession and Recovery

The AD-AS model provides a powerful framework for analysing Malaysia's 2020 COVID-19 recession. The pandemic caused a simultaneous negative AD shock (private consumption collapsed as lockdown restrictions prevented spending; investment fell sharply as business confidence evaporated; tourism exports fell to near zero) and a negative AS shock (supply chains were disrupted; production facilities were shut by the Movement Control Order; labour supply fell as workers were unable to commute). The combination of leftward AD shift and leftward SRAS shift produced a large fall in real output (Malaysia's GDP contracted by 5.6% in 2020 — the worst performance since the Asian Financial Crisis) and a slight fall in the aggregate price level. The government's response — the *Pemerkasa*, *Pemulihan*, and *Penjana* stimulus packages totalling approximately RM 350 billion in combined government spending and financial guarantees — was an attempt to shift AD rightward, mitigating the demand shortfall. BNM's OPR cuts (from 3.00% to 1.75% between January and July 2020) further stimulated aggregate demand through lower borrowing costs. The 2021–2022 recovery — as vaccinations progressed and restrictions lifted — saw AD shift sharply rightward, driving GDP growth of 8.7% in 2022, the strongest in over two decades.

2.6 Government Debt

Government debt (public debt) is the cumulative total of all past government budget deficits (minus any surpluses). A government runs a budget deficit when its expenditure exceeds its revenue in a given year; the deficit must be financed by borrowing, increasing the total stock

of public debt. The debt-to-GDP ratio is the standard measure of debt sustainability — it compares the debt burden to the economy's capacity to service it.

Malaysia's Federal Debt and Sarawak's Fiscal Position

Malaysia's federal government debt as a percentage of GDP rose from approximately 52% pre-pandemic (2019) to approximately 63–65% in 2022–2024, as massive COVID stimulus packages were financed through borrowing. The statutory debt ceiling under the Government Funding Act 2023 is 65% of GDP — the government has operated near this ceiling, prompting debate about fiscal consolidation. For managers in Malaysia, high public debt raises concerns about future tax increases (to service and repay debt), potential crowding out of private investment (as government borrowing competes with private borrowers for available funds), and the risk of credit rating downgrades that raise borrowing costs for the entire economy. Sarawak's own fiscal position is notably stronger than the federal government's: the state government consistently runs a surplus or near-surplus position, enabled by oil royalty revenues from Petronas (5% of the value of petroleum produced in Sarawak) plus PETROS dividends since 2019. The 2024 Sarawak State Budget was the largest in history at RM 12.3 billion — a direct reflection of the state's oil-revenue-funded fiscal capacity.

2.7 Monetary Policy

Monetary policy refers to the central bank's management of the money supply and interest rates to achieve macroeconomic objectives — typically price stability (low inflation), full employment, and economic growth. BNM conducts monetary policy primarily through the Overnight Policy Rate (OPR). Changes in the OPR transmit through multiple channels: the interest rate channel (higher OPR → higher commercial lending rates → reduced investment and consumption), the credit channel (higher OPR → tighter bank lending standards → reduced credit availability for businesses), the exchange rate channel (higher OPR → capital inflows → Ringgit appreciation → reduced export competitiveness), and the expectations channel (OPR signals BNM's inflation expectations to markets).

BNM's 2022–2024 Rate Cycle and Sarawak SME Impact

BNM raised the OPR five times between May 2022 and May 2023, from a historic low of 1.75% (set during the pandemic) to 3.00%. This tightening cycle was a response to rising inflation — Malaysia's CPI reached 4.7% in August 2022, driven by food prices, energy costs, and post-pandemic supply chain disruptions. For Sarawak SMEs — which typically rely on commercial bank loans for working capital and capital expenditure — each 25 basis point OPR increase raised borrowing costs meaningfully. A business with a RM 500,000 floating-rate loan faced approximately RM 6,250 more in annual interest payments for each 125 basis point cumulative increase. This tightening reduced credit demand from smaller businesses and contributed to some Sarawak SME financing difficulties in 2022–2023. Understanding the relationship between OPR and business borrowing costs is essential for financial planning: businesses should consider fixed-rate financing options when monetary tightening cycles are anticipated.

2.8 Fiscal Policy

Fiscal policy refers to the government's use of taxation and spending to influence macroeconomic conditions. Expansionary fiscal policy (increased government spending or reduced taxes) shifts the AD curve rightward, stimulating output and employment but risking inflation and increased debt. Contractionary fiscal policy (reduced spending or increased taxes) shifts AD leftward, reducing inflationary pressure but risking economic slowdown. The fiscal multiplier describes how a change in government spending amplifies through the economy: each ringgit of government spending generates more than one ringgit of additional GDP, because the initial spending creates income for workers and suppliers who then spend a portion of that income on other goods and services.

Sarawak State Budget 2024: Fiscal Federalism in Action

The 2024 Sarawak State Budget of RM 12.3 billion was the largest in Sarawak's history — a direct product of the state government's fiscal strength from oil royalties and PETROS revenues. Key fiscal priorities included: RM 4.1 billion for infrastructure development (road and bridge construction, rural electrification, clean water supply in the interior); RM 2.3 billion for education and human capital development (including STEM scholarships and the Sarawak Scholarship Fund); RM 1.8 billion for the digital economy (SDEC programmes, broadband

expansion); and RM 0.9 billion for healthcare. For businesses operating in Sarawak, this fiscal expansion has direct implications: government procurement opportunities across infrastructure, technology, and professional services; improved infrastructure reducing logistics costs; and an expanding skilled labour pool from education investments. The Sarawak state government's ongoing fiscal federalism dispute with the federal government — centred on the state's claim to a larger share of oil royalties and greater autonomy over federal taxes collected in Sarawak — reflects a broader tension in Malaysia's fiscal architecture between federal revenue centralisation and state development needs.

2.9 Saving and Investment

In a closed economy, saving equals investment by definition: $Y = C + I + G$ implies $I = Y - C - G = \text{Private Saving} + \text{Public Saving} = \text{National Saving}$. In an open economy, investment can be financed by both domestic saving and foreign capital inflows: $I = \text{National Saving} + \text{Net Capital Inflows}$. The loanable funds model describes the market for saving and investment, with the interest rate as the price that equilibrates saving supply and investment demand. A government budget deficit reduces national saving (since public saving is negative), shifting the supply of loanable funds leftward, raising the equilibrium interest rate, and crowding out some private investment — unless the deficit is financed by foreign capital inflows.

EPF, SME Finance, and Saving in Sarawak

Malaysia's Employees Provident Fund (EPF) is one of the world's largest pension funds, managing approximately RM 1.1 trillion in assets. EPF is a forced savings mechanism: Malaysian private sector employees contribute 11% of their salary and employers contribute 13% (or 12% for higher earners) to EPF accounts. This mandatory saving system generates a large pool of long-term capital that EPF invests in domestic equities, bonds, property, and infrastructure — channelling household savings into productive investment throughout the economy. For Sarawak, EPF investments in PETRONAS-linked companies, Sarawak infrastructure bonds, and Malaysian REITs with Sarawak property holdings provide indirect exposure to the state's economic growth. SME credit access in Sarawak remains a challenge: many small businesses — particularly indigenous-owned enterprises in rural areas — lack the collateral and credit history required for conventional bank loans. PUNB (Perbadanan

Usahawan Nasional Berhad), TEKUN Nasional, and SEDC's loan programmes provide alternative financing channels, but the financing gap for Sarawak SMEs is significant.

2.10 Sources of Growth

Economic growth — the sustained increase in a country's real GDP per capita over time — is the most important macroeconomic phenomenon for long-run living standards. The sources of economic growth can be decomposed using the framework of growth accounting: $\text{output growth} = \text{contribution from labour input growth} + \text{contribution from capital input growth} + \text{Total Factor Productivity (TFP) growth}$. TFP (also known as the Solow residual) captures the portion of output growth not explained by growth in labour or capital — it primarily reflects technological progress, improvements in resource allocation efficiency, better education and skills, improved institutional quality, and managerial innovation.

Malaysia's Growth Sources and Sarawak's Development Trajectory

Malaysia's historical growth from the 1970s through the 1990s was driven primarily by capital accumulation (physical capital investment) and labour force expansion — classic extensive growth. The Malaysian economy's transition toward more intensive growth (driven by TFP and human capital) has been a policy objective since the Multimedia Super Corridor (MSC) initiative in the late 1990s and has accelerated with the Malaysia Digital Economy Blueprint and the New Industrial Master Plan (NIMP) 2030. For Sarawak, TFP-driven growth is now a central policy theme: the SDEC aims to build digital economy capabilities (software development, cloud computing, data analytics) that raise productivity rather than simply adding more labour or capital. Investment in STEM education at UNIMAS, Curtin Sarawak, and i-CATS directly builds the human capital base required for TFP-driven growth. Sarawak's hydrogen export ambitions — building on its renewable energy base (hydroelectric, solar) to produce green hydrogen for export to Japan and South Korea — represent a potential frontier of TFP-intensive growth distinct from the commodity extraction economy of the past.

SUMMARY

Managerial Insight: Sarawak's Fiscal Advantage — Oil Royalties and Economic Resilience

Unlike most Malaysian states, Sarawak receives a 5% oil royalty from Petronas plus PETROS revenues from 2019 onwards, giving the state government exceptional fiscal capacity. The 2024 Sarawak State Budget of RM 12.3 billion was the largest ever — enabling massive investment in infrastructure, education, and digital economy development that directly reduces business operating costs and expands market opportunities. For businesses operating in Sarawak, this fiscal strength translates into: (a) infrastructure reliability — roads, electricity, broadband — that reduces logistics and operating costs; (b) competitive industrial electricity tariffs (RM 0.23/kWh for industrial users); (c) government procurement opportunities across multiple sectors; and (d) an expanding skilled labour pool from state education investments. Understanding fiscal policy — and Sarawak's unique fiscal position within the Malaysian federal system — is essential for managerial decision-making, particularly for businesses seeking government contracts, investment incentives, or infrastructure co-investment opportunities.

Self-Reflection



1. Review Sarawak's latest State Budget announcement (available on sarawak.gov.my). Identify two major fiscal policy measures in the budget. Using the AD-AS model, predict the macroeconomic effects of each measure on Sarawak's output, employment, and price level. Does the budget appear expansionary or contractionary? How does it compare to the federal government's fiscal stance?

Self-Reflection



2. How does the BNM Overnight Policy Rate (OPR) affect your personal financial decisions and your organisation's borrowing costs? When BNM raised the OPR from 1.75% to 3.00% in 2022–2023, what happened to mortgage payments, car loan rates, or business loan costs for people you know? Using the monetary policy transmission mechanisms discussed in subtopic 2.7, trace how this OPR increase worked its way through the economy to affect your daily financial environment.

SUMMARY

GDP measures the total market value of all final goods and services produced within a country's borders. GNP measures income earned by residents regardless of location. The distinction matters for resource-exporting economies like Sarawak, where some petroleum income is repatriated to non-resident owners.

Bank Negara Malaysia (BNM) is Malaysia's central bank, responsible for monetary and financial stability. The Overnight Policy Rate (OPR) is BNM's primary monetary policy tool — raising OPR tightens credit and reduces inflation; lowering OPR stimulates borrowing and economic activity.

Unemployment types include frictional (temporary job search), structural (skills mismatch from technological change), and cyclical (recession-driven). Sarawak's timber sector contraction created significant structural unemployment in rural communities, requiring reskilling responses.

The AD-AS model analyses economy-wide output and price levels. Leftward AD shifts (as in COVID-19) reduce output and prices; rightward AD shifts (government stimulus) increase both. The LRAS is vertical at potential output in the long run.

Fiscal policy (taxation and government spending) can shift AD. Sarawak's 2024 budget of RM 12.3 billion is a powerful example of expansionary state fiscal policy enabled by oil royalties.

Saving and investment are equilibrated through the loanable funds market. EPF is Malaysia's primary institutional savings vehicle, channelling household savings into productive investment.

Long-run economic growth depends on capital accumulation, labour growth, and Total Factor Productivity (TFP). Sarawak's transition from commodity dependence to digital economy aims to shift growth toward TFP-intensive sources.

Multiple Choice Questions (MCQs)

1. GDP measures:

- A) The total income earned by a nation's residents, regardless of location
- B) The total market value of all final goods and services produced within a country in a given year — ANSWER
- C) Total government spending only
- D) Net exports minus imports

2. The Overnight Policy Rate (OPR) in Malaysia is set by:

- A) The Ministry of Finance
- B) The Economic Planning Unit
- C) Bank Negara Malaysia (BNM) — ANSWER
- D) The Prime Minister's Department

3. Structural unemployment arises when:

- A) Workers are temporarily between jobs
- B) The economy is in recession
- C) Workers' skills no longer match available jobs due to technological or economic changes — ANSWER
- D) The minimum wage is set too high

4. In the AD-AS model, an increase in government spending causes:

- A) AD to shift leftward, reducing output
- B) AS to shift rightward, increasing output
- C) AD to shift rightward, increasing output and price level — ANSWER
- D) No change in equilibrium

5. If Malaysia's GDP is RM 1.8 trillion and GNP is RM 1.75 trillion, which of the following is true?

- A) Malaysian residents earn more abroad than foreigners earn in Malaysia
- B) Foreigners earn more in Malaysia than Malaysians earn abroad — ANSWER
- C) The economy has a trade surplus

D) The government has a budget surplus

6. Sarawak's unique fiscal strength compared to other Malaysian states is primarily due to:

A) Tourism revenue

B) Federal government grants

C) Oil royalties and PETROS revenue — ANSWER

D) Income tax collection

7. Quantitative easing (QE) by a central bank primarily involves:

A) Raising interest rates

B) Purchasing government bonds to increase money supply and stimulate the economy — ANSWER

C) Reducing government spending

D) Increasing the reserve requirement for banks

8. Malaysia's EPF (Employees Provident Fund) serves as a mechanism for:

A) Tax collection

B) Forced savings, channelling household savings into long-term investments — ANSWER

C) Government borrowing

D) Controlling inflation directly

Questions for Discussion

1. In 2020, Malaysia's GDP contracted by 5.6% — the worst recession since the Asian Financial Crisis. Using the AD-AS model, analyse the demand and supply-side shocks that caused this contraction, evaluate the government's fiscal and monetary policy responses, and assess how Sarawak's economic structure made it more or less vulnerable to the pandemic recession than Peninsular Malaysia. (CLO1)

2. Explain the relationship between BNM's Overnight Policy Rate (OPR) and (a) Sarawak SME borrowing costs, (b) the Ringgit exchange rate, and (c) household mortgage repayments. If you were a financial manager at a Sarawak-based construction company in 2022 when BNM began its tightening cycle, what steps would you take to manage the impact of rising interest rates on your company's financing costs? (CLO1)
3. Sarawak has one of Malaysia's strongest state fiscal positions, primarily due to oil royalties. Analyse the macroeconomic implications of Sarawak's RM 12.3 billion 2024 state budget for the state's aggregate demand, employment, and long-run growth prospects. What risks might arise from excessive dependence on oil royalty revenues as the global energy transition progresses? (CLO1)

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4. **Department of Statistics Malaysia (DOSM)** (2024). *National Accounts — Malaysia 2023* (Annual Publication). DOSM, Putrajaya.
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Formative Exercise: GDP Calculation

Summary

GDP Calculation — Malaysia's Expenditure Components (2023, RM billions): C (Private Consumption) = 980, I (Investment) = 340, G (Government Spending) = 320, X (Exports) = 1,100, M (Imports) = 950. Calculate: (a) GDP using the expenditure approach ($GDP = C + I + G + X - M$); (b) net exports ($NX = X - M$); (c) the proportion of GDP attributable

to exports. Based on your calculations, what does the high export share tell you about Malaysia's economic structure and its vulnerability to global demand shocks? How might a global recession affecting Malaysia's export partners impact Sarawak specifically?

This is a formative exercise — not graded. It prepares you for the macroeconomic data analysis components of CA1 and CA2.

Online Learning Resources

The following resources supplement your study of LU2. All are free or freemium.

Bank Negara Malaysia — Quarterly Economic Bulletin (Free, official Malaysian source)

<https://www.bnm.gov.my/publications/qb>

Khan Academy — Macroeconomics (Free interactive lessons)

<https://www.khanacademy.org/economics-finance-domain/macroeconomics>

World Bank — Malaysia Economic Monitor (Free research reports)

<https://www.worldbank.org/en/country/malaysia/publication/malaysia-economic-monitor>

DOSM — Malaysia GDP Data and National Accounts (Free, official)

<https://www.dosm.gov.my>

END OF TOPIC

TOPIC

3



MARKET STRUCTURES

*Lecturers: Dr. Colin Wong Koh King; Prof. Dr. Dato
Shazali Abu Mansor; Prof. Dr. Frank Kiong*

Topic 3: Market Structures

Learning Outcomes

By the end of this topic you should be able to:

- Analyse cost structures and their relationship to firm behaviour across market types
- Compare pricing and output decisions across perfect competition, monopoly, oligopoly, and monopolistic competition
- Evaluate the welfare implications of monopoly and oligopoly using deadweight loss analysis
- Apply factor market analysis to wage determination and labour allocation in Sarawak
- Explain income inequality and poverty in the Malaysian context using economic frameworks



INSPIRING MINDS, SHAPING FUTURE

LO Learning Objectives

By the end of this topic you should be able to:

- **Distinguish between competitive markets, monopoly, monopolistic competition, and oligopoly** and analyse their strategic implications for firms.
- **Apply cost and production theory** to evaluate firm behaviour under different market conditions.
- **Analyse labour markets and factor markets** including earnings determination and income inequality.
- **Evaluate market structure impacts** on pricing strategy, innovation, and consumer welfare in Malaysian markets.

KC Key Concepts and Definitions

Concept	Definition	Example / Notes
Perfect Competition	A market with many sellers, identical products, and free entry/exit.	<i>Agricultural commodity markets closely resemble perfect competition.</i>
Monopoly	A single firm controls the entire market with no close substitutes.	<i>PETRONAS holds a monopoly position in Malaysia's upstream oil and gas.</i>
Oligopoly	A market dominated by a small number of large firms with interdependent pricing.	<i>Malaysian telecommunications sector: Maxis, Celcom, Digi, Telekom.</i>
Monopolistic Competition	Many firms sell differentiated products with some pricing power.	<i>Sarawak's food and beverage retail sector — many choices, slight differentiation.</i>

Marginal Cost	The cost of producing one additional unit of output.	<i>A Sarawak bakery deciding whether to produce one more batch of bread.</i>
Price Discrimination	Charging different prices to different consumers for the same product.	<i>Airlines charge different fares for the same flight based on booking time.</i>

3.1 The Costs of Production (Advanced Analysis)

LU1 introduced the fundamental cost concepts — fixed and variable costs, average costs, and marginal cost. LU3 deepens this analysis to include the long-run cost curve, sunk costs, and the connection between cost structure and market conduct. These more advanced cost concepts are essential for understanding the behaviour of firms in different market structures and for making strategic investment decisions.

Short-Run vs Long-Run Cost: The Flexibility Distinction

In the short run (SR), at least one input is fixed — typically capital equipment or production capacity. This constrains the firm's ability to adjust output optimally and generates the U-shaped short-run ATC curve. In the long run (LR), all inputs are variable — the firm can adjust its scale of operations, enter or exit markets, and choose its optimal plant size. The long-run average cost (LRAC) curve is constructed as the envelope of all possible short-run ATC curves — it represents the minimum achievable ATC for each output level when the firm has fully adjusted all inputs. The LRAC curve is typically flatter and lower than any individual SRATC curve, reflecting the efficiency gains available when all inputs are freely adjustable. The minimum point of the LRAC defines the minimum efficient scale (MES) — the smallest output level at which the firm achieves the lowest long-run average cost.

Sunk Costs and the Decision Horizon

A sunk cost is a cost that has already been incurred and cannot be recovered regardless of the decision made going forward. The fundamental economic principle is: sunk costs should be ignored in forward-looking decisions. Only future (incremental) costs and benefits should

influence decisions. This principle has profound implications for business decision-making, yet sunk cost fallacy — allowing past irrecoverable expenditure to distort current decisions — is one of the most common and costly cognitive biases in business. Consider a Sarawak timber company that invested RM 50 million in a logging road to access an area that has since been designated as a protected forest reserve. The RM 50 million is a sunk cost — it cannot be recovered. The question of whether to continue any operations in the area should be based entirely on future incremental costs and revenues, not on the hope of "recovering" the sunk cost. Similarly, a restaurant that has paid non-refundable rent for the year should not continue losing money month-after-month simply to "cover the fixed cost" — if variable costs exceed revenue, it is better to close and minimise ongoing losses.

The MC = ATC Intersection and Market Implications

The proof that MC intersects ATC at the minimum of ATC has critical implications for market analysis. In a perfectly competitive market, long-run equilibrium drives price to the minimum of ATC — the efficient scale. This means competitive markets achieve productive efficiency: they minimise the cost per unit of output. In a monopoly, the profit-maximising output ($MR = MC$) is below the efficient scale, meaning monopolists produce less than the cost-minimising quantity — productive inefficiency is a cost of monopoly power. In a regulated natural monopoly (like SEB), regulators often aim to set price equal to ATC (average cost pricing) rather than marginal cost pricing, because setting $P = MC$ would generate losses for firms with declining cost curves (where $ATC > MC$ throughout the relevant output range). Understanding where the firm is on its cost curve relative to minimum ATC helps managers diagnose whether the firm is over- or under-capacity and whether expansion or contraction would reduce average costs.

3.2 Firms in Competitive Markets

Perfect competition is the benchmark market structure in economic analysis — the standard against which all other structures are compared. A perfectly competitive market has four defining characteristics: (1) many sellers, each too small to affect the market price; (2) homogeneous product — all sellers offer an identical product; (3) free entry and exit — no barriers prevent new firms from entering or existing firms from leaving; and (4) perfect

information — all buyers and sellers have complete information about prices, product quality, and production technology. Under these conditions, individual firms are price takers: they can sell any quantity at the prevailing market price, but nothing at a higher price (customers would simply buy from a competitor offering the identical product more cheaply).

Short-Run and Long-Run Equilibrium

In the short run, competitive firms may earn positive economic profit (if price exceeds ATC at the profit-maximising output level), earn zero economic profit (if price equals minimum ATC — the break-even point), or suffer economic losses (if price falls below ATC). Firms will temporarily continue operating as long as price exceeds AVC — covering variable costs — since fixed costs must be paid regardless. A firm should shut down if price falls below AVC, as continuing to produce would deepen losses beyond the fixed cost alone. In the long run, free entry eliminates economic profit: when existing firms earn positive economic profit, new firms enter, shifting the market supply curve rightward, driving price down until it equals minimum ATC and economic profit returns to zero. Conversely, when firms suffer economic losses, exit occurs, shifting supply leftward, raising price until losses are eliminated. Long-run competitive equilibrium is therefore characterised by zero economic profit (firms earn normal profit — sufficient to cover all opportunity costs) and productive efficiency (production at minimum ATC).

Sarawak Pepper: Near-Perfect Competition in Practice

Sarawak's pepper farming sector closely approximates perfect competition. Thousands of smallholder farmers — predominantly from the Bidayuh and Chinese communities in the Kuching and Serian divisions — grow Sarawak Black Pepper and Sarawak White Pepper. The product is largely homogeneous (though quality variations exist); individual farmers have no pricing power and accept the international commodity price as given; entry and exit are relatively straightforward (pepper cultivation requires moderate capital investment and agricultural land). The Sarawak Pepper Board provides market information and quality grading services that improve the information efficiency of the market. When international pepper prices rise — as they did in 2020–2022 following supply disruptions in Vietnam and India — Sarawak farmers respond by expanding production (moving along the supply curve) and new farmers enter the market (shifting the supply curve rightward), eventually driving prices back

toward the long-run equilibrium. This supply response dynamic — absent in monopolistic or oligopolistic markets — is a key implication of the perfectly competitive market structure.

3.3 Monopoly

A monopoly exists when a single firm is the sole seller of a product with no close substitutes. The monopolist faces the entire market demand curve (downward-sloping) and therefore has pricing power — it can choose its price, subject to consumer demand. The sources of monopoly power include: exclusive government licences (e.g., PETROS's exclusive onshore petroleum rights in Sarawak), control of key resources (e.g., a firm that owns the only mine of a particular mineral), economies of scale making single-firm production cheapest (natural monopoly — e.g., SEB electricity generation), and patents or intellectual property protection.

Monopoly Profit Maximisation and Deadweight Loss

The monopolist maximises profit at the output level where $MR = MC$. Since the monopolist faces a downward-sloping demand curve, $MR < P$ for all positive output levels (to sell more, the monopolist must lower the price on all units, so marginal revenue from the last unit is less than the price). The profit-maximising output is less than the competitive output (where $P = MC$), and the profit-maximising price is above the competitive price. The welfare consequences are significant: the monopolist earns economic profit (a transfer from consumers to the monopolist), but also creates deadweight loss — a reduction in total economic surplus that represents value destroyed by the restriction of output below the competitive level. Deadweight loss is the classic economic argument against monopoly: not only does monopoly redistribute income from consumers to the monopolist, it destroys value that could have been created if the monopolist produced the competitive output level.

Natural Monopoly: The Case of SEB

A natural monopoly arises when a single firm can supply the entire market at lower cost than multiple competing firms — due to economies of scale that persist across the entire relevant range of market output. Sarawak Energy Berhad (SEB) is a natural monopoly in electricity generation and transmission: the enormous fixed costs of hydroelectric dams and transmission grids mean it would be grossly inefficient to have multiple competing generation and

transmission networks. The economic justification for natural monopoly is straightforward: having a single firm avoids duplication of expensive fixed infrastructure. Government regulation is needed, however, to prevent the natural monopolist from exploiting its monopoly position. SEB is regulated by the Sarawak Energy Authority, which sets tariff levels. The regulatory objective is to allow SEB to earn a normal return on its invested capital (covering economic costs) while preventing monopoly pricing that would reduce consumer welfare and deter energy-intensive industrial investment.

PETROS: Government-Created Monopoly

PETROS (Petroleum Sarawak Bhd) was established in September 2018 by the Sarawak state government under the Petroleum Mining Ordinance 1969, which the state government reinterpreted to claim that the 1969 Ordinance (predating the 1974 Petroleum Development Act that gave Petronas national control) gives Sarawak authority over its own petroleum resources. PETROS has been granted exclusive rights to manage upstream petroleum exploration, development, and production in Sarawak — a monopoly position backed by state government legislation. From an economic welfare perspective, PETROS's monopoly position involves trade-offs: it keeps more petroleum revenue within Sarawak (reducing royalty leakage to the federal government) but may reduce exploration investment if PETROS is less efficient than the previous Petronas-dominated system, and raises the cost of gas inputs for Sarawak industrial users if PETROS uses its monopoly position to extract maximum rents.

3.4 Monopolistic Competition

Monopolistic competition describes markets that combine elements of both competitive markets and monopoly. Like perfect competition, monopolistic competition features many sellers and free entry and exit. Unlike perfect competition, products are differentiated — each firm sells a product that is similar but not identical to rivals'. Differentiation may be physical (ingredients, quality, size, packaging), locational (proximity to customers), or based on brand image and perceived quality. Product differentiation gives each firm a small degree of monopoly power — its own downward-sloping demand curve — because some customers have specific preferences for that firm's differentiated product and will continue to buy it even if the price rises slightly.

Short-Run Profit and Long-Run Zero Profit

In the short run, a monopolistically competitive firm maximises profit at $MR = MC$, earning positive economic profit if price exceeds ATC. However, free entry erodes this profit in the long run: new firms enter with similar but differentiated products, shifting existing firms' demand curves leftward (reducing each firm's market share) until price equals ATC and economic profit is zero. Long-run equilibrium in monopolistic competition is characterised by excess capacity — each firm produces below its minimum efficient scale, because the demand curve is tangent to the ATC curve on its downward-sloping portion, not at the minimum. This is the efficiency cost of differentiation: consumers get variety, but production costs are higher than in a perfectly competitive market producing a homogeneous product.

Kuching's F&B Sector and Food Delivery Platforms

Kuching's vibrant food and beverage sector is a canonical example of monopolistic competition. The city's hundreds of cafés, hawker centres, cloud kitchens, and restaurants offer differentiated products — each with a unique menu, ambiance, location, and social media identity. Sarawak laksa, kolo mee, nasi ayam kampung, and Foochow cuisine are served in countless variations across the city, with loyal customer bases for each established vendor. New cafés open regularly (low entry barriers), some earn profits in the short run while their novelty and social media buzz attract customers, but competition and imitation eventually erode those profits. The emergence of Grab Food and Foodpanda as digital food delivery platforms in Kuching has added a new competitive layer: these platforms aggregate multiple monopolistic competitors on a single digital marketplace, reducing search costs for consumers and intensifying price competition among restaurants. Restaurants that previously competed for local foot traffic now also compete in a digital marketplace with transparent pricing and customer reviews — a market structure evolution with significant implications for pricing strategy and customer relationship management.

3.5 Oligopoly

Oligopoly is a market structure in which a small number of firms dominate the market. The defining feature of oligopoly is strategic interdependence: each oligopolistic firm must account

for the anticipated reactions of its rivals when making pricing, output, and investment decisions. This interdependence makes oligopoly the most complex market structure to analyse — the outcome depends not just on costs and demand, but on the strategic interactions among firms, which game theory (covered in LU6) provides tools to model formally.

The Kinked Demand Curve and Price Rigidity

One classical model of oligopoly behaviour is the kinked demand curve model (Sweezy, 1939), which explains the empirical observation that oligopoly prices tend to be relatively rigid (sticky) compared to competitive market prices. The model assumes: if a firm raises its price, rivals will not follow (they will gain market share from the price-raising firm), so the price-raising firm faces highly elastic demand above the current price. If a firm lowers its price, rivals will match the reduction to avoid losing market share, so the price-cutting firm faces inelastic demand below the current price. This creates a "kink" in the demand curve at the current price, with a corresponding discontinuity in the marginal revenue curve. Prices are sticky because a wide range of changes in marginal cost will not change the profit-maximising output level (since MR is discontinuous at the current price).

Collusion, Cartels, and Game Theory

Oligopolists can collectively increase profits by colluding to raise prices and restrict output — behaving like a joint monopolist. A formal agreement to coordinate prices and output is a cartel. The Organisation of Petroleum Exporting Countries (OPEC) is the world's most famous cartel, coordinating production quotas among member countries to maintain oil prices above competitive levels. However, cartels face a fundamental instability: each member has an individual incentive to cheat — to secretly increase output above its quota and earn additional revenue at the high cartel price, while relying on other members to maintain their quotas. If all members cheat, the cartel collapses and prices fall to competitive levels. This "prisoner's dilemma" structure — each firm does better by cheating regardless of what others do — means cartels are inherently unstable and illegal in most jurisdictions (including Malaysia, under the Competition Act 2010).

Sarawak Case: The CelcomDigi Merger and Oligopoly Dynamics

The 2022 merger of Celcom Axiata and Digi.Com Berhad to form CelcomDigi Bhd created Malaysia's largest mobile operator by subscriber base (approximately 20 million subscribers). This merger reduced the effective number of major mobile network operators in Malaysia from four (Celcom, Digi, Maxis, and U Mobile) to three, increasing market concentration. The Malaysian Communications and Multimedia Commission (MCMC) approved the merger subject to conditions including: (a) maintaining service coverage in rural and underserved areas, including Sarawak's interior districts where CelcomDigi must maintain and expand network coverage; and (b) spectrum sharing obligations to ensure smaller operators retain competitive access. From an oligopoly theory perspective, the CelcomDigi merger illustrates how mergers reduce the number of strategic players, potentially facilitating tacit coordination and reducing price competition — the theoretical concern that motivated the MCMC's conditions. It also illustrates the regulator's role in balancing efficiency gains from consolidation against competition and consumer welfare concerns.

3.6 The Markets for Factors of Production

Factor markets are the markets for inputs used in production — labour, capital, land, and entrepreneurship. The economics of factor markets determines wages, interest rates, rents, and profits — the distribution of income across households. Just as goods markets are analysed using supply and demand, factor markets can be analysed using the same framework, though with some important distinctions: demand for factors is a derived demand (derived from the demand for the goods and services that factors help produce), and factor supply curves reflect household decisions about labour-leisure trade-offs, saving decisions, and investment choices.

The Demand for Labour: Marginal Revenue Product

A firm's demand for labour (and other factors) is determined by the value of marginal product (VMP): $VMP = \text{Marginal Physical Product (MPP)} \times \text{Output Price}$. The firm maximises profit by hiring labour up to the point where the wage rate equals the VMP — where the additional revenue from hiring one more worker exactly equals the cost of that worker. A competitive labour market sets wages equal to VMP across all firms, ensuring labour is allocated to its highest-value uses. In practice, labour markets are often not perfectly competitive: monopsony (a single dominant buyer of labour) gives employers wage-setting power analogous to a

monopolist's price-setting power, enabling employers to pay wages below VMP. Large employers in relatively small regional labour markets — such as PETROS in Sarawak's oil and gas sector or SEB in the energy sector — have some degree of monopsonistic power over specialist engineers and technical workers, since these workers have few alternative employers within Sarawak for their specific skills.

Sarawak Labour Market: Construction, Energy, and the Digital Sector

Sarawak's labour market reflects the state's economic structure. The construction sector has been a major employer due to the Pan Borneo Highway, Baleh Dam, and Coastal Road projects — creating high demand for both skilled (project managers, structural engineers) and unskilled (general labourers, machine operators) workers. Construction wages in Sarawak have risen as labour demand expanded faster than local supply, attracting internal migration from rural Sarawak communities and some cross-border workers. The energy sector (SEB, PETROS, upstream oil and gas) creates demand for highly specialised petroleum engineers, electrical engineers, and environmental scientists — typically commanding significantly higher wages than average due to their specialized human capital and the limited supply of qualified workers. The emerging digital economy creates demand for software developers, data analysts, and digital marketers — skills that Sarawak's universities (UNIMAS, Curtin Sarawak, i-CATS) are actively developing but which remain in short supply relative to SDEC's ambitious digital economy targets.

3.7 Earnings and Discrimination

Wage differences across workers reflect many factors: differences in human capital (education, skills, experience — the primary driver of wage differences in competitive markets), compensating differentials (workers in unpleasant, dangerous, or geographically isolated jobs require higher wages to compensate for non-monetary disamenities), efficiency wages (some employers pay above-market wages to attract and retain better workers, reduce turnover, and incentivise effort), and discrimination (wage differences based on characteristics unrelated to productivity, such as race, gender, or ethnicity).

The Economics of Discrimination

Wage discrimination occurs when workers who are equally productive are paid different wages based on characteristics unrelated to their productivity — race, gender, ethnicity, religion, or disability. Gary Becker's (1957) taste for discrimination model argues that discriminatory employers are effectively paying a premium for the "luxury" of satisfying their discriminatory preferences: they pass up profitable trades (hiring equally productive but discriminated-against workers at a lower wage) due to personal prejudice. In competitive markets, Becker argued, discrimination is costly for discriminatory employers and should be competed away by non-discriminating firms — which can hire equally productive workers at lower wages, earning higher profits. This competitive pressure argument suggests discrimination should decline over time in competitive markets, though institutional barriers, social norms, and imperfect information may slow this process.

Earnings and Discrimination in the Sarawak Context

Malaysia's gender wage gap in professional services remains significant, with female workers earning approximately 10–15% less than male workers in comparable roles, even after controlling for education and experience. Research by the Department of Statistics Malaysia consistently shows lower median wages for women in management and technical roles. In Sarawak specifically, the indigenous Dayak communities (Iban, Bidayuh, Orang Ulu) have historically faced barriers to formal sector employment — including inadequate rural educational infrastructure, limited access to professional networks, and geographic isolation from urban employment centres. Bumiputera affirmative action policies (primarily designed for Peninsular Malay Bumiputera) provide some protection and preference for indigenous Sarawakians in public sector employment and business licensing, but the evidence on whether these policies effectively reduce economic disadvantage for rural Sarawak Dayak communities is mixed. From a business management perspective, diversity and inclusion policies that reduce discriminatory hiring and promotion practices are not merely ethical obligations — they improve firm performance by expanding the talent pool, reducing costly turnover, and increasing organisational creativity and problem-solving capacity.

3.8 Income Inequality and Poverty

Income inequality describes the unequal distribution of income across individuals or households in a society. The Lorenz curve is the standard graphical representation of income distribution: it plots the cumulative percentage of income (vertical axis) against the cumulative percentage of the population ranked by income (horizontal axis). If income were perfectly equally distributed, the Lorenz curve would be a 45-degree straight line (the "line of perfect equality"). In practice, the Lorenz curve bows below this line — the greater the bow, the more unequal the income distribution.

The Gini Coefficient and Malaysia's Income Distribution

The Gini coefficient is the most widely used summary measure of income inequality, calculated as the ratio of the area between the Lorenz curve and the line of perfect equality to the total area below the line of perfect equality. A Gini coefficient of 0 indicates perfect equality (everyone has the same income); a coefficient of 1 indicates perfect inequality (one person has all the income). Malaysia's Gini coefficient is approximately 0.40 (as of the 2022 Household Income Survey — DOSM), indicating moderate-to-high income inequality by international standards. For comparison: Sweden $\approx$ 0.27, China $\approx$ 0.38, USA $\approx$ 0.41, Brazil $\approx$ 0.53. Malaysia's T20/M40/B40 framework — grouping households into the top 20%, middle 40%, and bottom 40% by income — is the government's primary policy framework for targeting welfare interventions. In 2022, the T20 household income threshold was approximately RM 15,000/month, the M40 range was approximately RM 5,000–RM 15,000/month, and the B40 income was below approximately RM 5,000/month.

Sarawak's Rural-Urban Income Divide

Sarawak exhibits a pronounced rural-urban income divide that exceeds the national average. Kuching and Miri — Sarawak's two largest urban centres — have household income profiles comparable to or exceeding national averages, driven by oil and gas industry employment, professional services, and public sector jobs. Rural Sarawak — the Baram, Limbang, Kapit, and Sri Aman divisions — has median household incomes significantly below Sarawak's own average, reflecting limited access to formal employment, dependence on subsistence agriculture and fishing, and inadequate infrastructure connectivity. The Sarawak government's significant infrastructure spending (rural electrification reaching 97.5% of households by 2023, rural broadband expansion, rural road construction) is an attempt to close this gap by improving economic connectivity and productivity in the interior. Transfer programmes — the federal

Sumbangan Tunai Rahmah (STR, successor to BRIM), Sarawak's own Sarawakku Sayang financial assistance scheme, and the government's food security programmes — provide income support to B40 households, reducing poverty headcount rates but not necessarily reducing the structural drivers of income inequality.

SUMMARY

Managerial Insight: The CelcomDigi Merger — Oligopoly Theory in Practice

The 2022 merger of Celcom and Digi into CelcomDigi Bhd — creating Malaysia's largest mobile operator — illustrates oligopoly theory in real corporate practice. The merger reduced the number of major mobile operators from four to three, increasing market concentration as measured by the Herfindahl-Hirschman Index (HHI). Economic theory predicts that increased concentration typically leads to higher prices and reduced output — the standard welfare cost of oligopolistic restriction. However, the merger also generated cost synergies through shared network infrastructure, consolidated spectrum holdings, and elimination of duplicated management overhead. The MCMC's approval with conditions — requiring rural coverage obligations specifically including Sarawak's interior districts — reflects the regulator's attempt to balance consumer welfare (preventing anti-competitive price increases) against national development goals (universal connectivity for rural Malaysia). For managers in industries considering M&A activity, this case illustrates the regulatory economics of merger control: market concentration, consumer welfare, and efficiency gains must all be balanced, and regulatory conditions may impose significant ongoing operational obligations.

Self-Reflection



1. Consider the Sarawak telecommunications market. Do you experience the characteristics of oligopoly in your daily life as a mobile or internet customer? Have you ever felt "locked in" to a provider? What switching costs — financial, time-related, or contractual — affect your ability to change providers? How does your personal experience as a consumer connect to the oligopoly theory in LU3? Do you think the CelcomDigi merger has been good or bad for consumers in Sarawak?

SUMMARY

The long-run average cost (LRAC) curve is the envelope of all short-run ATC curves. Sunk costs should be ignored in forward-looking decisions — only incremental future costs and benefits are relevant. MC intersects ATC at minimum ATC — the efficient scale.

Perfect competition (many sellers, homogeneous product, free entry/exit) drives long-run economic profit to zero and achieves productive efficiency. Sarawak pepper farming approximates this structure.

Monopoly (single seller, no close substitutes) enables pricing above MC and generates deadweight loss — unrealised gains from trade. Natural monopolies (SEB electricity, SEB transmission) are justified by economies of scale but require regulation to prevent welfare-reducing monopoly pricing.

Monopolistic competition (many sellers, differentiated products, free entry) generates zero long-run economic profit but with excess capacity and product variety. Kuching's F&B sector exemplifies this structure.

Oligopoly (few large sellers, strategic interdependence) is analysed using game theory and models such as the kinked demand curve. The CelcomDigi merger illustrates real-world oligopoly dynamics and regulatory economics.

Factor market wages are determined by the value of marginal product (VMP). Wage differences reflect human capital, compensating differentials, efficiency wages, and discrimination. Labour markets in Sarawak reflect the state's sectoral structure — energy, construction, and digital economy each create distinct patterns of labour demand.

Income inequality is measured by the Gini coefficient and the Lorenz curve. Malaysia's $Gini \approx 0.40$. Sarawak's rural-urban income divide is pronounced, with B40 rural communities in the interior facing significantly lower incomes than the Kuching and Miri urban areas.

Multiple Choice Questions (MCQs)

1. In a perfectly competitive market, individual firms are:

- A) Price makers who set the market price
- B) Price takers who accept the market price — ANSWER
- C) Monopolists with downward-sloping demand curves
- D) Able to earn long-run economic profits

2. A natural monopoly exists when:

- A) The government grants an exclusive licence
- B) One firm can supply the entire market at lower cost than multiple firms due to economies of scale — ANSWER
- C) There are no substitutes for the product
- D) The government owns the firm

3. Deadweight loss in a monopoly arises because:

- A) The monopolist earns too much profit
- B) The monopolist produces less than the competitive output, creating unrealised gains from trade — ANSWER
- C) Prices are too low
- D) The government imposes a tax

4. The CelcomDigi merger in 2022 reduced the Malaysian mobile telecom market to primarily:

- A) A monopoly
- B) Perfect competition
- C) An oligopoly of 3 major players — ANSWER
- D) A regulated public utility

5. Wage discrimination occurs when workers are paid differently based on:

- A) Productivity differences
- B) Education level
- C) Characteristics unrelated to productivity, such as race or gender — ANSWER

D) Experience level

6. Malaysia's Gini coefficient of approximately 0.40 indicates:

- A) Perfect income equality
- B) Moderate to high income inequality — ANSWER
- C) No income disparity
- D) Income inequality worse than Brazil

7. In monopolistic competition, long-run equilibrium results in:

- A) Economic profit due to differentiated products
- B) Zero economic profit because of free entry and exit — ANSWER
- C) Output where $MC > ATC$
- D) Price equal to marginal cost

8. The value of marginal product (VMP) determines:

- A) The price of final goods in competitive markets
- B) The firm's demand for a factor of production — ANSWER
- C) The government's minimum wage
- D) The Gini coefficient

Questions for Discussion

1. PETROS was established in 2018 as Sarawak's own petroleum company, effectively creating a monopoly position in the state's onshore oil and gas sector. Using monopoly theory, analyse: (a) the economic justification for PETROS's monopoly status; (b) the deadweight loss implications compared to a competitive upstream petroleum market; (c) whether and how government regulation can improve outcomes; and (d) the implications for businesses that rely on Sarawak gas as an input into their production processes. (CLO1)

2. Compare and contrast the market structures of the Sarawak pepper farming sector (near-perfect competition) and the Sarawak electricity supply sector (natural monopoly operated by SEB). For each sector, analyse: the number of sellers, product differentiation, barriers to entry, pricing power, long-run profit prospects, and implications for consumer welfare. What policy prescriptions would you recommend for each market structure? (CLO1)
3. Using the concepts of human capital theory, compensating differentials, and discrimination, explain why wage differences exist between (a) petroleum engineers and general construction labourers in Sarawak, (b) male and female professionals in Malaysian management roles, and (c) urban Kuching workers and rural Kapit Division workers. What economic and policy interventions could reduce unjustified wage differentials while preserving efficiency-enhancing wage differences? (CLO1)

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Formative Exercise: Market Structure Analysis

Summary

Market Structure Analysis: Select a specific industry in Sarawak or Malaysia. Classify it as one of the four market structures (perfect competition, monopolistic competition, oligopoly, or monopoly). Write a 400-word analysis that: (a) justifies your classification

using the criteria of number of sellers, product differentiation, and barriers to entry; (b) describes the pricing behaviour you would expect based on this structure; (c) analyses the welfare implications — does this market structure generate deadweight loss, excess capacity, or economic efficiency?; and (d) identifies any relevant government regulation affecting the market.

This is a formative exercise — not graded. It prepares you for the market analysis component of CA2: Case Study.

Online Learning Resources

The following resources supplement your study of LU3. All are free or freemium.

MRU — Market Structures (Free video series, Marginal Revolution University)

<https://mru.org/courses/principles-economics-microeconomics/market-structures>

Sarawak Economic Planning Unit (EPU) — Sarawak Labour Statistics (Free, official)

<https://www.epu.sarawak.gov.my>

DOSM — Household Income and Expenditure Survey (HIES) Malaysia (Free, official)

<https://www.dosm.gov.my>

Investopedia — Market Structures Explained (Free reference)

<https://www.investopedia.com/terms/m/market-structure.asp>

END OF TOPIC

TOPIC

4



MACROECONOMIC VARIABLES AND ECONOMIC DEVELOPMENT

Topic 4: Macroeconomic Variables and Economic Development

Learning Outcomes

By the end of this topic you should be able to:

- Apply national income accounting to measure economic welfare beyond GDP, including the HDI
- Analyse inflation measurement and its impact on business decisions in Malaysia
- Apply Solow growth model concepts to Malaysia and Sarawak's development trajectory
- Evaluate the role of financial systems in economic growth and capital allocation
- Recommend macroeconomic stabilisation policies for real Malaysian business scenarios



LO Learning Objectives

By the end of this topic you should be able to:

- **Measure and interpret national income accounts** including GDP and GNP using the income, expenditure, and output approaches.
- **Analyse the relationship between savings, investment, and the financial system** in supporting economic growth.
- **Evaluate theories of economic growth** and apply them to Sarawak's development trajectory.
- **Assess the role of government debt, inflation, and unemployment** as macroeconomic policy challenges for Malaysia.

KC Key Concepts and Definitions

Concept	Definition	Example / Notes
National Income Accounting	Systematic measurement of a country's economic activity using GDP, GNP, and NNP.	<i>Malaysia uses both GDP and GNI to report economic performance annually.</i>
Savings and Investment	The relationship between household/firm savings and capital investment.	<i>High savings rates in Malaysia support infrastructure investment in Sarawak.</i>
Economic Growth	Sustained increase in an economy's productive capacity over time.	<i>Sarawak's PCDS 2030 targets economic growth through digital and green sectors.</i>
Consumer Price Index (CPI)	Measures the average change in prices paid by consumers for a basket of goods.	<i>CPI tracks the cost of living changes for households in Sarawak.</i>

Financial System	The network of institutions that channel funds between savers and borrowers.	<i>Banks, capital markets, and insurance firms facilitate Malaysia's capital flows.</i>
Human Development	Broader measure of wellbeing including education, health, and income.	<i>HDI rankings place Malaysia above the ASEAN average, Sarawak improving.</i>

4.1 Measuring a Nation's Income

LU2 introduced the basic concepts of GDP measurement using the expenditure approach. LU4 deepens this analysis by examining the income and production approaches, reconciling the three measures, and critically evaluating the limitations of GDP as a welfare indicator. Advanced national income accounting is not merely an academic exercise — it is the foundation for understanding economic development, comparing welfare across countries and time, and designing effective economic policy.

The Income Approach and Production Approach

The income approach measures GDP by summing all income generated in the economy: wages and salaries (compensation of employees), gross operating surplus (profit, rent, and interest earned by firms and asset owners), and mixed income (income of the self-employed, which combines labour and capital elements). Gross National Income (GNI) — formerly called GNP — equals GDP plus net primary income from abroad. The production (value-added) approach measures GDP by summing the value added at each stage of production across all industries: value added = gross output minus intermediate inputs consumed. All three approaches should yield the same GDP in principle, subject to statistical discrepancies arising from data measurement challenges. Malaysia's DOSM reconciles these approaches in the national accounts, providing confidence in the overall GDP figure.

Beyond GDP: The Human Development Index (HDI)

GDP and GDP per capita are imperfect measures of economic welfare and human development. GDP is a measure of production volume, not of the distribution of income, the

quality of the environment, the degree of social security, or the non-market activities (such as voluntary work, household production, and community care) that contribute significantly to human wellbeing. The United Nations Development Programme's Human Development Index (HDI) provides a broader measure, combining three dimensions: (1) a long and healthy life, measured by life expectancy at birth; (2) knowledge, measured by mean years of schooling and expected years of schooling; and (3) a decent standard of living, measured by GNI per capita (in purchasing power parity terms). The HDI ranges from 0 to 1, with higher values indicating higher human development.

Sarawak Case: Resource Wealth Without Broad Human Development

Sarawak's experience illustrates why GDP alone is an insufficient welfare indicator. Sarawak contributes approximately 12% of Malaysia's GDP, and its oil and gas production adds disproportionately to the state's production statistics. However, GDP per capita in Sarawak is below the national average, and HDI indicators for rural Sarawak — particularly life expectancy, access to quality healthcare, and educational attainment in interior communities — lag significantly behind urban centres. The rural Iban and Orang Ulu communities in the Baram, Kapit, and Belaga divisions face limited access to secondary and tertiary education, specialist healthcare, clean water, and electricity — essential components of human development that oil production statistics do not capture. This divergence between GDP (a production measure) and human welfare (a distribution and access measure) is the "resource curse" in a soft form: high natural resource production enriches state coffers and national statistics without necessarily delivering broad-based human development to the population as a whole. The HDI framework helps reveal these divergences and should inform Sarawak's development policy alongside GDP growth targets.

4.2 Measuring the Cost of Living

Inflation is the general increase in the price level — the average level of prices across the economy. Deflation is a general decrease. Inflation erodes the purchasing power of money: the same amount of Ringgit buys fewer goods and services as prices rise. Accurately measuring inflation is therefore essential for: (a) adjusting wages and contracts for purchasing power changes; (b) setting monetary policy (BNM's OPR decisions depend on inflation forecasts); (c)

evaluating real economic growth (stripping out price effects from nominal GDP growth); and (d) understanding the distributional impact of price changes across different household income groups.

The Consumer Price Index (CPI): Construction and Interpretation

The Consumer Price Index (CPI) measures the cost of a fixed "basket" of goods and services representative of a typical household's consumption pattern. DOSM constructs Malaysia's CPI by surveying prices of approximately 460 items across 12 categories: Food and Non-Alcoholic Beverages; Alcoholic Beverages and Tobacco; Clothing and Footwear; Housing, Water, Electricity, Gas and Other Fuels; Furnishings, Household Equipment and Routine Household Maintenance; Health; Transport; Communication; Recreation Services and Culture; Education; Restaurants and Hotels; and Miscellaneous Goods and Services. Food and Non-Alcoholic Beverages has the highest weight in Malaysia's CPI basket (approximately 28.5%), reflecting the importance of food expenditure for Malaysian households — particularly B40 households for whom food constitutes a larger share of total spending than for higher-income households.

Malaysia's 2022–2024 Inflation Episode and Sarawak's Supply Chain

Malaysia's CPI inflation rose to 4.7% year-on-year in August 2022 — the highest rate in over a decade — driven by post-COVID supply chain disruptions, global commodity price increases (especially palm oil, cooking oil, and grains following the Russia-Ukraine conflict), and domestic demand recovery. For Sarawak, inflation was further amplified by the state's geographic supply chain constraints: many food and consumer goods consumed in Sarawak are either imported from Peninsular Malaysia (adding transport costs) or from abroad through the Port of Kuching and Senari Port in Miri. Remote divisions — Limbang, Baram, Kapit — face even higher food price inflation than urban Sarawak, as supply chain costs increase with distance and limited road access. The 2024 targeted subsidy rationalisation — removing the blanket RON95 petrol subsidy for higher-income households — had a measurable direct impact on Sarawak's CPI, particularly for the transport and food components (since food logistics in Sarawak are heavily dependent on diesel-fuelled road and river transport).

Real vs Nominal Values and Purchasing Power

Distinguishing between real and nominal values is one of the most important skills in applied economics. Nominal values are measured in current prices — they include the effects of inflation. Real values are adjusted for inflation — they measure purchasing power or production volume in constant prices. The real interest rate = nominal interest rate – expected inflation rate (the Fisher equation). The real wage = nominal wage / price level. A worker who receives a 5% nominal wage increase during a year when inflation is 4.7% has experienced a real wage increase of only 0.3% — their purchasing power has barely improved. For managers, the distinction between real and nominal values affects: capital budgeting (should discount rates be real or nominal?); wage negotiations (are workers' real wages keeping pace with productivity growth?); long-term contracts (should prices be fixed in nominal terms or indexed to inflation?); and financial reporting (are revenue and cost trends real or inflation-driven?). Sarawak logistics companies that saw revenue rise 10% in 2022 but experienced 8% cost inflation from diesel price increases earned a real revenue increase of only approximately 2% — the nominal figures were misleading without inflation adjustment.

4.3 Production and Growth

The Solow growth model (Robert Solow, 1956) is the foundational framework for understanding long-run economic growth. The model asks: what determines an economy's steady-state level of GDP per worker, and what drives improvements in that steady-state over time? The model provides clear answers: in the absence of technological progress, capital accumulation alone cannot sustain per-capita growth indefinitely — diminishing returns to capital ensure that the economy converges to a steady state at which investment exactly offsets depreciation. Sustained per-capita growth requires technological progress — improvements in total factor productivity (TFP) that raise the output produced per unit of capital and labour.

The Solow Model: Capital Accumulation and Steady State

The production function in the Solow model is $Y = A \times F(K, L)$, where Y is output, A is total factor productivity, K is the capital stock, and L is labour. Output per worker $y = Y/L = A \times f(k)$, where $k = K/L$ is the capital-labour ratio. Investment adds to the capital stock; depreciation reduces it. The economy reaches a steady state when investment per worker exactly equals depreciation per worker: $\Delta k = sy - \delta k = 0$, where s is the savings rate and δ is the depreciation

rate. At the steady state, capital per worker and output per worker are constant. Crucially, a higher savings rate raises the steady-state capital-labour ratio and steady-state output per worker — once. Only ongoing TFP growth (improvements in A) can generate sustained per-capita income growth above the steady state.

Conditional Convergence and Sarawak's Growth Trajectory

The Solow model predicts conditional convergence: economies with lower capital per worker (and therefore higher marginal productivity of capital) should grow faster than richer economies with more capital — provided they have similar steady states (same savings rate, population growth rate, and TFP). This convergence property partly explains why China and the East Asian economies grew faster than the United States for decades — they started from a lower capital base. For Sarawak, the conditional convergence prediction is relevant: if Sarawak can increase its savings rate (through EPF contributions, state investment), attract foreign capital (through SEB's low-cost energy advantage and SDEC incentives), and improve TFP (through education investment, digital economy development), it should grow faster than steady-state Malaysia — closing the income gap with wealthier states. The SCORE (Sarawak Corridor of Renewable Energy) initiative represents a deliberate capital deepening strategy: attracting energy-intensive industries (aluminium smelting, silicon production, data centres) that add capital stock to the Sarawak economy, raising the capital-labour ratio and output per worker.

4.4 Saving, Investment, and the Financial System

Economic growth requires investment — additions to the capital stock that increase the economy's productive capacity. Investment must be financed by saving: either domestic saving (households and businesses setting aside current income for future use) or foreign saving (capital inflows from abroad). The financial system — banks, capital markets, insurance companies, pension funds, and other financial intermediaries — performs the essential function of channelling saving into productive investment, allocating capital to its highest-value uses.

Malaysia's Dual Financial System: Conventional and Islamic Banking

Malaysia operates one of the world's most sophisticated dual banking systems, in which conventional banking (based on interest-bearing deposits and loans) and Islamic banking (based on Syariah-compliant, interest-free profit-sharing and cost-plus financing) operate under the same regulatory framework overseen by Bank Negara Malaysia. Malaysia's Islamic banking sector accounts for approximately 40% of total banking system assets — one of the highest proportions globally. For businesses and individuals in Sarawak's Malay and Melanau communities (and increasingly for non-Muslim businesses seeking Islamic finance products for ethical or reputational reasons), Islamic banking products include: murabahah (cost-plus financing for asset purchases), ijarah (Islamic leasing), musyarakah (equity partnership financing), and sukuk (Islamic bonds used to finance infrastructure and large capital projects). Sarawak's state government has issued sukuk instruments to finance infrastructure — including roads and utilities — providing investors with Syariah-compliant exposure to Sarawak's economic growth.

Financial Development and Economic Growth

Financial development — the depth, efficiency, and accessibility of the financial system — is strongly positively associated with economic growth across countries and over time. A well-developed financial system reduces transactions costs and information asymmetries in credit markets, enabling more productive investment projects to be financed. For Sarawak SMEs, particularly indigenous-owned businesses in rural areas, access to formal financial services remains a constraint on growth. The "financial inclusion gap" — the proportion of adults without bank accounts, insurance coverage, or access to affordable credit — is larger in rural Sarawak than in urban Malaysia. BNM's Financial Inclusion Framework, TEKUN Nasional's microfinancing programmes, and SDEC's digital payment infrastructure development are all directed at closing this gap, recognising that financial exclusion is both an equity issue and an economic efficiency issue — productive entrepreneurs cannot invest if they cannot access capital.

4.5 The Basic Tools of Finance

Financial economics provides managers and policy analysts with quantitative tools for evaluating investments, managing risk, and making resource allocation decisions across time.

Three fundamental tools are present value analysis, risk and return assessment, and portfolio diversification — all of which have direct managerial applications in Sarawak's capital-intensive economic environment.

Present Value, NPV, and Investment Evaluation

The concept of present value (PV) recognises that a sum of money available today is worth more than the same sum available in the future — because money available today can be invested to earn a return. The present value of a future cash flow of RM X received n years from now, at a discount rate of r , is: $PV = X / (1 + r)^n$. Net Present Value (NPV) of an investment is the sum of the present values of all future cash flows minus the initial investment cost: $NPV = -C_0 + \sum [CF_t / (1 + r)^t]$. A positive NPV indicates the investment creates value (the present value of future cash flows exceeds the cost); a negative NPV destroys value. NPV is the theoretically correct criterion for investment decisions: accept all projects with positive NPV, reject those with negative NPV.

Managerial Application: Baleh Dam NPV Calculation

The Baleh Dam hydroelectric project (760 MW capacity, total capital cost approximately RM 8.6 billion, with SEB targeting commissioning in 2026) provides a practical NPV illustration. Assume the dam generates RM 1.2 billion in annual electricity revenue (at RM 0.23/kWh industrial tariff $\times$ 5,217 GWh capacity), with annual operating and maintenance costs of RM 150 million, yielding annual net cash flow of approximately RM 1.05 billion. For a dam with a 50-year operating life and a discount rate of 6% per annum: $PV \text{ of annuity} = CF \times [1 - (1 + r)^{-n}] / r = RM\ 1.05B \times [1 - (1.06)^{-50}] / 0.06 \approx RM\ 1.05B \times 15.76 \approx RM\ 16.55 \text{ billion}$. $NPV = RM\ 16.55B - RM\ 8.6B = \text{approximately } RM\ 7.95 \text{ billion}$ — significantly positive. This illustrates why the Sarawak government views the Baleh Dam as a sound economic investment, even at RM 8.6 billion capital cost: the long-term revenue stream (underpinned by energy-intensive industrial demand for low-cost electricity) substantially exceeds the construction cost in present value terms. Managers evaluating long-term supply contracts and industrial investments in Sarawak can use the same NPV framework to assess the value of locking in SEB's low industrial electricity tariff for extended periods.

Risk, Return, and Portfolio Diversification

Risk and return are fundamental trade-offs in all investment decisions: higher expected returns generally require accepting higher risk. Diversification — spreading investment across multiple assets whose returns are not perfectly correlated — reduces portfolio risk without sacrificing expected return. This is the key insight of Modern Portfolio Theory (Markowitz, 1952): combining assets with low correlations reduces the variance of portfolio returns because when one asset performs poorly, others may perform well, offsetting the loss. For businesses in Sarawak, diversification considerations are directly relevant: a company heavily dependent on a single commodity (e.g., pepper exports) faces concentrated price risk; diversifying into additional agricultural products, processed goods, or services reduces the volatility of overall business income. Sarawak's government has explicitly pursued an economic diversification strategy — reducing dependence on oil and gas through SDEC digital economy investments, tourism development, and manufacturing incentives — applying portfolio diversification principles at the macroeconomic policy level.

4.6 Unemployment (Advanced Treatment)

LU2 introduced the basic typology of unemployment (frictional, structural, cyclical, seasonal) and the natural rate of unemployment. LU4 deepens this analysis with two advanced concepts: NAIRU (the Non-Accelerating Inflation Rate of Unemployment) and hysteresis in unemployment — both essential for understanding macroeconomic stabilisation policy and the long-run effects of recessions.

NAIRU: The Inflation-Unemployment Trade-off

The Phillips Curve describes the short-run inverse relationship between unemployment and inflation: as unemployment falls (the economy tightens), inflationary pressure rises; as unemployment rises (recession), inflationary pressure falls. This trade-off occurs because in tight labour markets, workers have more bargaining power and push for higher wages, which firms pass on as higher prices. The NAIRU (Non-Accelerating Inflation Rate of Unemployment) is the unemployment rate consistent with stable (non-accelerating) inflation. If actual unemployment falls below the NAIRU, inflation accelerates; if it rises above the NAIRU, inflation decelerates. The NAIRU is not a fixed number — it shifts with changes in structural factors like labour market institutions, unemployment insurance generosity, skill

mismatches, and demographic composition. Malaysia's NAIRU is estimated at approximately 3–3.5% by Bank Negara Malaysia economists, though with significant uncertainty. In 2022–2023, as unemployment fell toward this level following pandemic recovery, BNM's OPR hikes were in part motivated by concern that tightening labour markets would fuel wage-price spiral inflationary dynamics.

Hysteresis: When Recessions Have Permanent Effects

Hysteresis in unemployment refers to the phenomenon whereby a temporary cyclical increase in unemployment leads to a permanent (or persistent) increase in the NAIRU — the unemployment rate that is consistent with stable inflation drifts upward after a recession. The mechanism: during a prolonged recession, unemployed workers experience skill erosion (their human capital depreciates through non-use), their professional networks atrophy, and employers come to view long-term unemployment as a negative signal (statistical discrimination). These effects can permanently reduce workers' employment prospects even after the recession ends, raising the equilibrium unemployment rate. Hysteresis has significant implications for macroeconomic policy: it provides a stronger argument for aggressive counter-cyclical stimulus during recessions, because the cost of allowing unemployment to remain elevated is not just temporary income loss but permanent damage to the economy's productive capacity. In Sarawak's tourism sector — where the COVID-19 pandemic caused a prolonged period of near-zero tourism activity from 2020 to 2022 — hysteresis effects are visible: experienced hospitality workers who left the sector during the pandemic have not all returned, forcing hotels and tour operators to retrain new entrants at higher cost and lower initial productivity.

4.7 Government Debt

LU2 introduced the basic concept of government debt and Malaysia's fiscal position. LU4 deepens this analysis with the concepts of debt sustainability, the dynamics of debt accumulation, generational equity, and the distinction between federal and state debt structures — particularly relevant for Sarawak's unique fiscal position.

Debt Sustainability: The Interest Rate vs Growth Rate Dynamics

Government debt becomes unsustainable when the debt-to-GDP ratio rises indefinitely — meaning the government cannot eventually stabilise the ratio without large, sustained fiscal adjustments (tax increases or spending cuts). The key condition for debt sustainability is: the primary budget balance (revenue minus non-interest expenditure) must be sufficient to prevent the debt-to-GDP ratio from rising without limit. The critical dynamic is the relationship between the interest rate (r) and the GDP growth rate (g): if $r < g$ (the interest rate on government debt is lower than the economy's growth rate), the government can sustain a primary deficit indefinitely — economic growth "outpaces" debt accumulation. If $r > g$, the debt-to-GDP ratio will rise without a primary surplus. For Malaysia in 2024, with the OPR at 3.00% and nominal GDP growth at approximately 5–6%, the $r < g$ condition is approximately satisfied, suggesting Malaysian federal debt dynamics are manageable — though the margin is narrow and any growth slowdown or interest rate increase could reverse this.

Generational Equity and the Burden of Debt

Government debt raises intergenerational equity concerns: borrowing today effectively transfers the obligation to repay to future taxpayers — today's government consumption is financed by future generations' tax burden. However, this intergenerational transfer is not automatic or unambiguous: if the debt finances productive public investment (infrastructure, education, research) that raises future GDP, future generations may benefit enough from the higher productive capacity to more than offset the debt service burden. The key distinction is between debt-financed consumption (which imposes a net burden on future generations) and debt-financed productive investment (which may leave future generations better off in net terms). Sarawak's capital investments — Baleh Dam, Pan Borneo Highway, rural broadband — are the kind of infrastructure investments that economists would argue justify some degree of debt financing: the benefits (higher productivity, better connectivity, cheaper energy) accrue over decades, not just in the year of investment.

Malaysia's Debt Ceiling and Fiscal Reform Agenda

Malaysia's Government Funding Act 2023 sets a statutory federal government debt ceiling at 65% of GDP. As of 2024, federal government debt (direct debt plus government guarantees) is at approximately 63–65% of GDP — near but not yet at the ceiling. The government's medium-term fiscal consolidation plan aims to reduce the budget deficit from approximately 5% of GDP (2023) toward 3% by 2026, through subsidy rationalisation (RON95 targeted

subsidies, electricity subsidy restructuring), expanded tax revenues (expanded Sales and Services Tax, carbon tax proposals), and spending prioritisation. For businesses and investors in Malaysia, fiscal consolidation carries both risks (slower government spending reduces AD) and benefits (lower risk of credit rating downgrades, more stable long-run macroeconomic environment, lower risk premium on Malaysian interest rates).

4.8 Measuring GNP and GDP — Advanced Reconciliation

LU4's final subtopic revisits national income measurement with the analytical sophistication to reconcile the three approaches (expenditure, income, production), to address specific measurement challenges in the Malaysian and Sarawak context, and to understand how measurement gaps — particularly the informal economy — affect the accuracy of economic statistics.

Reconciling the Three Approaches

The three approaches to measuring GDP — expenditure, income, and production — are conceptually equivalent: every act of production generates income for the factor owners who contributed to that production, and income earned must eventually be either spent or saved. In practice, statistical discrepancies arise because data sources differ across the three approaches: the expenditure approach uses household surveys, enterprise surveys, and trade data; the income approach uses tax records and labour force surveys; the production approach uses industry output statistics. DOSM reconciles these discrepancies through adjustments. The existence of statistical discrepancies does not indicate errors — it reflects the inherent difficulty of comprehensively measuring all economic activity in a large, diverse economy. For managers and analysts, it is important to understand which approach a given GDP figure is based on and how measurement limitations might affect its reliability.

The Oil and Gas Sector: GDP Accounting Complexities

Malaysia's oil and gas sector illustrates several important GDP accounting complexities. Depletion of natural resources — the extraction of oil and gas from the ground — is treated as final output in conventional GDP accounting (the extraction generates revenue counted as production), but it also represents the running-down of a non-renewable natural asset. "Green

GDP" or "adjusted net savings" measures attempt to account for resource depletion by subtracting the value of depleted natural assets from conventional GDP, providing a more accurate picture of sustainable national income. For Sarawak, this matters significantly: a large portion of Sarawak's historical GDP growth was attributable to increasing oil and gas extraction — production that was simultaneously depleting a finite natural resource base. On a green GDP basis, Sarawak's net wealth creation from oil and gas was considerably less than the conventional GDP figures suggested, particularly in the decades of peak extraction from 1970 to 2010.

The Informal Economy and GDP Underestimation in Sarawak

The informal economy comprises economic activity that is not recorded in official statistics — including subsistence agriculture, unregistered small businesses, cash-in-hand labour, and community exchange networks. In Sarawak's rural interior, a significant proportion of economic activity is informal: Iban longhouse communities engage in subsistence rice cultivation and fishing; indigenous craft production is sold through informal community markets; informal labour exchanges (bedurok in Iban tradition) substitute for formal labour market transactions. This informal activity contributes to household welfare and community cohesion but is not captured in GDP statistics — leading to systematic underestimation of economic activity and welfare in rural Sarawak. Policy implications are significant: if Sarawak's formal GDP understates the true level of economic activity, investment in rural infrastructure and services may be lower than warranted by the actual size of the rural economy; and welfare statistics based on formal income may overstate poverty in communities with high levels of subsistence production and social capital.

SUMMARY

Managerial Insight: Present Value and Sarawak's Baleh Hydroelectric Dam

The RM 8.6 billion Baleh Dam (760 MW capacity, targeted completion 2026) is one of Sarawak's largest infrastructure investments. Using present value analysis: if the dam generates approximately RM 1.05 billion in annual net cash flow (revenue minus operating costs) for 50 years, at a discount rate of 6%, the NPV = $RM\ 1.05B \times [(1 - 1/1.06^{50}) / 0.06] \approx RM\ 1.05B \times 15.76 \approx RM\ 16.55\ billion$ — well above the RM 8.6

billion capital cost. The NPV of approximately RM 7.95 billion illustrates why the Sarawak government views this investment as economically justified despite the large upfront cost. For managers in energy-dependent industries in Sarawak — including aluminium smelting, silicon metal production, data centres, and large-scale manufacturing — understanding NPV analysis is essential for evaluating the long-term economic value of locking in SEB's low industrial electricity tariff through long-term energy supply agreements. A 20-year energy supply contract at RM 0.23/kWh, compared to the Peninsular Malaysia rate of RM 0.38/kWh, represents a cost advantage of RM 0.15/kWh — with a present value that can be a decisive factor in investment location decisions.

Self-Reflection



1. Sarawak's GDP per capita is below the national average despite being resource-rich in oil, gas, and timber. Using the concepts in LU4 — HDI limitations of GDP, the Solow growth model, capital deepening, TFP, and financial inclusion — construct a multi-factor explanation for why resource wealth has not fully translated into broad human development for Sarawak's rural communities. What specific policy interventions, grounded in economic theory, would you recommend to accelerate Sarawak's human development convergence with national averages?

SUMMARY

Advanced national income accounting uses three equivalent approaches — expenditure, income, and production — all yielding the same GDP. Understanding the limitations of each approach is essential for accurate economic analysis.

The Human Development Index (HDI) improves on GDP by incorporating health (life expectancy), education (schooling years), and income dimensions of welfare. Sarawak's HDI profile reveals that resource-sector GDP growth has not fully translated into broad human development for rural communities.

Malaysia's 2022 inflation spike (CPI 4.7%) was driven by food prices, energy costs, and supply chain disruptions. Sarawak faced amplified food inflation due to geographic

supply chain constraints. The distinction between real and nominal values is essential for accurate business performance analysis.

The Solow growth model explains long-run per-capita growth through capital accumulation, labour growth, and TFP. Sustained per-capita growth requires TFP improvements — not just capital deepening. Sarawak's SCORE initiative is a capital deepening strategy; SDEC's digital economy programmes aim to generate TFP growth.

Malaysia's dual financial system (conventional + Islamic banking) channels savings into productive investment. Financial inclusion — extending affordable financial services to rural Sarawak communities — is both an equity goal and an economic efficiency objective.

Present value (PV) and NPV are the theoretically correct tools for investment evaluation. The Baleh Dam NPV calculation ($\approx$ RM 7.95 billion positive NPV) illustrates how long-term infrastructure investments can be rigorously evaluated and justified.

NAIRU and hysteresis explain why unemployment may persist above the natural rate after recessions. Sarawak's tourism sector displayed hysteresis effects following the 2020–2022 pandemic, with implications for labour market policy and reskilling investment.

Malaysia's federal debt (approximately 63–65% of GDP in 2024) is near the statutory ceiling. Debt sustainability depends on the r vs g relationship. Sarawak's own fiscal position is unusually strong due to oil royalties, enabling continued capital investment without compromising fiscal sustainability.

Multiple Choice Questions (MCQs)

1. The Human Development Index (HDI) improves on GDP as a welfare measure by including:

- A) Only income per capita
- B) Income, health (life expectancy), and education (years of schooling) —

ANSWER

- C) Trade balance and government spending
- D) Environmental sustainability indicators only

2. If Malaysia's nominal GDP grew by 7% and inflation was 4%, then real GDP growth was approximately:

- A) 11%
- B) 4%
- C) 3% — ANSWER
- D) 7%

3. According to the Solow growth model, sustained long-run per-capita growth depends on:

- A) Capital accumulation alone
- B) Total Factor Productivity (TFP) and technological progress — ANSWER
- C) Government spending increases
- D) Trade surplus growth

4. The Baleh Dam investment decision is best evaluated using:

- A) Supply-demand analysis
- B) Game theory
- C) Net Present Value (NPV) calculation — ANSWER
- D) Income inequality measures

5. Malaysia's dual banking system means:

- A) Two central banks operate simultaneously
- B) Both conventional and Islamic banking systems operate under the same regulatory framework — ANSWER
- C) All banks must be government-owned
- D) Foreign banks are prohibited from operating in Malaysia

6. Hysteresis in unemployment refers to:

- A) Seasonal unemployment patterns

B) The phenomenon where high unemployment persists even after economic recovery, due to skill erosion and stigma — ANSWER

C) Only structural unemployment

D) The natural rate being zero

7. Government debt becomes unsustainable when:

A) The debt-to-GDP ratio exceeds 30%

B) The interest rate on debt persistently exceeds the GDP growth rate, causing debt to spiral upward — ANSWER

C) Any level of deficit spending occurs

D) International rating agencies downgrade the country

8. The informal economy in Sarawak's rural areas leads to:

A) GDP overestimation

B) GDP underestimation — economic activity not captured in national accounts — ANSWER

C) Higher measured inflation

D) Trade deficit

Questions for Discussion

1. Malaysia's CPI rose to 4.7% in 2022, primarily driven by food prices and fuel costs. For a Sarawak-based logistics company relying on diesel fuel and employing 200 workers: (a) analyse the impact of this inflation on the company's real operating costs; (b) explain the distinction between nominal and real wages and its implications for worker welfare and union negotiations; (c) recommend two specific business strategies to manage cost inflation, citing the economic models and concepts from LU4; and (d) discuss whether the company should fix or index its service contracts to the CPI. (CLO2)
2. Using the Solow growth model framework, compare the growth strategies of two hypothetical Sarawak economies: Economy A pursues high savings and capital investment (building dams, roads, and factories); Economy B pursues TFP growth

through education investment, digital economy development, and research and development. Using the steady-state analysis from LU4, predict the long-run growth trajectory of each approach, and recommend a balanced strategy for Sarawak's economic development. (CLO2)

3. Critically evaluate the claim that "GDP per capita is a sufficient measure of Sarawak's economic development." Using the concepts of HDI, informal economy measurement, resource depletion accounting, and income distribution, construct a comprehensive critique of GDP as a welfare indicator and propose a composite index more suited to measuring Sarawak's economic wellbeing. Your answer should reference at least two LU4 subtopics explicitly. (CLO2)

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Formative Exercise: Inflation Impact Analysis (CA1 Preparation)

Summary

Assignment: A Sarawak-based food manufacturer (e.g., belacan producer in Sibu, sago processing plant in Mukah, or pepper cooperative in Serian) reports the following data for 2022–2024: (i) Raw material costs rose 18% (palm oil, sago starch, fuel); (ii) Worker wages increased by 5% per annum; (iii) Selling prices increased by 8%; (iv) Bank loan interest rate rose from 3.5% to 4.5% following BNM OPR increases. Using the LU4 concepts of CPI, real vs nominal wages, present value, and financial system analysis: (a) Calculate the real wage change for workers over the two-year period; (b) Calculate the change in real borrowing cost; (c) Analyse whether the company's profit margins have expanded or contracted in real terms; (d) Recommend two strategies — one operational, one financial — that the company could implement to restore real profitability. Present your analysis in a structured 600-word business memo format.

This is a formative exercise — not graded. It is direct preparation for CA1 (Assignment) and CA2 (Case Study). Use real Malaysian inflation data from DOSM (<https://www.dosm.gov.my>) and BNM OPR data from Bank Negara Malaysia (<https://www.bnm.gov.my>) to add credibility to your analysis.

Online Learning Resources

The following resources supplement your study of LU4. All are free or freely accessible.

World Bank — Malaysia Open Data: GDP, inflation, HDI, and financial sector indicators (Free)

<https://data.worldbank.org/country/malaysia>

IMF — World Economic Outlook Database: Malaysia macroeconomic projections (Free)

<https://www.imf.org/en/Publications/WEO>

Bank Negara Malaysia — Annual Report and Monetary Policy Statements (Free, Official)

<https://www.bnm.gov.my/publications/ar>

United Nations Development Programme — HDI Database: Sarawak and Malaysian human development data (Free)

<https://hdr.undp.org/data-center>

Khan Academy — Macroeconomics: GDP, inflation, Solow growth model video tutorials (Free)

<https://www.khanacademy.org/economics-finance-domain/macroeconomics>

END OF TOPIC

TOPIC

5



**CURRENT ISSUES IN
WORLD ECONOMICS
— TRADE AND
INTERNATIONAL**

**Topic 5: Current Issues in World Economics — Trade and
International Finance**

Learning Outcomes

By the end of this topic you should be able to:

- **Explain the theory of comparative advantage and gains from trade**
- **Analyse trade policy instruments (tariffs, quotas, subsidies) and their welfare effects**
- **Apply the balance of payments framework to Malaysia's external position**
- **Evaluate exchange rate determinants and their impact on Sarawak/Malaysian businesses**
- **Recommend trade and FDI strategies for Malaysian firms entering ASEAN markets**



INSPIRING MINDS, SHAPING FUTURE

LO Learning Objectives

By the end of this topic you should be able to:

- **Apply the theory of comparative advantage** to explain gains from international trade.
- **Evaluate trade policy instruments** including tariffs, quotas, and subsidies, and their business implications.
- **Analyse exchange rate systems and their effects** on Malaysian business competitiveness and investment.
- **Assess Malaysia's trade structure, ASEAN integration, and FDI flows** in the context of global value chains.

KC Key Concepts and Definitions

Concept	Definition	Example / Notes
Comparative Advantage	A country's ability to produce a good at a lower opportunity cost than another.	<i>Sarawak has comparative advantage in palm oil, LNG, and timber production.</i>
Terms of Trade	The ratio of a country's export prices to its import prices.	<i>Favourable terms of trade boost Malaysia's national income and living standards.</i>
Exchange Rate	The price of one currency expressed in terms of another.	<i>USD/MYR exchange rate affects Sarawak exporters' revenue and import costs.</i>
Balance of Payments	A record of all economic transactions between a country and the rest of the world.	<i>Malaysia's current account surplus reflects strong export performance.</i>
Free Trade Agreement (FTA)	A pact between countries to reduce or eliminate trade barriers.	<i>Malaysia's FTAs with ASEAN and China benefit Sarawak's manufacturing exports.</i>

Foreign Direct Investment	Investment made by a company in another country through establishing operations.	<i>Petrochemical FDI in Bintulu, Sarawak generates employment and technology transfer.</i>
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5.1 Comparative Advantage and the Gains from Trade

International trade is one of the most powerful forces shaping national prosperity, and its theoretical foundation rests on a principle that remains as relevant today as when David Ricardo formalised it in 1817: comparative advantage. The principle states that a country gains from trade not by being absolutely better at producing everything, but by specialising in goods and services where it has the lowest opportunity cost of production relative to its trading partners. This insight — seemingly simple but profound in its implications — explains why trade between economies of very different sizes and productivity levels can benefit both parties.

Understanding Comparative Advantage

Suppose two countries, Malaysia and China, produce both palm oil and manufactured electronics. Even if China can produce both goods more efficiently in absolute terms (absolute advantage), trade is mutually beneficial if their opportunity costs differ. If Malaysia gives up fewer units of manufactured goods for each additional tonne of palm oil produced compared to China, then Malaysia has a comparative advantage in palm oil. China, by contrast, has a comparative advantage in manufactured goods. Both countries gain by specialising and trading: Malaysia produces palm oil and exchanges it for Chinese electronics; China produces electronics and exchanges them for Malaysian palm oil. Both countries consume beyond what they could produce alone.

The gains from trade under comparative advantage take two forms. Static gains arise from more efficient allocation of resources — each country focuses on activities where it is relatively most productive, raising total output for the same total input of labour and capital. Dynamic gains arise over time through economies of scale (specialised industries grow larger and become

more efficient), technology transfer, and exposure to international competition that stimulates productivity improvements.

Sarawak's Comparative Advantage: Palm Oil and LNG

Sarawak's economic structure vividly illustrates comparative advantage at work. The state possesses an abundance of suitable land, a tropical climate, and decades of accumulated expertise in palm oil cultivation and processing. These endowments give Sarawak a comparative advantage in palm oil production — it can produce palm oil at a lower opportunity cost than virtually any other product or any other region. Sarawak accounts for approximately 25% of Malaysia's total palm oil output, which in turn represents around 30% of global palm oil supply. The state's comparative advantage is structural — rooted in geography, climate, and agricultural capital built over generations.

Similarly, Sarawak's offshore natural gas reserves, estimated at over 30 trillion cubic feet, give it a comparative advantage in liquefied natural gas (LNG) production. Petronas LNG in Bintulu has been supplying LNG to Japan and Korea under long-term contracts for over four decades — a testament to the durability of resource-based comparative advantage when supported by quality infrastructure and reliable supply chains. Japan and Korea, lacking domestic natural gas reserves, specialise in high-value manufacturing (automobiles, electronics, semiconductors) and trade these products for LNG. Both parties benefit from this exchange.

BIMP-EAGA Trade Flows and Shifting Comparative Advantages

The Brunei-Indonesia-Malaysia-Philippines East ASEAN Growth Area (BIMP-EAGA) represents a regional framework for harnessing comparative advantages across neighbouring economies. Sarawak trades agricultural products and energy with Kalimantan (Indonesian Borneo) while receiving manufactured goods and consumer products from more industrialised BIMP-EAGA partners. These cross-border flows reflect the comparative advantage principle at the sub-regional level.

A critical strategic question for Malaysia — and Sarawak — is whether comparative advantage is static or dynamic. China's dramatic industrialisation over the past three decades represents a shift in comparative advantage: as Chinese manufacturing productivity has risen, China has

moved up the value chain into higher-technology goods. Malaysia's response, articulated through the National Investment Aspirations (NIA) 2021 and the Sarawak Digital Economy Blueprint 2030, is to develop dynamic comparative advantage in digital services, renewable energy, and advanced manufacturing — sectors where Sarawak's abundant clean energy and growing digital infrastructure create new opportunities.

Limitations of Comparative Advantage Theory

While comparative advantage remains theoretically powerful, it has well-documented limitations. It assumes constant opportunity costs (production possibility frontiers are straight lines), frictionless trade, and full employment of resources. In practice, specialisation in commodities can create vulnerability to price volatility — Sarawak's heavy reliance on palm oil and LNG exposes its economy to global commodity price swings. The terms of trade (the ratio of export prices to import prices) can shift unfavourably, eroding gains from trade. These limitations explain why trade policy — explored in the next subtopic — is not simply about removing barriers to free trade, but about managing the complex welfare and distributional consequences of international economic integration.

5.2 Trade Policy: Tariffs, Quotas, and Subsidies

While free trade theory suggests that open markets maximise global welfare, governments routinely intervene in trade through a range of policy instruments. Understanding these instruments — their mechanics, their welfare effects, and their strategic rationale — is essential for business managers operating in internationally connected economies like Malaysia's. Trade policy instruments fall into two broad categories: tariff barriers (direct taxes on imports or exports) and non-tariff barriers (regulatory measures that restrict trade without being explicit taxes).

Tariffs

A tariff is a tax imposed on imported goods at the point of entry. An import tariff raises the domestic price of the imported good above the world price, generating several economic effects. Domestic consumers pay more, reducing consumer surplus. Domestic producers, now

protected from lower-priced foreign competition, benefit — producer surplus rises. The government collects tariff revenue equal to the tariff rate multiplied by the volume of imports. However, the combined gain to domestic producers and government is less than the loss to consumers — a deadweight loss triangle emerges, representing the inefficiency created by the tariff-induced price distortion. Net welfare in the importing country falls unless the country has sufficient market power to improve its terms of trade by driving down the world price (a 'terms of trade tariff' — the economists' justification for strategic trade policy).

Malaysia uses tariffs selectively to protect strategically important domestic industries. Import duties on motor vehicles protect the national automotive industry (Proton and Perodua) from foreign competition, maintaining domestic employment but at a cost to Malaysian consumers who pay substantially more for cars than in free-trade economies like Singapore. Import tariffs on certain agricultural products protect domestic food producers.

Import Quotas

A quota places a quantitative limit on the volume of a good that can be imported during a specified period. Like a tariff, a quota raises the domestic price of the imported good — but instead of government revenue, the 'quota rent' (the price premium created by the quantity restriction) accrues to whoever holds the right to import (importers, exporters, or the government, depending on how quota licences are allocated). Malaysia has used import quotas on steel products to protect the domestic steel industry, including Megasteel (now Malaysia Steel Works) which serves important national strategic and employment objectives in the Klang Valley industrial corridor.

Export Subsidies and Support Programmes

Export subsidies are government payments to domestic producers for each unit exported, enabling them to sell at a lower price in world markets. Export subsidies are generally prohibited under WTO rules for manufactured goods, though limited exceptions exist for agricultural products. In Malaysia, export support takes softer forms: MATRADE (Malaysia External Trade Development Corporation) provides market intelligence, trade missions, and export facilitation services to Malaysian SMEs; EXIM Bank Malaysia provides export

financing and insurance that reduces the risk of exporting to emerging markets; and sectoral development programmes support the upgrading of export-oriented industries.

The EU Deforestation Regulation (EUDR) as Non-Tariff Barrier

Perhaps the most significant trade policy challenge currently facing Sarawak is not a traditional tariff but a regulatory non-tariff barrier: the European Union Deforestation Regulation (EUDR), which entered into force in 2023 and is scheduled to be fully effective from 2025. The EUDR requires all palm oil, timber, soy, beef, cocoa, coffee, and rubber products sold in the EU market to be certified as 'deforestation-free' — meaning the products must originate from land not converted from forest after December 31, 2020. Producers must provide geolocation coordinates for every plot of land and supply chain documentation to demonstrate compliance.

For Sarawak's palm oil and timber exporters — particularly small and medium-sized planters and sawmillers — the EUDR imposes substantial compliance costs estimated at RM 500–1,000 per tonne for small farmers who lack the technology, data management systems, and staff to collect and submit the required geolocation and supply chain traceability documentation. The regulation is formally a market access requirement rather than a tariff, but its economic effect is equivalent to a prohibitive non-tariff barrier for producers who cannot demonstrate compliance. The Malaysian government has protested the EUDR at the WTO as potentially discriminatory, arguing that it conflates Sarawak's regulated, certificated palm oil industry with unmanaged deforestation in other regions. This dispute illustrates how trade policy analysis — grounded in LU5 concepts — is directly relevant to Sarawak business strategy.

5.3 Malaysia's Trade Structure and ASEAN Integration

Malaysia is a deeply trade-dependent economy, with total trade (exports plus imports) exceeding 130% of GDP — among the highest ratios in the world. This trade intensity reflects Malaysia's position as a manufacturing export hub, a significant commodity exporter, and a critical node in regional and global supply chains. Understanding Malaysia's trade structure — its main exports and imports, its principal trading partners, and the institutional frameworks

that govern its trade relationships — is essential context for any business manager operating in Malaysia or Sarawak.

Malaysia's Key Export Sectors

Malaysia's export base combines manufacturing and commodities. Electrical and electronic (E&E) products — semiconductors, printed circuit boards, disk drives — account for approximately 38% of merchandise exports, making Malaysia the world's seventh largest exporter of semiconductors and a critical player in global electronics supply chains. Petroleum and LNG exports account for a further 15–18% of total exports, with Sarawak's Bintulu LNG complex being a primary contributor. Palm oil and palm oil products — approximately 7–8% of total exports — are dominated by Sarawak and Peninsular Malaysia plantations. Other significant exports include chemicals, machinery, and rubber products.

Principal Trading Partners

Malaysia's most important trading partners are China (its largest trading partner since 2009), Singapore, the United States, Japan, Hong Kong, and South Korea. China receives approximately 14% of Malaysia's total exports and supplies 22% of its imports. The US remains Malaysia's largest export destination for E&E products. Singapore functions as both a trading partner and a major transshipment hub through which Malaysian goods reach global markets. Japan and Korea are among the most important destinations specifically for Sarawak's LNG, with decades-long supply relationships underpinning energy security agreements.

Sarawak LNG: A Case Study in Long-Term Trade Relationships

Sarawak's LNG export sector provides a particularly instructive example of how comparative advantage, trade infrastructure, and long-term institutional relationships create durable trade relationships. Petronas LNG began exports to Japan in 1983 under a 20-year contract — one of the first major LNG supply agreements in Asia. Since then, the relationship has deepened and expanded to include Korea, China, and other buyers. These long-term contracts provide Sarawak and Malaysia with stable export revenues but also limit flexibility to respond to price spikes in spot markets. The LNG price spike of 2022–2023 — driven by Russia's invasion of Ukraine and Europe's scramble for non-Russian gas — generated windfall revenues for LNG

exporters globally, including Sarawak, illustrating how commodity price volatility creates both opportunities and risks.

ASEAN Free Trade Area (AFTA) and Regional Integration

Malaysia is a founding member of ASEAN and a signatory to the ASEAN Free Trade Area (AFTA), which has progressively reduced tariffs on intra-ASEAN trade since 1992. Under the ASEAN Trade in Goods Agreement (ATIGA), tariffs on most goods traded among ASEAN members are zero or near-zero, dramatically expanding intra-regional trade flows. ASEAN also has free trade agreements with major trading partners including China (ACFTA), India, Japan, Korea, Australia-New Zealand, and Hong Kong. Malaysia's participation in the Regional Comprehensive Economic Partnership (RCEP) — the world's largest trade agreement by GDP, covering ASEAN plus China, Japan, Korea, Australia, and New Zealand — further expands Malaysia's trade framework.

For Sarawak firms, ASEAN integration creates both market access opportunities and competitive threats. Access to a market of 670 million people with reducing tariffs benefits Sarawak exporters — palm oil processors, timber product manufacturers, and digital service providers. At the same time, AFTA exposes Sarawak's protected domestic industries to competition from lower-cost ASEAN producers. Understanding Malaysia's trade surplus position — exports consistently exceed imports, generating a positive trade balance that contributes to Malaysia's current account surplus — is essential for analysing the macroeconomic context in which Sarawak businesses operate.

5.4

Balance of Payments (Current Account, Capital Account, Financial Account)

The balance of payments (BOP) is a systematic record of all economic transactions between residents of a country and the rest of the world during a specified period, typically one year. It provides the most comprehensive picture of a country's international economic position — how much it earns and spends abroad, how much foreign investment it attracts, and how its foreign

reserves change. For any business manager dealing with international trade, investment, or finance, understanding the BOP framework is indispensable.

Structure of the Balance of Payments

The BOP is divided into three main accounts. The Current Account records flows of goods, services, income, and current transfers. It comprises: (1) the trade balance (goods exports minus goods imports — the most visible component, reported monthly); (2) the services balance (services exports minus imports, including tourism, financial services, and transport); (3) the primary income account (wages earned abroad by residents minus wages earned in Malaysia by non-residents, plus investment income — interest, dividends); and (4) the secondary income account (current transfers — including worker remittances sent to Malaysia by Malaysian workers abroad, and outflows of remittances by foreign workers in Malaysia). A current account surplus means a country earns more from abroad than it spends abroad.

The Capital Account records transfers of non-produced, non-financial assets (such as debt forgiveness and transfers of intellectual property rights) — generally a small account for most countries. The Financial Account records transactions in financial assets and liabilities between residents and non-residents — encompassing foreign direct investment (FDI), portfolio investment (equity and bonds), other investment (loans and deposits), and changes in official reserve assets (foreign exchange reserves held by Bank Negara Malaysia). By accounting identity, $\text{Current Account} + \text{Capital Account} + \text{Financial Account} = 0$ (with statistical discrepancy). A current account surplus is financed by net capital and financial account outflows (the country is accumulating foreign assets) or reserve accumulation.

Malaysia's Current Account Position

Malaysia has maintained a current account surplus for most of the past two decades, driven by its trade surplus in goods (especially E&E products, LNG, and palm oil) that more than offsets its services deficit and net income outflows. In 2023, Malaysia's current account surplus was approximately 1.3% of GDP, supported by strong commodity export revenues. The surplus has been smaller than historical norms due to higher import costs (especially energy imports for Peninsular Malaysia during periods of low domestic gas production) and higher income outflows as foreign investors repatriate dividends.

Sarawak's Contribution to Malaysia's External Position

Sarawak makes a disproportionately large contribution to Malaysia's current account surplus through its LNG, palm oil, and timber exports. The Sarawak Corridor of Renewable Energy (SCORE) industrial development corridor has attracted substantial foreign direct investment — primarily into the financial account as FDI inflows. Key SCORE investments include Press Metal's aluminium smelting complex (Sarawak-listed, but drawing on foreign technology and export markets), and more recently data centre FDI from Microsoft, Google, and AWS, all of which register as FDI inflows in Malaysia's financial account. Bank Negara Malaysia (BNM) manages Malaysia's official foreign exchange reserves — which stood at approximately USD 110 billion in early 2024 — as a buffer against external shocks, maintaining confidence in the MYR and providing liquidity for external debt obligations.

The BOP Framework and Macroeconomic Policy

The BOP framework links closely to macroeconomic policy. The national income identity $Y = C + I + G + NX$ can be rearranged to show that the current account balance (NX) equals the difference between national saving ($S = Y - C - G$) and domestic investment (I): $NX = S - I$. A current account surplus means national saving exceeds domestic investment — Malaysians collectively save more than they invest at home, lending the surplus to the world. A current account deficit means the reverse — domestic investment exceeds saving, requiring net borrowing from abroad. This savings-investment identity shows why trade surpluses and deficits are deeply connected to domestic macroeconomic conditions — a theme that bridges LU5 trade theory with the macroeconomic concepts of LU2.

5.5

Exchange Rates: Determination, Regimes, and Business Implications

The exchange rate — the price of one currency in terms of another — is among the most watched macroeconomic variables in an open economy. For Sarawak and Malaysian businesses engaged in international trade, investment, or financing, the exchange rate is not merely an abstract monetary phenomenon but a direct determinant of costs, revenues,

competitiveness, and profitability. The MYR/USD exchange rate, in particular, has enormous practical significance for Sarawak's export-oriented economy.

Determinants of Exchange Rates

Exchange rates in floating regimes are determined by supply and demand in foreign exchange markets, influenced by several key factors. Inflation differentials: A country with higher inflation than its trading partners will generally see its currency depreciate over time — the purchasing power parity (PPP) hypothesis. Interest rate differentials: Higher domestic interest rates attract foreign capital seeking higher returns, increasing demand for the domestic currency and causing appreciation — the interest rate parity condition. Current account balance: A persistent current account deficit implies excess demand for foreign currency to pay for imports, putting depreciation pressure on the domestic currency. Capital flows: FDI and portfolio investment inflows increase demand for the domestic currency; capital flight causes depreciation. Market sentiment and speculation: In short to medium runs, speculative capital flows driven by risk appetite, geopolitical events, and market expectations can dominate fundamental determinants.

MYR/USD Movements: 2023–2024

The Malaysian Ringgit experienced significant volatility in 2023–2024 that directly impacted Sarawak businesses. In 2023, the MYR depreciated substantially against the USD, weakening from approximately RM 4.40 to RM 4.75 — a depreciation of approximately 8% — driven by a combination of factors: global USD strengthening as the US Federal Reserve raised interest rates aggressively to combat inflation, while BNM maintained a more moderate interest rate stance to protect Malaysia's economic recovery; concerns about Malaysia's fiscal deficit; and reduced investor confidence in emerging market currencies generally. In 2024, the MYR recovered to approximately RM 4.35, supported by a reversal of US interest rate trajectory (Federal Reserve began cutting rates in September 2024) and improved Malaysian growth and fiscal consolidation narratives.

Business Implications for Sarawak Firms

Exchange rate movements create asymmetric impacts across Sarawak's economy. Sarawak palm oil exporters — whose revenues are largely denominated in USD or USD-linked

currencies — benefit directly from a weaker MYR. A tonne of palm oil priced at USD 900 generates RM 3,960 when the exchange rate is 4.40, but RM 4,275 when the rate is 4.75 — a revenue increase of approximately 8% with no change in the underlying commodity price. This exchange rate windfall partially offset the 2023 palm oil price decline from its 2022 highs. LNG exporters benefit similarly from USD-denominated long-term contracts. In contrast, importers suffer from a weak MYR: a Kuching electronics retailer sourcing goods from China or Korea pays more MYR for the same USD or CNY invoice. Malaysian families planning overseas travel face higher effective costs. Companies with USD-denominated debt face higher RM repayment costs.

Exchange Rate Regimes

Countries operate under various exchange rate regimes. Fixed (pegged) regimes tie the domestic currency to another currency at a fixed rate, providing exchange rate certainty for businesses but requiring large foreign reserves to defend the peg. Malaysia famously pegged the MYR at RM 3.80/USD during the 1997–1998 Asian financial crisis, a controversial decision that successfully stabilised the economy but required capital controls. Floating regimes allow market forces to determine the exchange rate, providing adjustment flexibility but introducing volatility. Malaysia currently operates a managed float — BNM intervenes in foreign exchange markets to smooth excessive volatility but does not target a specific rate. Understanding the exchange rate regime is essential for businesses assessing currency risk and designing hedging strategies (forward contracts, options) to manage MYR exposure.

5.6 Foreign Direct Investment and Global Value Chains

Foreign Direct Investment (FDI) and global value chains (GVCs) represent two of the most transformative forces in the modern international economy. FDI occurs when a firm establishes or acquires productive capacity in a foreign country — building a factory, acquiring a subsidiary, or establishing a joint venture — creating a lasting interest and degree of control in the foreign enterprise. This contrasts with portfolio investment (purchasing shares without controlling interest) or loans. FDI is distinguished by its relatively long-term, strategic nature and its direct involvement in the productive activities of the host economy.

Why Countries Seek FDI

Host countries seek FDI for multiple reasons that extend well beyond the simple balance of payments contribution of capital inflows. Productive FDI brings employment creation, technology transfer (access to advanced production techniques, management practices, and proprietary know-how that domestic firms cannot easily develop independently), linkages to international markets (export market access through the multinational's global distribution network), and human capital development through training. These spillover benefits are particularly valuable for developing economies seeking to upgrade their industrial capabilities. Critics, however, note that FDI can also create negative effects: profit repatriation reduces net income flows; multinational firms may use transfer pricing to minimise tax obligations; and resource-extractive FDI may generate enclave economies with few domestic linkages.

FDI into Sarawak's SCORE Corridor

The Sarawak Corridor of Renewable Energy (SCORE), established in 2008, is a major government-led FDI attraction initiative leveraging Sarawak's most distinctive comparative advantage: abundant, low-cost, renewable hydroelectric power. The strategy is simple in conception but profound in economic logic — use cheap, clean electricity as a magnet for energy-intensive industries that have difficulty competing globally from countries with expensive or fossil-fuel-based electricity. Press Metal Aluminium Holdings, now Malaysia's largest aluminium producer and one of the largest in the Asia-Pacific, established its Sarawak smelting operations precisely because Sarawak's SEB hydropower tariffs (approximately RM 0.23/kWh for industrial users) are among the lowest in Southeast Asia. The aluminium smelting industry is intensely energy-intensive — electricity accounts for approximately 35% of production costs — making Sarawak's power cost advantage a decisive location factor.

The Digital Economy FDI Wave

In 2024, a new wave of FDI into Sarawak emerged from the digital economy sector. Microsoft announced a RM 10.5 billion investment in Malaysia's AI and cloud infrastructure — one of the largest foreign technology investments in Malaysian history — with Sarawak identified as a priority data centre location due to its hydroelectric power advantage. Data centres are extremely electricity-intensive: a large hyperscale data centre can consume as much electricity as a small city. The ability to power AI computing infrastructure with clean, renewable, low-

cost hydroelectricity represents a major competitive advantage for Sarawak in the global race to attract data centre investment. Google and Amazon Web Services (AWS) made similar Malaysia investments in 2024. This digital FDI creates construction jobs, IT employment, and infrastructure development, though the direct employment ratio per billion ringgit invested is lower than manufacturing FDI.

Global Value Chains and Malaysian Manufacturing

Global value chains (GVCs) — the cross-border networks of firms that design, produce, assemble, market, and distribute a final product — are the dominant organising structure of modern international manufacturing. Malaysia's E&E industry is deeply embedded in GVCs: semiconductor chips are designed in the US, manufactured in Taiwan or Korea, packaged and tested in Malaysia (particularly in Penang), and assembled into consumer electronics in China before being sold globally. Sarawak's manufacturing sector is less integrated into GVCs than Penang or Selangor, but the SCORE corridor's industrial development and the growing digital economy provide pathways for deeper GVC participation. Understanding GVC economics — the importance of logistics, reliability, quality standards, and trade facilitation — is essential for Malaysian manufacturers seeking to climb the value chain from labour-intensive assembly to higher value-added design and advanced manufacturing.

5.7 Trade Shocks and Malaysian Business Vulnerability

Open economies benefit from international trade — greater market access, comparative advantage gains, technology spillovers — but openness also creates vulnerability to external shocks. A trade shock is any sudden, unexpected change in the external trade environment that significantly affects domestic economic outcomes. For Malaysia and Sarawak, with total trade exceeding 130% of GDP, trade shocks can rapidly translate into employment effects, income volatility, and macroeconomic instability. Understanding the sources, transmission mechanisms, and business implications of trade shocks is essential for managers developing resilient business strategies.

The US-China Trade War: Trade Diversion Effects on Malaysia

Beginning in 2018, the US-China trade war — characterised by escalating tariff increases on hundreds of billions of dollars of bilateral trade — created significant ripple effects across global supply chains. For Malaysia, the trade war created a trade diversion opportunity: as US tariffs on Chinese goods made China-based manufacturing more expensive for US buyers, multinational companies began relocating or diversifying production away from China to alternative locations including Malaysia, Vietnam, Thailand, and Mexico. Malaysian exports of E&E products to the US increased notably from 2019 onwards, as semiconductor and electronics manufacturers shifted capacity to Malaysia. Sarawak's benefits were more indirect — through increased Malaysian manufacturing activity and commodity demand — but the general principle illustrates how geopolitical trade conflicts create both threats (supply chain disruption) and opportunities (production relocation) for third-country economies.

Commodity Price Shocks: LNG Price Volatility

Commodity price shocks represent a distinct form of trade shock particularly relevant to Sarawak's resource-dependent economy. The LNG price spike of 2022–2023 — in which Asian LNG spot prices rose from approximately USD 10 per MMBtu in 2021 to over USD 70 per MMBtu at the peak in August 2022, before falling back to USD 12–15 in 2023 — illustrates both the magnitude of commodity price volatility and its complex implications for Malaysian businesses. Sarawak LNG exporters with flexible spot market exposure benefited enormously from the price spike. However, Malaysia simultaneously faced higher energy import costs for Peninsular Malaysia's gas-fired power plants that depend on imported LNG. The distributional effects of commodity price shocks are asymmetric — exporters benefit, importers suffer, and net effects depend on the relative magnitudes.

Supply Chain Disruption Risks

The COVID-19 pandemic (2020–2021) demonstrated how supply chain trade shocks — sudden disruption to the physical flow of goods across borders — can devastate trade-dependent economies. Malaysian factories supplying global automotive and electronics supply chains faced mandatory closures under MCO restrictions, causing delivery failures that rippled through international production networks. Sarawak palm oil exporters faced logistics disruptions as shipping container shortages and port congestion increased freight costs dramatically. The pandemic experience accelerated corporate supply chain resilience strategies

— nearshoring, dual-sourcing, and inventory building — that have permanently altered the economics of global supply chain management.

Strategies for Managing Trade Vulnerability

For Malaysian and Sarawak businesses, managing trade vulnerability requires deliberate strategic choices. Export market diversification — Sarawak palm oil exporters who had developed markets across Asia, India, and Africa faced lower exposure when the EU EUDR regulations threatened the EU market. Product diversification — moving from commodity palm oil to higher-value refined palm products, palm oleochemicals, and palm-based biofuels reduces commodity price vulnerability. Financial hedging — using forward exchange contracts and commodity futures to lock in prices reduces exposure to exchange rate and commodity price volatility. Supply chain localisation — developing domestic suppliers for key inputs reduces exposure to import supply disruptions. These strategic responses, informed by trade theory and macroeconomic analysis, represent the managerial application of LU5 concepts.

Managerial Insight: The EU Deforestation Regulation — Trade Policy as Environmental Weapon

The EU's EUDR (effective 2025) requires all palm oil, timber, soy, beef, cocoa, coffee, and rubber products sold in the EU to be 'deforestation-free.' For Sarawak's palm oil and timber exporters, this is a major non-tariff barrier: producers must demonstrate via geolocation data that their products come from land not deforested after 2020. Compliance costs are estimated at RM 500–1,000 per tonne for small farmers. This illustrates how environmental trade policy — analysed through the lens of trade theory — can fundamentally alter business strategy for Sarawak exporters. Managers must understand that trade policy is increasingly shaped by environmental and social governance (ESG) considerations in major importing markets, not just traditional protectionist economic interests. The EUDR effectively forces Sarawak palm oil producers to choose between investing in costly compliance systems or losing EU market access entirely — a strategic decision with profound implications for the entire Sarawak palm oil value chain. The Malaysian government's response — lobbying at the WTO, accelerating the Malaysian Sustainable Palm Oil (MSPO) certification programme, and engaging in diplomatic dialogue with EU officials — illustrates how trade policy analysis must inform both corporate and government strategy.

Self-Reflection



1. The Malaysian Ringgit fell to RM 4.75 against the USD in 2023 before recovering. How did this affect: (a) a Sarawak palm oil company earning USD revenues, (b) a Kuching retailer importing electronics from China, (c) a Malaysian family planning overseas travel? Use the exchange rate concepts from LU5 to structure your analysis.

Questions for Discussion

1. Malaysia's trade surplus with the rest of the world coexists with a significant budget deficit (fiscal deficit). Using the national income identity ($Y = C + I + G + NX$), the balance of payments framework, and trade policy concepts from LU5, explain how these macroeconomic relationships are connected — and recommend two policies a Sarawak-based export company should adopt to manage exchange rate and trade policy risks. (CLO2)

Summary

LU5 examined the theory and practice of international trade and finance through the lens of Malaysia's and Sarawak's real-world trade relationships. Comparative advantage theory — rooted in Ricardo's insight that specialisation based on opportunity cost differences creates mutual gains from trade — explains why Sarawak exports palm oil and LNG while importing manufactured goods, and why this exchange raises living standards on both sides.

Trade policy instruments — tariffs, quotas, and non-tariff barriers — distort these gains by protecting domestic producers at the expense of consumers and overall welfare. The EU Deforestation Regulation (EUDR) exemplifies how modern non-tariff barriers, framed in environmental terms, create significant compliance costs and market access challenges for Sarawak exporters, requiring strategic adaptation at both firm and government levels.

Malaysia's current account surplus reflects its strong export performance, particularly in E&E products, LNG, and palm oil — with Sarawak contributing disproportionately. The

balance of payments framework connects trade flows to capital account movements, exchange rate dynamics, and foreign reserve management. MYR/USD exchange rate volatility — the depreciation to 4.75 in 2023 and recovery to 4.35 in 2024 — creates asymmetric winners and losers among Sarawak businesses, benefiting USD-revenue exporters while hurting importers. FDI into Sarawak's SCORE corridor and the new wave of digital economy FDI (Microsoft, Google, AWS) demonstrate how comparative advantage in renewable energy creates new strategic opportunities in the global digital economy. Managers who master these trade and international finance concepts are equipped to navigate the complex external environment that shapes Sarawak business performance.

Multiple Choice Questions (MCQs)

1. The theory of comparative advantage states that countries gain from trade when they:
 - A) Produce everything more efficiently than their trading partners
 - B) Specialise in producing goods with lower opportunity costs and trade for others
 - C) Impose tariffs on all imports
 - D) Maintain a fixed exchange rate

2. A tariff on imported goods will:
 - A) Benefit domestic consumers and harm domestic producers
 - B) Benefit domestic producers and harm domestic consumers; government collects revenue
 - C) Increase total social welfare
 - D) Have no effect on imports

3. Malaysia's LNG exports to Japan and Korea benefit from:
 - A) Import quotas imposed by Japan
 - B) Long-term supply contracts and Malaysia's comparative advantage in LNG production

- C) MYR appreciation
- D) EU trade agreements

4. A current account surplus means:

- A) Government spending exceeds tax revenue
- B) A country exports more goods and services than it imports, earning net foreign assets
- C) Foreign investment exceeds domestic investment
- D) The exchange rate is overvalued

5. If the MYR depreciates against the USD, Sarawak palm oil exporters will:

- A) Earn lower RM revenues
- B) Earn higher RM revenues (USD proceeds convert to more RM)
- C) Face higher input costs
- D) See export volumes fall

6. Foreign Direct Investment (FDI) into Sarawak's data centre sector (e.g., Microsoft) primarily benefits Sarawak through:

- A) Reducing trade deficit
- B) Creating employment, transferring technology, and building digital infrastructure
- C) Lowering the exchange rate
- D) Increasing tax on companies

7. The EU Deforestation Regulation (EUDR) is an example of a:

- A) Import tariff
- B) Export subsidy
- C) Non-tariff barrier
- D) Exchange rate manipulation

8. In the national income identity ($Y = C + I + G + NX$), an increase in net exports (NX) with all else equal will:

- A) Reduce GDP
- B) Increase GDP
- C) Reduce government spending
- D) Have no effect on GDP

Online Learning Resources

The following free resources supplement your study of this unit.

WTO — Trade Data and Statistics (Free)

https://www.wto.org/english/res_e/statis_e/statis_e.htm

Bank Negara Malaysia — Balance of Payments Statistics (Free, Malaysian)

<https://www.bnm.gov.my/data/bop>

MATRADE — Malaysia Trade and FDI Data (Free, Malaysian)

<https://www.matrade.gov.my>

ASEAN Statistical Database — ASEAN Stats (Free, Regional)

<https://data.aseanstats.org>

END OF TOPIC

TOPIC

6



GAME THEORY IN DECISION MAKING

*Lecturers: Dr. Colin Wong Koh King; Prof. Dr. Dato
Shazali Abu Mansor; Prof. Dr. Frank Kiong*

Topic 6: Game Theory in Decision Making

Learning Outcomes

By the end of this topic you should be able to:

- Define game theory and identify its core components (players, strategies, payoffs)
- Apply dominant strategy analysis to find Nash Equilibria in 2×2 games
- Analyse the Prisoner's Dilemma and its implications for business competition and cooperation
- Apply extensive-form game analysis to sequential business decisions
- Evaluate the role of repeated games and reputation in sustaining cooperation
- Apply game theory to oligopoly pricing, auction design, and negotiation



INSPIRING MINDS, SHAPING FUTURE

LO Learning Objectives

By the end of this topic you should be able to:

- **Apply Nash equilibrium concepts** to analyse strategic interactions among competing firms.
- **Use extensive-form game trees** to evaluate sequential decision-making in business negotiations.
- **Analyse signalling, reputation formation, and repeated games** to understand long-term strategic behaviour.
- **Apply game-theoretic frameworks** to real-world business strategy, pricing decisions, and competitive dynamics.

KC Key Concepts and Definitions

Concept	Definition	Example / Notes
Nash Equilibrium	A stable state where no player can improve their outcome by unilaterally changing strategy.	<i>Two competing telcos both choosing aggressive pricing — neither can gain by changing alone.</i>
Dominant Strategy	A strategy that yields the best outcome regardless of what the opponent does.	<i>A retailer always advertising aggressively regardless of competitor behaviour.</i>
Prisoner's Dilemma	Two rational players fail to cooperate despite mutual benefit from doing so.	<i>Two Sarawak firms both undercut prices, harming profitability for both.</i>
Extensive Form Game	A game tree representing sequential decisions with payoffs at each endpoint.	<i>Negotiating a timber concession with sequential bids — modelled as a game tree.</i>

Signalling	An action taken to credibly convey private information to another party.	<i>A Sarawak startup's ISO certification signals quality to potential investors.</i>
Repeated Game	A game played multiple times, enabling cooperation through repeated interaction.	<i>Long-term supplier relationships reward honest behaviour over multiple rounds.</i>

6.1

Solution Concepts for Static Games (Dominant Strategies, Nash Equilibrium)

Game theory is the mathematical study of strategic interaction — situations where the outcome for each participant depends not only on their own choices but on the choices of others. Developed formally by John von Neumann and Oskar Morgenstern in their 1944 work *Theory of Games and Economic Behavior*, and extended decisively by John Nash in the 1950s, game theory has become one of the most powerful analytical tools in economics, business strategy, political science, biology, and computer science. For MBA students, game theory provides a systematic framework for thinking about competitive and cooperative business situations where strategic interdependence matters.

Core Components of a Game

- **Players:** The decision-makers in the strategic situation — could be firms, individuals, governments, or any entity making choices. In business contexts, players are typically competing firms, negotiating parties, or bidders in an auction.
- **Strategies:** The complete set of actions available to each player. In a static (simultaneous-move) game, each player chooses a strategy without knowing what the other players have chosen. A strategy can be a price to charge, a quantity to produce, whether to enter a market, or any other choice variable.

- **Payoffs:** The outcomes (profits, utilities, values) each player receives for each possible combination of strategies. Payoffs are the fundamental inputs to the analysis — they describe what each player cares about and how much.

Worked Example: CelcomDigi vs Maxis — Price War Game

The following payoff matrix illustrates a pricing game between two leading telecommunications providers in Sarawak's mobile market:

	Maxis: High Price	Maxis: Low Price
CelcomDigi: High Price	(80, 80)	(40, 90)
CelcomDigi: Low Price	(90, 40)	(50, 50)

Payoffs represent profit in RM millions. Reading the matrix: if both firms charge a High Price, each earns RM 80M — the cooperative outcome. If CelcomDigi cuts price while Maxis holds high, CelcomDigi earns RM 90M (gains market share) while Maxis earns only RM 40M. If both cut price, each earns RM 50M — the competitive outcome.

Dominant Strategy Analysis: CelcomDigi's dominant strategy is Low Price — if Maxis plays High Price, CelcomDigi earns 90 (Low) versus 80 (High), so Low is better; if Maxis plays Low Price, CelcomDigi earns 50 (Low) versus 40 (High), so Low is still better. Low Price dominates High Price regardless of what Maxis does. By symmetric reasoning, Maxis's dominant strategy is also Low Price. The Nash Equilibrium — the outcome where neither player wants to deviate given the other's strategy — is (Low, Low) = (50, 50). Crucially, this Nash Equilibrium is worse for both firms than the cooperative outcome (80, 80)! This is the classic Prisoner's Dilemma applied to business competition: individual rationality leads to collective irrationality. The implication for Sarawak's telecom market is that price competition will tend to push margins down to the Nash Equilibrium, unless the firms can find credible mechanisms to sustain cooperation (explored in LU6.5).

Nash Equilibrium: Formal Definition and Properties

A Nash Equilibrium is a combination of strategies — one for each player — such that no player can improve their payoff by unilaterally changing their strategy, given the other players' strategies. Every finite game has at least one Nash Equilibrium (possibly in mixed strategies). Nash Equilibria are self-enforcing — once reached, no player has an incentive to deviate. However, Nash Equilibria are not necessarily efficient — as the Prisoner's Dilemma illustrates, rational individual behaviour can lead to collectively suboptimal outcomes. This gap between individual rationality and collective optimality is one of the most important insights game theory offers to business managers.

6.2

Solution Concepts for Extensive-Form Games (Backward Induction, Subgame Perfect Equilibrium)

Many important business decisions are sequential rather than simultaneous — one party moves first, then another responds, knowing what the first player did. These strategic situations require a different analytical framework: the extensive-form game, represented as a game tree showing all possible decision nodes, actions, and payoffs in chronological order. Extensive-form analysis is directly applicable to entry deterrence, negotiation, merger decisions, and the strategic licensing of natural resources.

Backward Induction

The solution concept for extensive-form games with perfect information is backward induction — reasoning backwards from the final decision node to determine the optimal strategy at each earlier node. The key insight is that rational players anticipate future optimal play when making current decisions. Starting from the last decision node, identify the optimal choice for the player moving at that node. Working backwards, at each earlier node, the player moving there should choose the action that leads to the best outcome given the anticipated optimal future play of all players.

PETROS Licensing Round: A Sequential Game Example

Consider PETROS (Petroleum Sarawak Berhad) offering an oil block exploration licence to three oil companies (Company A, B, and C) in a sequential negotiation. PETROS first

announces the minimum acceptable bid. Company A decides whether to bid or pass. If A bids, Company B observes A's bid and decides whether to top it or pass. If B tops A's bid, Company C decides whether to bid even higher or accept defeat. Backward induction reveals the equilibrium: Company C will bid up to its maximum valuation; Company B, anticipating C's response, will only bid if its value exceeds C's likely bid; Company A, anticipating both B's and C's responses, will only bid if its value is high enough to survive subsequent competition. The sequential structure allows each company to infer information about rivals' valuations from their bidding behaviour — a form of information revelation through strategic action.

Subgame Perfect Equilibrium

The subgame perfect equilibrium (SPE) — introduced by Reinhard Selten — refines the Nash Equilibrium concept for extensive-form games by requiring that strategies constitute a Nash Equilibrium in every subgame (every portion of the game tree starting from a particular node). SPE eliminates non-credible threats — commitments a player claims to make that would be irrational to actually carry out if called upon. For example, an incumbent firm might threaten to engage in a ruinous price war if a new competitor enters. If this threat is not credible (carrying it out would also hurt the incumbent), a rational entrant will ignore it. SPE analysis focuses on credible, equilibrium behaviour throughout the game — a critical tool for understanding strategic competition in Sarawak's industrial markets.

Entry Deterrence and First-Mover Advantages

Extensive-form analysis has important applications in entry deterrence strategy. An incumbent firm that can credibly commit to capacity expansion, exclusive contracts with customers, or long-term supply agreements before a potential entrant makes its entry decision can deter entry by making the post-entry market less attractive. In Sarawak's aluminium smelting market, Press Metal's massive capital investment in smelting capacity and its long-term power purchase agreements with SEB function as commitment devices that raise barriers to entry for potential competitors — a strategic use of sequential game logic.

6.3 Equilibrium Concepts for Games with Imperfect Information

Most real-world strategic interactions involve asymmetric information — situations where different players have access to different information relevant to the game. A seller knows the quality of what they are selling; a buyer does not. An employee knows their own work ethic and ability; an employer does not. A palm oil farmer knows the quality of this season's harvest; a trader may not. These informational asymmetries fundamentally alter the strategic environment and require new equilibrium concepts beyond the basic Nash Equilibrium.

Bayesian Nash Equilibrium

In games of incomplete information — where some players have private information unknown to others — the relevant solution concept is the Bayesian Nash Equilibrium (BNE). In a BNE, each player maximises their expected payoff given their own private information and given beliefs about the probability distributions over other players' types (private information). A 'type' captures the player's private information — their cost structure, quality, valuation, or strategic intent. The key insight is that players with private information respond to that information differently, and rational uninformed players form beliefs about informed players' types based on observable actions.

Sarawak Palm Oil Price Negotiations: Asymmetric Information

Consider the palm oil price negotiation between Sarawak smallholder farmers and commodity traders at a Serian weekly market. The farmer has private information about this season's harvest quality — whether the fresh fruit bunches are high-grade (high oil extraction rate, RM 850/tonne value) or lower-grade (RM 720/tonne value). The trader cannot perfectly observe quality before purchase. This is a classic 'lemons problem' (Akerlof, 1970): if the trader offers an average price, high-quality farmers may reject it (receiving below their product's value), while low-quality farmers accept. Over time, the market may only attract low-quality produce — adverse selection. The trader's equilibrium strategy involves offering a price schedule that reflects quality uncertainty and potentially investing in quality assessment technology (sampling, laboratory testing) to reduce information asymmetry.

Perfect Bayesian Equilibrium

The Perfect Bayesian Equilibrium (PBE) extends BNE to dynamic games with private information by requiring that beliefs be updated consistently with Bayes' rule wherever

possible. In a PBE, players have both strategies (contingent plans for every possible information set) and beliefs (probability distributions over which node they are at within each information set), and both must be consistent with each other. PBE is the standard solution concept for signalling games (LU6.4) and dynamic games with incomplete information — including sequential bargaining, auctions with private values, and strategic entry under uncertainty. While mathematically sophisticated, the core insight for managers is straightforward: rational players will use all available information — including what the timing and nature of others' actions reveal about their private information — to form strategic responses.

6.4 Signalling and Forward Induction

Signalling theory — developed by Michael Spence in his seminal 1973 paper on job market signalling — addresses how informed parties can credibly communicate private information to uninformed parties. The fundamental challenge is that any announcement of private information is cheap talk unless accompanied by a costly action that only high-quality types would be willing to take. A signal is credible only when it is costly enough that low-quality types would not find it profitable to mimic.

The Logic of Credible Signalling

Consider educational credentials in the labour market. A highly productive worker can signal productivity by obtaining a university degree — not necessarily because the degree directly increases productivity (it may or may not), but because the opportunity cost of obtaining the degree (time, tuition, effort) is lower for productive workers (who learn more quickly) than for less productive workers. If the signalling cost differential is large enough, a separating equilibrium emerges: productive workers obtain degrees, less productive workers do not, and employers rationally infer high productivity from degree holders. The key condition is the single-crossing property — the cost of signalling must be lower for the high-type (high-quality) party than for the low-type.

SEB's Low Tariff as a Commitment Signal to FDI

Sarawak Energy Berhad (SEB) faces a signalling problem in attracting foreign direct investment into energy-intensive industries. A data centre investor committing RM 3–5 billion to Sarawak needs confidence that SEB will maintain low industrial electricity tariffs over the 15–20 year lifespan of the investment. SEB's low tariff commitment is credible precisely because it is backed by massive sunk cost investment in renewable hydroelectric generation capacity — the Baleh Dam (2,400 MW, under construction) and the existing Batang Ai, Bakun, and Murum dams represent billions of ringgit of invested capital whose returns depend on attracting industrial electricity customers. This capital investment functions as a costly signal: SEB has put its own balance sheet behind the commitment, making future tariff increases costly to SEB itself.

CelcomDigi's 5G Rollout Signal

In the Sarawak telecommunications market, CelcomDigi's public commitment to 5G network rollout in Kuching and Bintulu by 2025 functions as a signalling strategy toward multiple audiences. Toward customers, it signals service quality leadership. Toward potential entrants, it signals that CelcomDigi will be well-positioned to defend its market share with superior technology. Toward Maxis, it signals an intention to compete aggressively on network quality, potentially triggering a 'signalling arms race' of infrastructure investment commitments. The credibility of this signal depends on whether CelcomDigi actually follows through with capital expenditure — consistent with the separating equilibrium logic of costly signalling.

Forward Induction

Forward induction is a solution concept for extensive-form games that reasons forward from past play to make inferences about future intentions. If a player's past action was dominated unless they planned to pursue a particular future strategy, rational opponents infer that future strategy. For example, if SEB publicly declines a profitable short-term tariff increase opportunity, potential FDI investors use forward induction to infer that SEB intends to maintain low tariffs long-term — the past action (declining profit) is rationalisable only by the long-term commitment interpretation. Forward induction complements backward induction by allowing observed strategic behaviour to communicate credible intentions.

6.5 Repeated Games and the Folk Theorem

The static Prisoner's Dilemma shows that one-shot strategic interaction tends toward non-cooperative equilibria even when cooperation would benefit all parties. But most real business relationships are not one-shot — firms compete repeatedly, governments and concession holders interact over long periods, and employer-employee relationships recur. Repeated game theory examines how the possibility of future interaction changes the strategic calculation and creates the potential for cooperative equilibria that would not be sustainable in a single period.

Trigger Strategies and Cooperation

The key mechanism enabling cooperation in repeated games is the credible threat of future punishment. A trigger strategy specifies cooperative behaviour as long as all players cooperate, but switches to punishment behaviour if any player defects. The simplest trigger strategy is the grim trigger: cooperate until anyone defects; punish forever after. A less extreme version is tit-for-tat: cooperate in the first period; in every subsequent period, do whatever your opponent did in the previous period. The critical condition for cooperation to be sustainable is that the present value of future gains from cooperation must exceed the short-term gain from defecting. This condition depends on the discount factor δ (a measure of how much players value future payoffs relative to immediate payoffs) — the more patient the players (high δ), the wider the range of cooperative outcomes that can be sustained.

Sarawak Timber Concessions: Repeated Government-Industry Interaction

The Sarawak state government's relationship with timber concession holders provides a real-world illustration of repeated game dynamics. A timber company granted a 25-year concession faces a recurring choice: comply with sustainable harvesting requirements (cooperate) or over-harvest beyond permitted volumes (defect). In a one-shot interaction, defection might seem attractive — maximise short-term revenue before the relationship ends. But the Sarawak Forestry Corporation monitors concession compliance and controls concession renewal decisions. A timber company that defects risks losing its concession renewal — the permanent grim trigger punishment that makes defection unattractive for companies valuing continued access to Sarawak's timber resources. The credibility of the government's commitment to

enforce penalties is therefore essential to sustaining compliance in this repeated game framework.

The Folk Theorem

The Folk Theorem is the foundational result of repeated game theory. It states that if players are sufficiently patient (discount factor δ is close to 1), any outcome that gives each player at least their minimax payoff (the worst the other players can force on them) can be sustained as a subgame perfect equilibrium of an infinitely repeated game. This powerful result has a profound implication: in ongoing relationships, virtually any cooperative outcome — including collusive pricing among competitors, sustained compliance with regulatory requirements, or long-term supply partnerships — can in principle be sustained if the players are patient enough and the threat of punishment is credible. The Folk Theorem does not say cooperation will happen — only that it can be an equilibrium. Selecting among multiple possible equilibria requires additional considerations such as communication, focal points, and institutional design.

6.6 Reputation Formation

Reputation is the expectation that others form about a player's future behaviour based on their past behaviour. In game theory, reputation is a strategic asset — it shapes what others believe about your type (cooperative vs. exploitative, high-quality vs. low-quality, reliable vs. unreliable) and therefore influences the opportunities and terms available to you in future strategic interactions. Reputation formation is especially important in markets with asymmetric information, long-term relationships, and repeated interactions.

The Economics of Reputation

The economic theory of reputation was formalised by Kreps and Wilson (1982) and Milgrom and Roberts (1982), who showed that in finite repeated games with incomplete information about player types, reputation effects can sustain cooperation far longer than the simple finite game logic would suggest. The key is that if there is even a small probability that a player is a 'committed type' (e.g., a firm that is genuinely committed to quality or honest behaviour),

rational opponents will treat the player as likely to be the committed type as long as they have maintained that behaviour. This creates a reputational incentive for even self-interested players to behave as if they were the committed type — because defection destroys the valuable reputation that has been built up over many periods.

Hock Seng Lee (HSL) Berhad: Reputation as Strategic Asset

Hock Seng Lee Berhad (HSL), established in Sarawak in 1973 and listed on Bursa Malaysia, is one of Sarawak's most prominent construction and infrastructure companies. Over five decades of operating in Sarawak, HSL has built a reputation for delivering government infrastructure projects — roads, bridges, drainage systems, marine works — on time and within budget. This reputation functions as a strategic asset in the competitive government tendering market. In repeated game terms, the Sarawak government knows from decades of experience that HSL is a reliable contractor. When HSL submits a tender, this reputation serves as a credibility premium — the government can bid adjust its evaluation of HSL's tender by a discount for execution risk relative to less-established competitors.

Reputation and Sarawak SMEs

For any Sarawak SME seeking to establish long-term relationships with government clients, major corporations, or international buyers, the game-theoretic lesson is direct: invest in reputation early and protect it fiercely. Every delivered project that exceeds expectations, every promise kept, every quality standard maintained contributes to the reputational capital that creates competitive advantages in future interactions. Conversely, a single high-profile quality failure or contract dispute can destroy decades of accumulated reputational capital. The asymmetry between reputation building (slow, costly, incremental) and reputation destruction (fast, often irreversible) implies that risk management in business relationships is as much about protecting reputational capital as about protecting financial capital.

6.7 Supermodular Games

Supermodular games — developed formally by Topkis (1978) and extended by Milgrom and Roberts (1990) — are games in which strategies are strategic complements: one player's

increase in their strategy raises other players' incentives to also increase their strategies. This contrasts with strategic substitutes (where one player's increase reduces others' incentives to act). The concept of supermodularity provides a powerful framework for understanding coordination, complementarity, and the economics of ecosystems — including the industrial development ecosystems that Sarawak is building through its SCORE corridor and digital economy strategy.

Strategic Complements and Complementary Investments

In a supermodular game, strategies are strategic complements when increasing one player's strategy raises the marginal returns to other players' strategies. In economic terms, investments are complementary when each investment makes the others more valuable. The defining mathematical condition is a positive cross-partial derivative of payoffs: $\partial^2 \pi_i / (\partial a_i \partial a_j) > 0$ — the marginal benefit of player i's action increases as player j takes a higher action. This generates powerful coordination dynamics: if one player increases their action, all others have incentive to increase theirs, potentially generating a virtuous cycle of mutually reinforcing investments.

SCORE Corridor: Supermodular Complementary Investments

Sarawak's SCORE corridor industrial development provides a vivid illustration of supermodular investment complementarities. The Sarawak state government invested in the Bakun Dam (2,400 MW) and transmission grid infrastructure. This power infrastructure made it viable for Press Metal to invest in aluminium smelting capacity. The smelting capacity created demand for a deepwater port at Samalaju. The port made it viable for other heavy industries to locate at Samalaju Industrial Park. Each investment made the others more valuable — the power station is more valuable with industrial customers, the smelter is more valuable with port access, the port is more valuable with multiple industrial users. This is a supermodular investment ecosystem: the payoff to any single investment depends positively on the level of all other investments.

Equilibrium Selection in Supermodular Games

Supermodular games generally have multiple equilibria — including a 'low activity' equilibrium (no one invests because no one else is investing) and one or more 'high activity'

equilibria (everyone invests because everyone else is investing). Coordination failure — getting stuck in the low-activity equilibrium — is the central challenge. Governments can play a catalytic role by making public investments that signal commitment to the high-activity equilibrium, coordinating private investors' expectations. Sarawak's government investment in the Baleh Dam, the SCORE master plan, and targeted FDI incentives represents exactly this role: using public commitment to trigger the coordination cascade that generates the high-investment supermodular equilibrium in Sarawak's industrial and digital economy.

6.8 Global Games

Global games — introduced by Carlsson and van Damme (1993) and developed by Morris and Shin (1998, 2003) — provide a framework for analysing strategic situations where agents face uncertainty about the fundamental state of the world and also uncertainty about what other agents believe and will do. Global games have become particularly important in the economics of currency crises, financial contagion, bank runs, and investment under macroeconomic uncertainty — all highly relevant to Sarawak and Malaysian business decision-making in an internationally connected economy.

The Problem of Coordination Under Global Uncertainty

In many coordination games — bank runs, currency attacks, mass investment in an emerging market — the outcome depends on how many other agents act in the same way. Whether a bank run materialises depends not just on whether the bank is fundamentally insolvent, but on whether depositors expect other depositors to run. This creates a self-fulfilling prophecy problem: if everyone expects a run, there will be a run regardless of the bank's fundamentals; if no one expects a run, there will not be a run even if the bank has some weakness. Standard game theory, with its multiple equilibria, provides limited guidance for predicting which outcome will occur.

Global Games Solution: Unique Threshold Equilibrium

The global games framework resolves this multiplicity by introducing a small amount of private information uncertainty: each agent observes a private signal about the fundamental

state (e.g., the true health of the bank's balance sheet) with a small amount of noise. This private information heterogeneity, even in a vanishingly small amount, typically yields a unique equilibrium — a threshold strategy — where agents act (run, attack, invest) if and only if their private signal falls below (or above) a threshold value. The critical insight is that strategic uncertainty about what others will do is resolved through the threshold structure: each agent asks 'what would I need to believe about the fundamental for my action to be optimal, given that others are following the same threshold rule?'

Application to Sarawak Investment Decisions Under Global Uncertainty

For Sarawak and Malaysian investment decisions, the global games framework illuminates how international investor sentiment — shaped by noisy private signals about Malaysian economic fundamentals, political stability, and growth prospects — can trigger sudden shifts in capital flows. The MYR depreciation episode of 2023 reflects global games dynamics: each investor's decision to sell MYR assets was influenced by their private assessment of Malaysian fundamentals and their beliefs about what other investors would do. Government policy responses — BNM's foreign reserve management, the government's fiscal consolidation narrative — aim to shift the threshold at which investors switch from 'stay' to 'exit', making the 'stay' equilibrium more likely to prevail.

6.9 Cooperative Games

The game theory frameworks examined in LU6.1–6.8 are largely non-cooperative: they analyse strategic interaction where players choose independently, without binding agreements. Cooperative game theory takes a different approach: it analyses situations where players can form binding coalitions and asks how the total value created by cooperation should be divided among the coalition members. Cooperative game theory is directly applicable to joint ventures, trade negotiations, corporate alliances, and the structure of federal-state economic relationships.

Characteristic Function Form and Coalition Values

A cooperative game in characteristic function form specifies, for each possible coalition of players, the total value (or payoff) that coalition can guarantee itself. The central question is: given the potential surpluses from different coalition configurations, how should the grand coalition (all players) distribute the total value? Solution concepts include: (1) the Core — the set of allocations where no coalition can profitably deviate and do better on their own; (2) the Shapley Value — a unique allocation formula (proposed by Lloyd Shapley, who won the 2012 Nobel Prize in Economics) that distributes total value proportional to each player's average marginal contribution to all possible coalitions; (3) the Nucleolus — an allocation that lexicographically minimises the maximum dissatisfaction of any coalition.

ASEAN Trade Negotiations: Coalition Formation

ASEAN trade negotiations provide a natural application of cooperative game theory. In negotiating the RCEP agreement, ASEAN members collectively had more bargaining power as a bloc than any individual country would have in bilateral negotiations with China, Japan, or Korea. The cooperative game insight is that ASEAN's value in the grand coalition (ASEAN + major Asian economies) exceeded the sum of individual bilateral negotiating values — creating a rationale for collective action. The distribution of RCEP benefits across ASEAN members, however, reflects the Shapley Value logic: larger, more trade-exposed economies (Singapore, Malaysia, Thailand) with greater marginal contributions to the coalition's negotiating value extract larger shares of the total gains.

Sarawak-Federal Government Cooperation Frameworks

The complex relationship between the Sarawak state government and the Malaysian federal government can be analysed as a cooperative game. Sarawak's constitutional rights over its petroleum resources (asserted through the establishment of PETROS in 2017 and the Gas Supply Ordinance Sarawak 2016) have significantly altered the value distribution in the Sarawak-federal coalition. Before PETROS, virtually all petroleum value was captured by the federal government through Petronas; after PETROS, Sarawak retains a greater share of the petroleum surplus through royalty arrangements and PETROS's participation rights. The cooperative game framework predicts that the allocation will reflect each party's outside option (what they can achieve without cooperation) and their relative bargaining power — insights

that illuminate the ongoing negotiation dynamics between Sarawak and Putrajaya on matters ranging from petroleum revenue sharing to the Malaysia Agreement 1963 review process.

Managerial Insight: Reputation as a Strategic Asset in Sarawak

Construction

Hock Seng Lee (HSL) Berhad is one of Sarawak's largest listed construction companies. In repeated-game terms, HSL has built a reputation over decades for completing government infrastructure projects on time and within budget — a reputation that functions as a strategic asset. In each new government tender, HSL's bid carries an implicit credibility premium: the government knows from repeated interaction that HSL will deliver. This illustrates how reputation formation (LU6.6) translates into a competitive advantage in repeated business relationships. The game-theoretic lesson extends directly to Sarawak SMEs: a company that consistently delivers on its commitments — even at short-term cost — is building a reputational capital stock that compounds over time, generating superior access to contracts, partners, and financing. Conversely, a single high-profile failure can trigger a 'defect' inference by clients, triggering punitive responses (loss of future contracts) that destroy far more value than any short-term gain from cutting corners. Reputation management is therefore a strategic investment decision with calculable game-theoretic logic, not merely a soft management consideration.

Self-Reflection



1. Think about a negotiation you have been involved in — a salary negotiation, a purchase, or a business deal. Identify the players, strategies, and payoffs. Was it a one-shot game or a repeated game? Did reputation play a role? How would game theory have helped you make a better decision?

Questions for Discussion

1. PETROS awards oil block exploration licences through competitive bidding. Three oil companies (Company A, B, C) are each deciding whether to bid aggressively (high bonus offer) or conservatively (low bonus offer). Using game theory concepts from

LU6, construct a simplified payoff matrix for this bidding game, identify the Nash Equilibrium, and explain how PETROS as the 'auctioneer' can design the bidding mechanism to maximise state revenue (e.g., sealed-bid first-price auction vs English auction). (CLO3)

Summary

LU6 introduced game theory as the analytical framework for strategic decision-making under interdependence — situations where outcomes depend on the choices of multiple actors. The core concepts of dominant strategies, Nash Equilibrium, and the Prisoner's Dilemma (LU6.1) explain why competitive markets tend toward non-cooperative equilibria, with real-world illustrations from Sarawak's telecommunications industry. Extensive-form game analysis and backward induction (LU6.2) extend the framework to sequential decisions — relevant to PETROS licensing rounds and entry deterrence strategies.

Games with imperfect information (LU6.3) and signalling (LU6.4) address the reality that most business interactions involve asymmetric information, requiring players to make inferences about opponents' private knowledge from observed behaviour. Repeated games and the Folk Theorem (LU6.5) demonstrate how ongoing relationships enable cooperative outcomes that are impossible in one-shot interactions, while reputation formation (LU6.6) shows how past behaviour creates strategic assets that shape future opportunities — powerfully illustrated by HSL Berhad's position in Sarawak's construction market.

Supermodular games (LU6.7) explain how complementary investments in Sarawak's SCORE corridor create self-reinforcing development dynamics, while global games (LU6.8) illuminate coordination failures under uncertainty — directly relevant to exchange rate dynamics and investment decisions. Cooperative game theory (LU6.9) provides tools for analysing coalition formation in ASEAN trade negotiations and the Sarawak-federal government partnership. Together, these frameworks equip managers with a rigorous vocabulary for diagnosing strategic interactions and designing mechanisms — including auctions, contracts, and governance structures — that achieve better outcomes.

Multiple Choice Questions (MCQs)

1. A Nash Equilibrium is a situation where:
 - A) Both players achieve maximum joint payoff
 - B) No player can improve their payoff by unilaterally changing their strategy, given the other player's strategy
 - C) Players cooperate to maximise total welfare
 - D) One player dominates the game

2. In the Prisoner's Dilemma, the Nash Equilibrium results in:
 - A) The socially optimal outcome
 - B) An outcome that is worse for both players than the cooperative solution
 - C) One player winning and one losing
 - D) No equilibrium

3. Backward induction is used to solve:
 - A) Static games
 - B) Cooperative games
 - C) Sequential (extensive-form) games by working backwards from the last decision
 - D) Games with complete information only

4. Signalling in economics involves:
 - A) All players having the same information
 - B) An informed party taking costly actions to credibly convey private information to uninformed parties
 - C) Government price controls
 - D) Random strategy selection

5. In a repeated game, cooperation can be sustained if:

- A) Players have no future interactions
- B) Players are sufficiently patient and the threat of future punishment (e.g., trigger strategy) deters deviation
- C) One player has a dominant strategy
- D) The game is zero-sum

6. The Folk Theorem in repeated games states that:

- A) Cooperation is impossible in repeated interactions
- B) A wide range of outcomes (including cooperative ones) can be sustained as equilibria if players are patient enough
- C) Nash Equilibrium is always Pareto optimal
- D) Only one equilibrium exists in any repeated game

7. Supermodular games are characterised by:

- A) Strategic substitutes — one player's action reduces others' incentives to act
- B) Strategic complements — one player's action increases others' incentives to act in the same direction
- C) Zero-sum payoffs
- D) No Nash Equilibrium

8. HSL's long-term reputation in Sarawak government contracts is best explained by which game theory concept?

- A) One-shot Prisoner's Dilemma
- B) Reputation formation in repeated games creating a credibility premium
- C) Dominant strategy equilibrium
- D) Zero-sum competition

Online Learning Resources

The following free resources supplement your study of this unit.

Game Theory Online — Yale/Stanford (Coursera, Freemium)

<https://www.coursera.org/learn/game-theory-1>

MRU — Game Theory Videos (Free)

<https://mru.org/courses/game-theory>

NRICH (Cambridge) — Game Theory Interactive Exercises (Free)

<https://nrich.maths.org/game-theory>

Nobel Prize — Game Theory Explainers (Nash, Aumann) (Free)

<https://www.nobelprize.org/prizes/economic-sciences/>

END OF TOPIC

TOPIC

7



DIGITAL AND KNOWLEDGE ECONOMY

Lecturers: Dr. Colin Wong Koh King; Prof. Dr. Dato

Topic 7: Digital and Knowledge Economy

Learning Outcomes

By the end of this topic you should be able to:

- Explain the economic dimensions of the Fourth Industrial Revolution (IR4.0)
- Analyse the impact of digital technologies on market structures, labour markets, and productivity
- Evaluate environmental and governance implications of the digital economy
- Apply knowledge economy concepts to Sarawak's digital transformation strategy
- Anticipate the economic implications of IR5.0 and the future of work in Malaysia



INSPIRING MINDS, SHAPING FUTURE

LO Learning Objectives

By the end of this topic you should be able to:

- **Explain the core characteristics of Industry 4.0 and the digital economy** and their implications for business management.
- **Analyse the economic consequences of digital transformation** including workforce displacement, productivity, and market disruption.
- **Evaluate governance and regulatory challenges** arising from the digital and knowledge economy in Malaysia.
- **Apply insights from the digital economy** to develop future-ready business strategies aligned with IR5.0 principles.

KC Key Concepts and Definitions

Concept	Definition	Example / Notes
Industry 4.0 (IR4.0)	The fourth industrial revolution driven by AI, IoT, automation, and digitalisation.	<i>Smart manufacturing facilities in Sarawak's Samajaya Technology Park.</i>
Digital Economy	Economic activity based on digital technologies, data, and internet-enabled transactions.	<i>Sarawak's 2030 Digital Economy Blueprint aims to digitise all government services.</i>
Knowledge Economy	An economy where knowledge, innovation, and intellectual capital drive growth.	<i>Sarawak's BioNexus clusters leverage knowledge assets in biotechnology and agri-tech.</i>
Artificial Intelligence (AI)	Simulation of human intelligence by machines to perform tasks and make decisions.	<i>AI-powered analytics improve palm oil yield predictions for Sarawak estates.</i>

Disruptive Innovation	An innovation that displaces established products, services, or markets.	<i>Grab disrupted Sarawak's taxi industry; e-wallets disrupted traditional banking.</i>
Platform Economy	An economic system built around digital platforms connecting buyers and sellers.	<i>Shopee and Lazada platforms enable Sarawak SMEs to reach global consumers.</i>

7.1 Introduction to IR4.0

The Fourth Industrial Revolution (IR4.0) — a term coined and popularised by Klaus Schwab, founder of the World Economic Forum, in his 2016 book of the same name — represents the current wave of technological transformation characterised by the fusion of physical, digital, and biological systems. Unlike previous industrial revolutions, which were defined by discrete technological shifts (steam power in IR1.0, electricity and mass production in IR2.0, computers and automation in IR3.0), IR4.0 is characterised by the convergence and interaction of multiple technologies across previously distinct domains: artificial intelligence and machine learning, the Internet of Things (IoT), robotics and advanced automation, big data analytics, cloud computing, additive manufacturing (3D printing), advanced biotechnology, and quantum computing.

The Distinguishing Features of IR4.0

Three characteristics distinguish IR4.0 from earlier technological waves. First, velocity: the pace of technological change is unprecedented — Moore's Law (the doubling of computing power every two years) has driven exponential growth in computing capability for six decades. AI capabilities have progressed from narrow task automation to large language models capable of sophisticated reasoning within a single decade. Second, breadth: IR4.0 technologies are general-purpose — they apply across all industries and sectors simultaneously, unlike industry-specific technologies of earlier periods. Third, depth: IR4.0 fundamentally changes not just production processes but also consumer behaviour, market structures, labour market relationships, governance systems, and even human cognition and biology (through neurotechnology, genetic editing, and human-machine interfaces).

Malaysia's IR4.0 Policy Framework

Malaysia has responded to IR4.0 with a comprehensive national policy architecture. The National Fourth Industrial Revolution (4IR) Policy 2021, developed by the Ministry of Science, Technology and Innovation (MOSTI), provides the overarching strategic framework — setting out six strategic thrusts: talent and culture development; infrastructure; regulation; funding; technology adoption; and policy coordination. Malaysia's MyDigital initiative targets digital economy contribution of 25.5% of GDP by 2025, up from approximately 23% in 2024. The National Industry4WRD policy specifically targets the manufacturing sector's digital transformation, while the MDEC (Malaysia Digital Economy Corporation) coordinates digital economy development across sectors.

Sarawak's Digital Economy Blueprint 2030 and IR4.0 Deployment

Sarawak has developed its own state-level digital economy strategy: the Sarawak Digital Economy Blueprint 2030, managed by the Sarawak Digital Economy Corporation (SDEC). The blueprint targets digital economy contribution of 30% of Sarawak's GDP by 2030, with priority focus on digital infrastructure, digital government services, digital industry adoption, and digital talent development. Sarawak's 5G deployment — with Kuching and Bintulu among the first cities in East Malaysia to receive 5G coverage — provides the connectivity infrastructure for IoT applications in agriculture, manufacturing, and services. The state's investment in smart city infrastructure, including Sarawak's Smart City initiative for Kuching, deploys IR4.0 technologies for urban traffic management, public service delivery, and environmental monitoring. For Sarawak businesses, understanding IR4.0 is not optional — it is the strategic context in which all competitive decisions must now be made.

7.2 Technological Consequences

The technological consequences of IR4.0 are wide-ranging and fundamentally alter production possibilities across all economic sectors. At the macroeconomic level, IR4.0 technologies are general-purpose technologies (GPTs) — innovations that diffuse across the entire economy and transform production processes across many industries simultaneously. Historical GPTs

include the steam engine, electricity, and the internet. The economic literature suggests that GPTs generate productivity gains, but with long and variable lags: it takes decades for organisations, workers, and institutions to restructure around new technological possibilities, which explains why productivity statistics often lag behind the visible pace of technological change.

Artificial Intelligence in Sarawak Industry

Artificial intelligence is transforming multiple industries in Sarawak. PETROS and its production partners have deployed AI-powered well monitoring systems that analyse seismic data, pressure readings, and production rates in real time to optimise extraction, predict equipment failures, and reduce downtime — replacing what previously required teams of engineers reviewing data manually. AI-assisted analysis of well log data accelerates geological interpretation, reducing the time and cost of exploration decision-making. In agriculture, Sarawak's smart farming initiatives are applying AI-powered image recognition to monitor pepper plant health in Serian and Bau, identifying pest infestations and nutrient deficiencies through drone-captured images with accuracy exceeding trained agronomists in controlled trials.

Automation and Smart Grids at SEB

Sarawak Energy Berhad (SEB) has invested in smart grid technology that transforms electricity distribution from a passive one-way system (generator to consumer) to an active, data-driven network. IoT sensors throughout the distribution network provide real-time visibility of electricity flows, faults, and demand patterns. Machine learning algorithms forecast demand by time of day, season, and weather conditions, enabling optimised dispatch of generation capacity. Smart meters in industrial and commercial customers enable dynamic pricing and demand response — reducing peak demand stress on the grid. These technologies collectively reduce distribution losses, improve reliability, and enable SEB to integrate variable renewable energy sources more effectively.

Fintech Disruption in Malaysian Banking and Payments

Financial technology (fintech) has disrupted Malaysia's banking and payments sector with direct impacts on Sarawak consumers and businesses. GrabPay, Touch 'n Go (TNG) eWallet,

and BigPay — all major Malaysian digital payment platforms — have achieved widespread adoption in Kuching and Miri, enabling cashless transactions that bypass traditional banking infrastructure. These platforms are particularly significant in rural Sarawak, where physical bank branch density is low: a Kapit-based trader can now receive digital payment from a Kuching buyer instantly via QR code, whereas a bank transfer might have required a branch visit previously. Bank Negara Malaysia's digital banking framework (five digital bank licences issued in 2022) is accelerating fintech adoption further, with digital banks targeting underserved segments including Sarawak's rural and informal sector population.

7.3 Economic Consequences

The economic consequences of IR4.0 are profound and multidimensional. At the aggregate level, digital technologies are reshaping GDP composition, productivity growth, market structures, and income distribution. Economists debate the timing and magnitude of the productivity dividend from digital technologies — some noting a 'productivity paradox' (Solow, 1987: 'you can see the computer age everywhere except in the productivity statistics') while others identify an emerging wave of AI-driven productivity gains in the 2020s as large language models and AI automation begin to measurably increase output per worker in knowledge-intensive occupations.

Malaysia's Digital Economy: Contribution to GDP

Malaysia's digital economy contributed approximately 23% of GDP in 2024 — representing a significant increase from approximately 19% in 2019 — reflecting accelerated digitalisation during the pandemic and strong post-pandemic growth in e-commerce, digital financial services, and data centre investment. DOSM (Department of Statistics Malaysia) uses the OECD framework for measuring the digital economy, capturing e-commerce transactions, ICT-sector GDP, and digitally-enabled services. The government targets 25.5% digital economy share of GDP by 2025 under the Malaysia Digital initiative, with Sarawak targeting a 30% share by 2030 under its state-level blueprint.

Platform Economy and Winner-Takes-All Dynamics

The platform economy — economic activity mediated by digital platforms connecting buyers and sellers, producers and consumers — is one of the defining economic phenomena of IR4.0. In Malaysia, dominant platforms include Grab (ride-hailing and food delivery), Shopee and Lazada (e-commerce), AirAsia (travel), and BigPay (payments). These platforms exhibit strong network effects: the more users (buyers and sellers) a platform has, the more valuable it becomes to each additional user, creating positive feedback loops that tend toward market concentration. The winner-takes-all (or winner-takes-most) dynamic describes how markets with strong network effects tend to produce dominant monopolistic or near-monopolistic outcomes — the largest platform captures a disproportionate share of total market value. From an economic welfare perspective, this creates a tension between efficiency gains (single large platform offers low transaction costs and vast selection) and competition concerns (dominant platform has pricing power and can foreclose competitors).

Data Centre FDI and the Digital Infrastructure Economy

The surge of data centre foreign direct investment into Malaysia in 2024 — with Microsoft committing RM 10.5 billion, Google committing RM 9.4 billion, and AWS investing substantially — represents a new form of industrial FDI driven by digital economy demand for AI computing infrastructure. Data centres are the physical infrastructure of the digital economy: massive facilities housing servers, networking equipment, and cooling systems that store, process, and transmit the world's digital information. The economics of data centre location are dominated by electricity cost (power accounts for 40–50% of operating cost), connectivity (fibre broadband network proximity), land availability, political stability, and regulatory environment. Malaysia — and specifically Sarawak — offers globally competitive electricity costs from renewable hydroelectric power, making it a preferred location for green data centre development serving the Southeast Asian digital economy.

7.4 Workforce and Employment Implications

The labour market implications of IR4.0 are among the most widely debated issues in contemporary economics and policy. The central tension is between automation-driven job displacement (technology substituting for human labour in existing tasks) and innovation-driven job creation (technology creating new tasks, new industries, and new forms of human-

machine complementarity). Historical evidence from previous industrial revolutions suggests that technological waves ultimately create more jobs than they destroy — but the transition can be painful, prolonged, and unevenly distributed across skills, industries, and regions. For Sarawak's labour market, with its particular combination of resource industry employment, emerging digital sector, and large rural workforce with lower formal education levels, the IR4.0 workforce transition raises acute policy questions.

Task-Based Framework: Automation Risk by Job Type

The academic literature on automation risk, beginning with Frey and Osborne's (2013) influential Oxford study, uses a task-based framework to assess job vulnerability: occupations intensive in routine, codifiable tasks (data entry, basic assembly, routine clerical work) are most vulnerable to automation; occupations intensive in non-routine cognitive tasks (complex analysis, creative work, interpersonal skills, judgment under uncertainty) and non-routine manual tasks (fine motor coordination in novel environments) are least vulnerable. Applying this framework to Sarawak's labour market, timber industry jobs involving predictable, repetitive tasks (machine operators, sawmill workers) face significant automation risk; while jobs requiring community relationships, cultural knowledge, and contextual judgment (agricultural extension officers, community health workers, indigenous language teachers) are much more resilient to automation.

Gig Economy in Sarawak

The gig economy — platform-mediated work arrangements where individuals provide services on a task-by-task basis without traditional employment relationships — has expanded significantly in Kuching and Miri. GrabCar drivers, Grab Food and Food Panda delivery riders, and freelance digital service providers constitute a growing segment of Sarawak's urban labour market. From an economic welfare perspective, gig work offers flexibility and supplementary income to participants, but raises important questions about social protection: gig workers typically lack EPF contributions, SOCSO coverage, paid leave, and employment security. This creates a 'precariat' — workers with unstable incomes and inadequate social protection — as a structural feature of the digital labour market. BNM's policy research and Malaysia's Employment Act 1955 amendments have begun addressing gig economy regulation, but significant gaps remain.

Reskilling and Digital Talent Development in Sarawak

SDEC has invested substantially in digital talent development to supply the human capital needed for Sarawak's digital economy aspirations. The SDEC Academy offers coding bootcamps, data analytics training, digital marketing, and cybersecurity courses — targeting fresh graduates, career-changers, and SME owners seeking to upgrade digital competencies. Malaysia's HRD Corp (Human Resource Development Corporation) provides levy-funded training grants enabling companies to subsidise employee digital skills development. MDEC's Malaysia Digital economy programmes support tech talent development nationally, with Sarawak-specific components. The World Economic Forum's Future of Jobs Report 2025 estimates that 60% of all workers globally will need reskilling by 2027 — a challenge of enormous scale requiring coordinated government, education, and employer responses across Malaysia and Sarawak.

7.5 Environmental and Sustainability Dimensions

The relationship between digital technology and environmental sustainability is complex and multidirectional. IR4.0 technologies offer powerful tools for environmental monitoring, resource efficiency optimisation, and clean energy management. At the same time, the digital economy is itself a major and growing source of energy consumption and carbon emissions — creating what economists term a 'digital rebound effect' where efficiency gains from digitisation are offset by increased overall consumption of digital services. For Sarawak, with its exceptional renewable energy endowment and strategic aspiration to become a green digital economy hub, this tension has direct relevance to development planning.

Data Centres and Energy Consumption

Global data centres consumed approximately 460 terawatt-hours (TWh) of electricity in 2022 — roughly 2% of global electricity consumption — and this share is growing rapidly as AI training workloads become more compute-intensive. Training a large language model like GPT-4 is estimated to consume approximately 10,000 MWh of electricity — equivalent to the annual electricity consumption of roughly 1,000 Malaysian households. The rapid proliferation

of AI applications will substantially increase data centre energy demand over the 2025–2030 period. Sarawak's competitive advantage — cheap, clean hydroelectric power from SEB — positions it ideally for this demand wave. A data centre powered by Sarawak's hydroelectric grid has a carbon footprint significantly lower than one powered by coal (as in parts of China) or natural gas (as in Singapore), making Sarawak a genuinely 'green' computing location.

Malaysia's Carbon Market and Green Economy Initiatives

Malaysia is developing the institutional infrastructure for a carbon market and green economy transition. Bursa Carbon Exchange (BCX), launched in 2022, is Southeast Asia's first voluntary carbon market, enabling Malaysian companies to buy and sell verified carbon credits from forestry, renewable energy, and other low-carbon projects. Sarawak's extensive managed forests and expanding renewable energy capacity position it as a significant supplier of carbon credits — a potential new revenue stream for the state. The government's National Energy Transition Roadmap (NETR) targets 70% renewable energy in the national electricity mix by 2050, with Sarawak's hydroelectric contribution being a critical component. Additionally, Sarawak has initiated a green hydrogen development strategy, leveraging its cheap hydroelectric power to produce green hydrogen (via electrolysis using clean electricity) for export to Japan and Korea — markets that have committed to hydrogen as a cornerstone of their decarbonisation strategies.

IoT for Environmental Monitoring and Precision Agriculture

IR4.0 technologies offer significant environmental management benefits for Sarawak. IoT sensor networks can monitor river water quality in Sarawak's major rivers (the Rejang, Baram, Sarawak rivers) in real time, detecting pollution from industrial, agricultural, and urban sources. Satellite imagery combined with AI analysis enables near-real-time monitoring of forest cover changes, enabling both the state forestry department and external verifiers (required under EUDR compliance) to track deforestation with unprecedented accuracy and cost efficiency. Precision agriculture technologies — soil sensors, drone-based canopy analysis, AI-powered yield prediction — enable palm oil farmers to optimise fertiliser and pesticide application, reducing input costs and environmental impacts simultaneously. These technologies represent a powerful convergence of digital economy and environmental sustainability that aligns with Sarawak's strategic positioning.

7.6 Governance and Regulatory Consequences

The digital economy creates profound governance challenges. Traditional regulatory frameworks were designed for physical products, geographic markets, and clearly identifiable legal entities — none of which map neatly onto digital platforms operating across jurisdictions, intangible digital goods, and data flows that traverse borders instantaneously. Governments worldwide are grappling with how to regulate digital markets to protect consumers, maintain competition, ensure data privacy, prevent harmful content, and collect fair taxes — while avoiding regulatory overreach that stifles the innovation that generates digital economy value.

Malaysia's PDPA 2010: Adequacy and Reform

Malaysia's Personal Data Protection Act (PDPA) 2010 provides the primary legal framework for data governance, establishing principles for lawful data processing: informed consent, stated purpose, data minimisation, accuracy, storage limitation, and security. However, the PDPA has been widely criticised as inadequate for the contemporary digital economy: it does not cover data processing by government agencies; its enforcement mechanisms are relatively weak; it does not address the rights of data subjects (no right to erasure, no data portability); and its scope does not fully address cross-border data transfers. A PDPA amendment bill has been under development, but as of 2024, Malaysia lags significantly behind the EU's GDPR (General Data Protection Regulation, 2018) in the comprehensiveness and enforcement power of its data protection framework.

The EU AI Act and Implications for Malaysian Tech Companies

The EU Artificial Intelligence Act (AI Act), which entered into force in 2024, establishes a risk-based regulatory framework for AI systems sold or deployed in the EU — the world's most comprehensive AI regulation to date. High-risk AI applications (those used in employment decisions, credit scoring, law enforcement, healthcare, and critical infrastructure) face strict requirements for transparency, explainability, accuracy, and human oversight. For Malaysian tech companies exporting AI-powered products or services to European markets, the AI Act creates significant compliance obligations. Companies developing AI-powered hiring tools,

credit scoring algorithms, or medical diagnostic systems must ensure their systems meet EU transparency and fairness standards — or face market exclusion from the EU's 450 million consumer market.

Sarawak's Digital Governance Landscape

Sarawak has taken independent initiatives in digital governance, reflecting the state government's proactive approach to managing the digital transition. Discussions around a Sarawak Data Protection Bill have been ongoing, recognising the inadequacy of the federal PDPA for the specific needs of Sarawak's digital economy development context. MCMC (Malaysian Communications and Multimedia Commission) provides the national regulatory framework for telecommunications and internet services, but Sarawak's unique telecommunications challenges — rural connectivity, satellite internet deployment for interior communities, spectrum allocation for 5G — require engagement with both federal and state-level governance. For managers operating in Sarawak's digital economy, maintaining compliance with the PDPA (and its eventual replacement), MCMC regulations, and emerging sector-specific digital regulations is a legal and strategic imperative.

7.7 Ethical and Humanitarian Concerns

The rapid deployment of digital technologies across the economy raises profound ethical and humanitarian questions that economists, social scientists, and policymakers are urgently grappling with. While IR4.0 offers enormous potential to improve welfare, productivity, and quality of life, its benefits and harms are not evenly distributed — creating new forms of inequality, vulnerability, and power asymmetry that require active policy and managerial attention.

Digital Divide: Sarawak's Rural Connectivity Challenge

Digital inequality — the gap between those with access to digital technologies and the connectivity, skills, and devices to use them effectively, and those without — is one of the most acute humanitarian dimensions of IR4.0. In Sarawak, the digital divide maps closely onto geography and ethnicity: urban centres (Kuching, Miri, Sibul, Bintulu) have strong 4G and

emerging 5G connectivity, while inland districts including Kapit, Limbang, Belaga, and interior Baram communities have limited to no reliable broadband connectivity. This connectivity gap means that rural Sarawakians are systematically excluded from digital economy opportunities: online banking, e-commerce, digital government services, telehealth, and online education are effectively inaccessible. The Malaysian government's National Fiberisation and Connectivity Plan (NFCP) and Jendela initiative have accelerated rural broadband rollout, but significant gaps remain in Sarawak's interior.

AI Bias and Algorithmic Fairness

AI systems trained on historical data can perpetuate and amplify existing biases — a phenomenon with direct humanitarian implications. Hiring algorithms trained on historical employment data may systematically disadvantage applicants from underrepresented ethnic groups, reflecting historical discriminatory patterns in the training data rather than genuine ability differences. Credit scoring algorithms using alternative data sources (social media activity, mobile phone usage patterns, location data) may disadvantage rural Sarawakians whose data profiles differ from the urban majority on which models were trained. These AI bias concerns are not hypothetical — they have been documented in real-world systems in the US, UK, and increasingly in Southeast Asian financial services contexts. For Malaysian businesses deploying AI in human resource management and customer credit decisions, auditing for algorithmic fairness is both an ethical obligation and an emerging regulatory requirement.

Surveillance Capitalism and Data Sovereignty

Surveillance capitalism — the economic logic described by Shoshana Zuboff in her 2019 book *The Age of Surveillance Capitalism* — describes a business model where personal data, extracted from users without full informed consent, is processed into behavioural prediction products sold to advertisers and other clients. This model, pioneered by Google and Facebook, creates fundamental tensions with individual privacy, autonomy, and democratic governance. For ASEAN economies including Malaysia, there is an additional dimension of digital colonialism: if the primary platforms mediating access to the digital economy are owned and controlled by US or Chinese technology corporations, the data of Malaysian consumers flows offshore to generate profits in the US or China, raising questions of data sovereignty and

economic dependency. Sarawak's aspiration to build locally owned digital platforms — through SDEC's digital ecosystem development programmes — reflects a strategic response to these concerns.

7.8 Future Trends and the Fifth Industrial Revolution (IR5.0)

The Fifth Industrial Revolution (IR5.0) is an emerging conceptual framework that responds to the limitations and negative externalities identified in IR4.0's efficiency-focused paradigm. Where IR4.0 prioritised automation, digitalisation, and economic efficiency, IR5.0 shifts emphasis toward human-centric design: technology that augments human capabilities rather than simply replacing them; resilient and sustainable industrial systems; and production models that serve broad societal well-being rather than narrow shareholder value maximisation. The European Commission's Industry 5.0 vision (2021) defines it around three pillars: human-centricity, sustainability, and resilience.

Collaborative Robotics and Human-Machine Complementarity

A defining technology of IR5.0 is collaborative robotics (cobots) — robots designed to work alongside human operators in shared workspaces, complementing human capabilities (dexterity, judgment, creativity, social interaction) rather than replacing them entirely. Unlike the industrial robots of IR4.0 (which operate in caged, fully automated environments), cobots are designed to be safe, flexible, and adaptable to human-directed tasks. In Sarawak's manufacturing context, cobots could support assembly operations in the aluminium and petrochemical sectors — handling heavy, repetitive, or hazardous tasks while human workers focus on quality inspection, problem-solving, and machine management. The economic implication is that IR5.0 may generate a more complementary labour-technology relationship, sustaining or growing human employment while dramatically enhancing productivity, compared to IR4.0's more substitutive automation.

Malaysia's National Industry 4WRD Policy and IR5.0 Transition

Malaysia's National Industry 4WRD policy, launched in 2018 and extended in 2021, provides the strategic framework for manufacturing sector digitalisation. While initially focused on

IR4.0 adoption (smart manufacturing, IoT integration, automation), the policy has evolved to incorporate IR5.0 elements: emphasis on sustainable manufacturing, human capital upgrading alongside automation, and resilience-building in supply chains. The Industry4WRD Readiness Assessment tool allows Malaysian manufacturers — including Sarawak-based producers in food processing, timber, and petrochemicals — to benchmark their IR4.0 adoption status and identify priority areas for digital investment.

Sarawak's Vision: Digital-Green Economy Convergence in IR5.0

Sarawak's strategic position at the IR4.0/IR5.0 intersection is arguably unique among Malaysian states. Its renewable energy advantage (hydroelectric, solar, potentially green hydrogen) aligns perfectly with IR5.0's sustainability pillar. Its large, talent-hungry digital infrastructure sector (data centres, AI computing) drives demand for the human-centric, high-skill workforce development that IR5.0 envisions. Its culturally diverse, community-oriented society aligns with IR5.0's emphasis on inclusive, equitable development rather than efficiency-only optimisation. The Sarawak government's Post COVID-19 Development Strategy (PCDS 2030) and Digital Economy Blueprint 2030 collectively envision an economy where digital technology, renewable energy, and social inclusion reinforce each other — a genuinely IR5.0 development model that could serve as a template for resource-rich developing economies globally.

Quantum Computing and the Next Horizon

Quantum computing — based on quantum mechanical phenomena (superposition and entanglement) enabling computation far beyond the capabilities of classical computers for certain problem types — represents the next horizon of technological transformation. While full-scale, fault-tolerant quantum computers remain under development, quantum computing is expected to achieve practical commercial applications in cryptography, drug discovery, materials science, and complex optimisation problems within the 2030–2040 timeframe. For Sarawak and Malaysia, quantum computing has two strategic implications: first, current cryptographic systems protecting financial transactions, government data, and private communications will become vulnerable to quantum decryption, requiring migration to quantum-resistant cryptography in the 2025–2035 window; second, quantum machine learning

algorithms may dramatically accelerate AI capabilities, opening new economic applications but also amplifying workforce disruption risks beyond current IR4.0 projections.

Managerial Insight: Sarawak's Green Data Centre Strategy — IR4.0 Meets Renewable Energy Economics

In 2024, Microsoft announced a RM 10.5 billion investment in Malaysia's AI and cloud infrastructure, with Sarawak being identified as a priority site due to its abundant renewable hydroelectric power — the greenest and cheapest in Southeast Asia. This convergence of IR4.0 demand (AI computing power) with Sarawak's comparative advantage (cheap, clean electricity from SEB) illustrates how digital economy economics creates new strategic opportunities for resource-rich but non-industrial regions. Power Usage Effectiveness (PUE) — the ratio of total data centre power consumption to IT equipment power consumption — is the primary efficiency metric in data centre economics. A hyperscale data centre in Sarawak using SEB hydroelectric power achieves a carbon intensity of approximately 0.032 kg CO₂ per kWh, compared to Singapore's natural gas grid at approximately 0.42 kg CO₂ per kWh — a 13-fold carbon intensity advantage. This 'green premium' is increasingly valuable to global technology companies facing net-zero commitments from shareholders and regulators. Managers who understand the economics of digital infrastructure — data centre power costs, cloud computing pricing, AI compute economics — are positioned to spot investment opportunities others miss.

Self-Reflection



1. Assess your own workplace (current or most recent) on the IR4.0 readiness spectrum. Which technologies (AI, IoT, cloud, automation, big data) are already in use? Which of your current skills are at risk of automation within 10 years? What specific digital competencies would you prioritise developing to remain economically relevant in the IR5.0 era?

Questions for Discussion

1. A traditional Sarawak timber company is considering transforming into a precision forestry and carbon credit business using digital technology, IoT sensors, and satellite

imagery. Using the digital economy framework from LU7, analyse: (a) the economic opportunities and threats from IR4.0 transformation, (b) the workforce and environmental implications, (c) the governance and regulatory challenges under Malaysia's current digital framework, and (d) the competitive positioning of the transformed company in an IR5.0 economy. (CLO3)

Summary

LU7 examined the digital and knowledge economy through eight interrelated dimensions, using Sarawak's digital transformation as the running contextual thread. IR4.0 — characterised by the fusion of physical, digital, and biological systems through AI, IoT, robotics, and big data — is not a distant future scenario but the present competitive environment in which Sarawak and Malaysian businesses must operate. The SDEC Digital Economy Blueprint 2030 and Malaysia's MyDigital initiative provide the policy framework within which businesses must locate their digital strategies.

The technological consequences of IR4.0 (AI in PETROS operations, smart grids at SEB, fintech disruption) translate into economic consequences (platform economy dynamics, winner-takes-all market concentration, data centre FDI surge) that fundamentally alter competitive landscapes. Workforce implications are the most sensitive: automation displaces routine tasks while creating new digital roles, generating a reskilling imperative that SDEC Academy and HRD Corp programmes are beginning to address. Environmental dimensions are strategically important for Sarawak — its renewable energy advantage positions it as an ideal location for green data centres serving the AI compute demand of the 2025–2030 decade.

Governance challenges (PDPA inadequacy, EU AI Act compliance implications) and ethical concerns (digital divide in rural Sarawak, AI bias risks, surveillance capitalism) require active managerial and policy responses. IR5.0 offers an evolved vision — human-centric, sustainable, resilient — that aligns naturally with Sarawak's development priorities, suggesting that the state is well-positioned to be an IR5.0 model economy if it manages the digital transition with both strategic ambition and social inclusiveness.

Multiple Choice Questions (MCQs)

1. IR4.0 is characterised primarily by:
 - A) Steam power and mechanisation
 - B) Mass production and assembly lines
 - C) The fusion of physical, digital, and biological systems through technologies like AI, IoT, and robotics
 - D) Nuclear energy and space exploration

2. The 'winner-takes-all' dynamic in digital markets occurs because:
 - A) Digital products are expensive to produce
 - B) Network effects — the value of a platform increases with users — create high barriers to competition
 - C) Government regulation favours large firms
 - D) Digital products are easily copied

3. Sarawak's comparative advantage for green data centres is primarily:
 - A) Low labour costs
 - B) Abundant, cheap renewable hydroelectric power from SEB
 - C) Proximity to China
 - D) Low corporate tax rates

4. The gig economy primarily challenges traditional labour economics by:
 - A) Creating full-time employment with benefits
 - B) Creating a category of workers outside traditional employee-employer relations, with implications for benefits, wages, and regulation
 - C) Reducing total employment
 - D) Increasing union membership

5. Under Malaysia's PDPA 2010, businesses collecting personal customer data must:
 - A) Share data freely with government agencies

- B) Obtain informed consent, state the purpose, and protect data from unauthorised access
- C) Delete all data after 30 days
- D) Report data to Bank Negara Malaysia monthly

6. IR5.0 differs from IR4.0 in that it emphasises:

- A) Greater automation and efficiency only
- B) Human-centric technology design — resilience, sustainability, and human well-being alongside productivity
- C) Replacing all human workers with robots
- D) Complete digitalisation of government services

7. Microsoft's RM 10.5 billion investment in Malaysia's AI/cloud infrastructure is best classified as:

- A) Portfolio investment
- B) Foreign Direct Investment (FDI) in the digital economy
- C) A government grant
- D) An export subsidy

8. Digital inequality in Sarawak's rural interior primarily manifests as:

- A) Too many technology options
- B) Poor internet connectivity and digital literacy gaps, limiting economic participation for rural communities
- C) Excessive digital spending
- D) Competition from urban businesses

Online Learning Resources

The following free resources supplement your study of this unit.

**SDEC — Sarawak Digital Economy Blueprint 2030
(Free, Sarawak)**

<https://www.sdec.com.my/digital-economy-blueprint>

**World Economic Forum — Future of Jobs Report
2025 (Free)**

<https://www.weforum.org/publications/the-future-of-jobs-report-2025>

**MDEC — Malaysia Digital Economy Data (Free,
Malaysian)**

<https://mdec.my>

**MIT Technology Review — IR4.0 and IR5.0 Articles
(Freemium)**

<https://www.technologyreview.com>

END OF TOPIC

CONSOLIDATED GLOSSARY OF KEY ECONOMICS TERMS

The following terms are defined for quick reference. Definitions are written in plain language for MBA students; formal mathematical definitions are provided in Appendix B.

Scarcity: The fundamental economic condition in which human wants and needs exceed the available resources to satisfy them, necessitating choices about how to allocate limited resources among competing uses.

Opportunity Cost: The value of the next-best alternative foregone when a decision is made; the true economic cost of any choice is the value of what must be given up to obtain it.

Production Possibility Frontier (PPF): A curve showing all combinations of two goods that an economy can produce when all resources are fully and efficiently employed; points inside the frontier represent inefficiency, points outside are unattainable with current resources.

Price Elasticity of Demand (PED): A measure of the responsiveness of quantity demanded to a change in price; $PED = \% \Delta Q_d / \% \Delta P$; absolute values greater than 1 indicate elastic (price-sensitive) demand, less than 1 indicate inelastic demand.

Supply: The quantity of a good or service that producers are willing and able to offer for sale at various prices during a specified period, other things held constant; represented by an upward-sloping supply curve.

Demand: The quantity of a good or service that consumers are willing and able to purchase at various prices during a specified period; represented by a downward-sloping demand curve.

Equilibrium Price: The price at which the quantity demanded by buyers exactly equals the quantity supplied by producers, leaving no persistent shortage or surplus in the market.

Consumer Surplus: The difference between the maximum price consumers are willing to pay for a good and the price they actually pay; a measure of the net benefit consumers receive from market participation.

Producer Surplus: The difference between the price producers receive for a good and the minimum price they would accept (their marginal cost); a measure of the net benefit producers receive from market participation.

Deadweight Loss: The loss of total economic surplus (consumer plus producer surplus) caused by a market distortion such as a tax, price control, or monopoly pricing; represents welfare that is destroyed rather than transferred.

Price Ceiling: A legally imposed maximum price set below the market equilibrium price, typically to make goods affordable; creates persistent shortages as quantity demanded exceeds quantity supplied at the controlled price.

Price Floor: A legally imposed minimum price set above the market equilibrium price, typically to support producer incomes; creates persistent surpluses as quantity supplied exceeds quantity demanded at the controlled price.

Fixed Cost: A cost that does not change with the level of output in the short run; examples include rent, insurance, and depreciation on plant and equipment; fixed costs must be paid whether the firm produces zero or positive output.

Variable Cost: A cost that changes directly with the level of output; examples include raw materials, direct labour, and energy; variable costs are zero when output is zero.

Marginal Cost: The additional cost of producing one more unit of output ($MC = \Delta TC / \Delta Q$); the marginal cost curve is U-shaped and crosses the AVC and ATC curves at their minimum points; the profit-maximising output rule for competitive firms is $P = MC$.

Average Total Cost (ATC): Total cost divided by output ($ATC = TC/Q$); the U-shaped ATC curve reflects the interaction of fixed and variable costs; the minimum point of ATC is the 'efficient scale' of production.

Economies of Scale: The reduction in average cost per unit that occurs when a firm increases its scale of production; driven by specialisation, division of labour, indivisibilities, and bulk purchasing; a key source of competitive advantage for large firms.

Perfect Competition: A market structure with many small firms selling identical products, free entry and exit, and perfect information; individual firms are price-takers; long-run equilibrium occurs where $P = MC = ATC$, maximising allocative and productive efficiency.

Monopoly: A market structure with a single seller of a good or service with no close substitutes; monopolists are price-makers, restrict output below the competitive level, charge prices above marginal cost, and generate deadweight loss; examples include SEB's electricity distribution in Sarawak.

Oligopoly: A market structure characterised by a small number of interdependent firms whose pricing and output decisions affect each other; competitive outcomes range from perfect competition (Bertrand) to monopoly (collusion); examples include Malaysia's telecommunications sector.

Nash Equilibrium: A combination of strategies in a game where no player can improve their payoff by unilaterally changing their strategy, given the strategies chosen by all other players; the central solution concept of game theory.

Prisoner's Dilemma: A game where individually rational choices lead to collectively suboptimal outcomes; both players have dominant strategies (defect) that yield a Nash Equilibrium worse for both than the cooperative outcome; widely applicable to price competition, arms races, and environmental agreements.

Gross Domestic Product (GDP): The total monetary value of all final goods and services produced within a country's borders during a specified period, typically measured quarterly and annually; the primary measure of national economic output and size.

Gross National Product (GNP): The total monetary value of all goods and services produced by a country's residents, regardless of where production occurs; $GNP = GDP + \text{Net Factor Income from Abroad}$; relevant for measuring national income versus domestic production.

Consumer Price Index (CPI): A measure of the average price level of a fixed basket of consumer goods and services, used to track inflation and adjust nominal values for purchasing power changes; published monthly by DOSM in Malaysia.

Inflation: A sustained increase in the general price level of goods and services in an economy over time, eroding purchasing power; measured by the rate of change in the CPI; managed primarily through monetary policy (OPR adjustments by BNM).

Monetary Policy: Actions taken by a central bank (Bank Negara Malaysia) to manage the money supply, interest rates, and credit conditions to achieve macroeconomic objectives — price stability, full employment, and economic growth; key instruments include the Overnight Policy Rate (OPR).

Fiscal Policy: Government decisions on taxation and public spending designed to influence aggregate demand, income distribution, and economic growth; expansionary fiscal policy (higher G or lower T) stimulates demand; contractionary fiscal policy reduces it.

Aggregate Demand (AD): The total demand for goods and services in an economy at a given price level, comprising consumption (C), investment (I), government spending (G), and net exports (NX); the AD curve is downward-sloping due to wealth, interest rate, and exchange rate effects.

Aggregate Supply (AS): The total quantity of goods and services that firms in an economy are willing and able to produce at various price levels; short-run AS slopes upward; long-run AS is vertical at potential output (LRAS), determined by resource endowments and technology.

Comparative Advantage: A country (or firm) has a comparative advantage in producing a good if it can produce it at a lower opportunity cost than its trading partner; the basis for mutually beneficial international trade even when one party has absolute advantage in all goods.

Balance of Payments: A systematic record of all economic transactions between residents of a country and the rest of the world; comprises the current account (trade, services, income), capital account, and financial account; must sum to zero by accounting identity.

Exchange Rate: The price of one currency expressed in terms of another; for Malaysia, typically quoted as MYR per USD; appreciation means fewer MYR per USD (stronger MYR); depreciation means more MYR per USD (weaker MYR); affects export competitiveness and import costs.

Foreign Direct Investment (FDI): Investment by a firm or individual in productive capacity in a foreign country, involving a lasting interest and degree of management control; contrasts with portfolio investment; brings capital, technology, employment, and market access to host countries.

IR4.0: The Fourth Industrial Revolution — characterised by the fusion of physical, digital, and biological systems through AI, IoT, big data, robotics, and advanced manufacturing; represents the current wave of technological transformation reshaping economies and industries worldwide.

Digital Economy: The component of economic activity driven by digital technologies, including e-commerce, digital financial services, cloud computing, AI applications, and data-intensive services; contributed approximately 23% of Malaysia's GDP in 2024.

Total Factor Productivity (TFP): A measure of the efficiency with which an economy uses all its inputs (labour and capital) to produce output; TFP growth represents improvements in technology, organisation, and management that increase output beyond what growth in inputs would predict.

Gini Coefficient: A statistical measure of income or wealth inequality ranging from 0 (perfect equality — everyone has the same income) to 1 (perfect inequality — one person has all income); Malaysia's Gini coefficient was approximately 0.401 in 2022 (DOSM), reflecting moderate inequality.

Human Development Index (HDI): A composite index developed by UNDP measuring average achievement in three dimensions of human development: long and healthy life (life expectancy), knowledge (education years), and a decent standard of living (GNI per capita); Malaysia's HDI was 0.803 in 2023 (high human development category).

NAIRU: Non-Accelerating Inflation Rate of Unemployment — the unemployment rate consistent with stable inflation; also called the 'natural rate' of unemployment; includes frictional and structural unemployment but not cyclical; Malaysia's estimated NAIRU is approximately 3–3.5%.

Net Present Value (NPV): The sum of the present values of all cash flows from an investment, discounted at the appropriate cost of capital; $NPV > 0$ indicates an investment creates value; used for capital investment appraisal including major infrastructure projects such as Sarawak's Baleh Dam.

Game Theory: The mathematical study of strategic interaction among rational decision-makers, analysing how each player's optimal choice depends on the expected choices of others; central to understanding competition, negotiation, auction design, and cooperation in business contexts.

Backward Induction: A solution method for sequential (extensive-form) games that works backwards from the final decision node to determine optimal strategies at each earlier node; the foundation of subgame perfect equilibrium analysis.

Supermodular Games: Games where strategies are strategic complements — one player's increase in their strategy raises other players' incentive to increase their strategies; generates coordination dynamics and multiple equilibria; applicable to complementary investment decisions in Sarawak's SCORE corridor.

PETROS: Petroleum Sarawak Berhad — Sarawak's state-owned petroleum company, established in 2017 following Sarawak's assertion of constitutional rights over its petroleum resources; responsible for regulating and participating in Sarawak's oil and gas sector, including upstream licensing and downstream gas supply.

SDEC: Sarawak Digital Economy Corporation — the state agency responsible for driving Sarawak's digital economy transformation, including the Sarawak Digital Economy Blueprint 2030; manages digital infrastructure, digital talent development, and startup ecosystem programmes.

DOSM: Department of Statistics Malaysia — the primary federal government agency responsible for collecting, processing, and disseminating official statistical data including GDP, CPI, employment, household income, and trade statistics.

BNM (Bank Negara Malaysia): The central bank of Malaysia, responsible for monetary policy (OPR setting), financial system stability, foreign exchange reserve management, and regulation of the banking, insurance, and payment systems; publishes the Annual Report and quarterly BOP statistics.

Answer Key for Self-Check Multiple Choice Questions

The correct answers for all Self-Check MCQs in MBA6664 are listed below. Students are encouraged to attempt all questions before consulting this key.

Learning Unit	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
LU1	B	B	B	B	B	C	C	B
LU2	B	C	C	C	B	C	B	B
LU3	B	B	B	C	C	B	B	B
LU4	B	C	B	C	B	B	B	B
LU5	B	B	B	B	B	B	C	B
LU6	B	B	C	B	B	B	B	B
LU7	C	B	B	B	B	B	B	B

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Assessment Overview

Assessment Component	Detail
Type	Individual Written Assignment
CLO Alignment	CLO1 (primary) — Analyse microeconomics and macroeconomics models; CLO2 — Recommend models to real-world situations
Word Count	1,500–2,000 words (excluding title page, diagrams, and references)
Due Date	Week 8 — submit via i-CATS LMS by 11:59 PM
Submission Format	Microsoft Word (.docx) or PDF uploaded to i-CATS LMS
Assessment SLT	30 hours self-learning time (included in CA total SLT)

Assessment Scenario

"The Malaysian government announced in June 2024 that targeted fuel subsidies would replace the blanket RON95 petrol subsidy, with Sarawak consumers potentially paying higher prices. As an economic analyst at a Kuching-based consulting firm, you have been asked to prepare an economic analysis report for a logistics company client that operates a fleet of 80 vehicles across Sarawak, serving the timber, palm oil, and retail sectors."

Required Sections

1. Market Analysis (approximately 400 words): Use supply-demand diagrams to analyse the impact of subsidy removal on Sarawak's petrol market. Your written analysis must describe: (a) the original market equilibrium with the subsidy (showing the subsidy effect on the supply curve and the resulting lower consumer price); (b) the new equilibrium after subsidy removal (higher price, lower quantity demanded); and (c) the size of the consumer surplus loss and government fiscal saving. Describe the diagrams

clearly in words — include a description of what each axis represents, where the curves shift, and what the new equilibrium represents.

2. Elasticity Analysis (approximately 400 words): Calculate and interpret the price elasticity of demand (PED) for petrol in Sarawak, given the following data: Before subsidy removal — price: RM 2.05/litre, daily sales: 2.4 million litres; After subsidy removal — price: RM 3.35/litre, daily sales: 2.1 million litres. Use the midpoint formula: $PED = [(Q2 - Q1) / ((Q1 + Q2)/2)] / [(P2 - P1) / ((P1 + P2)/2)]$. Interpret the result. How does PED differ between urban Kuching consumers (more alternatives, lower PED in absolute value) and rural consumers in Kapit and Limbang (fewer alternatives, higher PED in absolute value)?
3. Macroeconomic Impact (approximately 400 words): Using AD-AS framework analysis, evaluate the macroeconomic implications of the fuel price increase — effects on household consumption (C), the CPI (inflation), and GDP growth. How does a 63% increase in petrol prices (RM 2.05 to RM 3.35) ripple through the AD-AS model for Sarawak's economy?
4. Policy Recommendation (approximately 400 words): Recommend two alternative government policies that would achieve fiscal savings while minimising welfare losses for low-income Sarawak consumers. Use economic theory (elasticity, deadweight loss, income effects) to justify each recommendation and discuss the equity-efficiency trade-off involved.

AI and Academic Integrity Note

Students may use AI tools (e.g., ChatGPT, Microsoft Copilot) as learning aids to brainstorm, structure arguments, or check grammar. However, students must: (1) Verify all AI-generated content against course readings and their own analysis; (2) Not submit AI-generated text as their own original work; (3) Include an 'AI Tools Used' declaration at the end of the submission stating which tools were used and for what purpose; (4) Not upload confidential, personal, or proprietary information to external AI platforms, in compliance with Malaysia's PDPA 2010 and i-CATS academic integrity policies.

Assessment Rubric — CA1

Criterion	CLO	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Correct Model Application	CLO1	Diagrams described accurately; supply-demand and AD-AS both used correctly; subsidy removal effects correctly traced	Good model use with minor errors in diagram description or curve shifts	Basic models applied; some conceptual gaps in diagram or analysis	Models absent or incorrectly applied; diagrams missing
Elasticity Analysis	CLO1	PED correctly calculated using midpoint formula; result correctly interpreted; rural vs urban differentiation insightful and accurate	Correct calculation; limited rural/urban differentiation analysis	Basic elasticity analysis; numerical calculation attempted but some errors	Elasticity concept missing, incorrect formula, or no numerical work
Macro-Micro Integration	CLO2	Seamlessly connects micro subsidy analysis to	Good connection between micro and	Both levels addressed but in separate	Only one level (micro OR macro)

		macro AD-AS implications; multiplier effects considered	macro levels; minor analytical gaps	sections without integration	addressed; no integration
Policy Recommendation Quality	CLO2	Two realistic, well-justified policies; equity-efficiency trade-off explicitly addressed; Sarawak context specific	Two policies with reasonable justification; some attention to trade-offs	One policy with limited justification; equity or efficiency addressed but not both	No recommendation, unrealistic proposals, or no economic justification
Quantitative Evidence & Communication	CLO1+CLO2	PED calculation shown; data used throughout; APA 7th citations; professional analytical writing standard	Good use of data; minor writing or citation issues	Some data used; writing adequate; limited citations	No data, no calculations shown; poor writing; missing citations

Assessment Overview

Assessment Component	Detail
Type	Individual Case Study Analysis
CLO Alignment	CLO2 (primary) — Recommend economic models to real-world situations; CLO3 — Apply appropriate models in business decision making
Word Count	1,500–2,000 words (excluding title page and references)
Due Date	Week 12 — submit via i-CATS LMS by 11:59 PM
Submission Format	Microsoft Word (.docx) or PDF uploaded to i-CATS LMS
Assessment SLT	30 hours self-learning time

Case Scenario: Sarawak Energy Berhad — Pricing Strategy Decision

"Sarawak Energy Berhad (SEB) operates the only electricity grid in Sarawak, making it a natural monopoly. In 2024, SEB faces two strategic decisions: (1) whether to maintain its low industrial tariff (RM 0.23/kWh) to attract more data centre and aluminium smelting FDI, or increase it to RM 0.30/kWh to fund grid expansion into Sarawak's interior; (2) whether to enter the retail telecommunications market (broadband for rural areas) using its existing fibre infrastructure, potentially competing with CelcomDigi and Maxis.

Data provided in the case:

Data Item	Value
Current industrial customers	45 companies (Press Metal, data centres, heavy industry)
Current annual revenue	RM 2.1 billion/year
Grid expansion cost	RM 3.5 billion over 5 years

Expected new FDI if tariff maintained at RM 0.23	RM 8 billion (3 data centres + 1 smelter)
Expected FDI if tariff raised to RM 0.30	RM 4 billion (1 data centre only)
Rural broadband market size	250,000 potential subscribers in Sarawak
Estimated cost of rural broadband infrastructure	RM 800 million investment

Required Sections

1. Market Structure Analysis (approximately 350 words): Using monopoly and natural monopoly theory (LU3), analyse SEB's market position in Sarawak's electricity sector. What pricing power does SEB have? What are the regulatory implications of its natural monopoly status? How does its pricing decision affect consumer surplus and total welfare in Sarawak?
2. Pricing Decision Analysis (approximately 450 words): Use cost-benefit economic analysis to evaluate the tariff increase option. Calculate: (a) the revenue impact of raising tariff from RM 0.23 to RM 0.30 on existing 45 industrial customers (assume total consumption of 9 billion kWh/year); (b) the foregone FDI under the higher tariff option (RM 8B vs RM 4B — what are the long-run economic multiplier effects?); (c) a simplified NPV comparison of the two options over 10 years at a 6% discount rate. Which option maximises Sarawak's long-run economic welfare?
3. Market Entry Analysis (approximately 450 words): Using game theory from LU6, analyse SEB's potential entry into the rural broadband market as a new player competing against CelcomDigi and Maxis. Construct a simplified 2×2 payoff matrix showing SEB's entry decision versus CelcomDigi's response (maintain coverage vs expand rural coverage). Identify the Nash Equilibrium and explain how SEB can use first-mover advantage and its existing fibre infrastructure to credibly commit to market entry.
4. Recommendation (approximately 300 words): Provide a clear, economically justified recommendation for both of SEB's strategic decisions. Address the equity-efficiency trade-off: lower tariffs and rural broadband access are socially valuable but require

financial trade-offs. What is the optimal strategy for Sarawak's long-run economic development?

AI and Academic Integrity Note

Students may use AI tools (e.g., ChatGPT, Microsoft Copilot) as learning aids to brainstorm, structure arguments, or check grammar. However, students must: (1) Verify all AI-generated content against course readings and their own analysis; (2) Not submit AI-generated text as their own original work; (3) Include an 'AI Tools Used' declaration at the end of the submission stating which tools were used and for what purpose; (4) Not upload confidential, personal, or proprietary information to external AI platforms, in compliance with Malaysia's PDPA 2010 and i-CATS academic integrity policies.

Assessment Rubric — CA2

Criterion	CLO	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Market Structure Diagnosis	CLO2	Accurate natural monopoly analysis; regulatory implications (price regulation, access pricing) clearly addressed; welfare effects correctly assessed	Good analysis with minor gaps in regulatory or welfare dimensions	Basic market structure identification; limited analysis of implications	Wrong or absent market structure analysis; no welfare discussion
Quantitative Decision Analysis	CLO3	Revenue/NPV calculations correct and	Good calculations with minor errors;	Some numbers used; calculation methodology	No quantitative analysis; case data not used

		explicit; methodology clearly explained; both tariff options quantitatively compared with assumptions stated	methodology mostly clear	unclear or incomplete	
Game Theory Application	CLO3	Correct payoff matrix constructed for SEB entry decision; Nash Equilibrium correctly identified; first- mover and commitment strategy implications clearly explained	Payoff matrix present with minor errors; Nash Equilibrium identified	Game theory concept referenced but matrix absent or incorrect	Game theory applied to broadband en decision
Quality of Recommendation	CLO2+CLO3	Clear, specific, economically grounded recommendations for both decisions; equity- efficiency trade- off explicitly acknowledged and resolved	Good recommendations with reasonable economic justification	Basic recommendation; weak or missing justification	No recommendation or recommendation unsupported economic analysis

Communication & Evidence	CLO2+CLO3	Case data used correctly throughout; professional analytical structure; APA 7th citations accurate	Good communication; minor data or citation issues	Adequate structure; some data used; limited citations	Poor structure; case data not used; missing citations
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Assessment Overview

Assessment Component	Detail
Type	Take-Home Individual Assignment (3 structured questions)
CLO Alignment	CLO1 + CLO2 + CLO3 (Integrative — all three CLOs)
Word Count	2,000–2,500 words total across all three questions
Release Date	Week 13 — distributed through i-CATS LMS
Submission Deadline	As specified by programme coordinator — typically 7–14 days after release
Submission Format	Microsoft Word (.docx) or PDF uploaded to i-CATS LMS
Assessment SLT	30 hours self-learning time

Assignment Questions

Question 1 (700–800 words, CLO1+CLO2)

"The Malaysian government is considering three policy options to manage the 2024 inflation surge (CPI at 3.5%) without triggering recession: (A) BNM raises OPR by 50 basis points; (B) Government reduces fuel subsidies further; (C) Government increases development spending in Sarawak and Sabah. Using both microeconomic (supply-demand, price controls) and macroeconomic (AD-AS, monetary and fiscal policy multiplier) models, analyse each policy option — including the trade-offs, unintended consequences, and distributional impacts (who gains and who loses in Sarawak specifically). Recommend the best policy combination and justify your recommendation with economic theory."

Question 2 (600–700 words, CLO2+CLO3)

"The CelcomDigi-Maxis-U Mobile oligopoly in Sarawak telecommunications is under scrutiny. Using market structure theory (LU3) and game theory (LU6), analyse: (a) whether the current industry structure maximises consumer welfare in Sarawak, particularly for rural consumers, (b) the strategic incentives for firms to collude vs compete on price and 5G coverage investment, (c) how a government regulator should design the 5G spectrum auction to maximise social welfare while incentivising rural coverage investment. Construct a 2×2 payoff matrix to support your game theory analysis of firm behaviour."

Question 3 (700–800 words, CLO3)

"Sarawak is betting on a digital-green economy convergence — using IR4.0 technologies and renewable energy to leapfrog industrial-age development constraints. Drawing on ALL concepts covered in this course (microeconomics, macroeconomics, market structures, trade theory, game theory, digital economy), develop a comprehensive economic framework for Sarawak's 2025–2030 development strategy. Your framework must: (a) identify Sarawak's comparative advantages and market structure opportunities, (b) recommend macroeconomic policies to support the digital transition, (c) address workforce and inequality challenges using labour market economics, and (d) apply game theory to explain how Sarawak should negotiate with the federal government, ASEAN partners, and multinational investors to maximise state-level economic outcomes."

AI and Academic Integrity Note

Students may use AI tools (e.g., ChatGPT, Microsoft Copilot) as learning aids to brainstorm, structure arguments, or check grammar. However, students must: (1) Verify all AI-generated content against course readings and their own analysis; (2) Not submit AI-generated text as their own original work; (3) Include an 'AI Tools Used' declaration at the end of the submission stating which tools were used and for what purpose; (4) Not upload confidential, personal, or proprietary information to external AI platforms, in compliance with Malaysia's PDPA 2010 and i-CATS academic integrity policies.

Assessment Rubric — Final Assignment

Criterion	CLO	Excellent (4)	Good (3)	Satisfactory (2)	In
Integration of Micro & Macro	CLO1	Fluently integrates micro and macro levels throughout; each question demonstrates clear micro-macro linkage with appropriate models cited	Good integration with minor gaps; mostly correct model use	Addresses both levels but in separate sections without analytical integration	On (m ad th
Depth of Economic Analysis	CLO2	Sophisticated multi-model analysis; models applied correctly, insightfully, and with reference to Sarawak data and context; logical chain of reasoning clear throughout	Good analysis; models mostly correctly applied; some insight	Basic analysis; limited model depth; descriptive elements present	Su pu de ec ab m
Applied Judgement & Recommendation	CLO3	Specific, practical, implementable recommendations grounded in economic theory; trade-offs and unintended consequences	Good recommendations with solid justification; most trade-offs addressed	Basic recommendations; limited economic grounding; some trade-offs missed	No re or re no ec

		explicitly discussed for Sarawak context			
Sarawak/Malaysian Contextualisation	CLO2+CLO3	Deep, specific knowledge of Sarawak's economic context; current data (2023–2024) and specific examples cited throughout all three questions	Good contextualisation; some specificity; mostly current references	General Malaysian context; limited Sarawak specificity; some dated references	No co ge irr ex
Academic Quality & Communication	CLO1+CLO2+CLO3	Excellent analytical structure; clear, precise writing at Masters level; APA 7th citations accurate throughout; word count respected	Good writing; minor structural or citation issues; generally clear	Adequate writing; some structural weaknesses; limited citations	Pe cla cit un

APPENDIX A: ECONOMICS AND ANALYTICS TOOLS GUIDE

This appendix provides a curated guide to free and low-cost online tools, databases, and resources relevant to each Learning Unit in MBA6664. All tools listed are either fully free or offer free access sufficient for academic study purposes. Priority is given to Malaysian and Sarawak-specific resources, then regional, then global.

LU1–LU2: Foundations of Microeconomics and Macroeconomics

Tool / Resource	Link	Cost	How to Use for LU1–LU2
Khan Academy Economics	https://www.khanacademy.org/economics-finance-domain	Free	Use for interactive supply-demand and elasticity exercises; ideal for reinforcing LU1 market analysis and LU2 macroeconomic concepts at your own pace.
Marginal Revolution University (MRU)	https://mru.org	Free	Watch the LU1 and LU2 video series by Tyler Cowen and Alex Tabarrok — the best free economics video content available; short 5–10 minute

			videos organised by topic.
Bank Negara Malaysia Statistics	https://www.bnm.gov.my/monetary-stability	Free (Malaysian)	Download Malaysia OPR, CPI, and money supply data for empirical LU2 analysis of monetary policy and inflation.
DOSM — Malaysia Statistics Portal	https://www.dosm.gov.my	Free (Malaysian)	Access real GDP, CPI, labour force, and household income data for LU1 and LU2 empirical analysis; primary source for Malaysian economic statistics.

LU3: Market Structure, Labour Economics, and Inequality

Tool / Resource	Link	Cost	How to Use for LU3
Sarawak Economic Planning Unit (EPU)	https://www.epu.sarawak.gov.my	Free (Sarawak)	Access Sarawak-specific labour statistics, income data, and economic development indicators

			for LU3 earnings and inequality analysis.
DOSM — HIES (Household Income Survey)	https://www.dosm.gov.my	Free (Malaysian)	Download Gini coefficient, income decile, and household expenditure data for Malaysia and Sarawak for LU3 income distribution analysis.
Bursa Malaysia — Company Filings	https://www.bursamalaysia.com	Free	Analyse listed Sarawak companies' (HSL, Press Metal, SEB-linked GLCs) cost structures, revenue data, and market positioning for LU3 case studies.
Competition Commission of Malaysia (MyCC)	https://www.mycc.gov.my	Free (Malaysian)	Read Malaysian competition law decisions and market investigation reports — essential for understanding how monopoly and oligopoly theory applies in Malaysian legal practice.

LU4: Development Economics and Macroeconomic Policy

Tool / Resource	Link	Cost	How to Use for LU4
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World Bank Open Data — Malaysia	https://data.worldbank.org/country/malaysia	Free	Download HDI, GDP per capita, poverty rates, and development indicators for Malaysia and ASEAN peers for LU4 development economics analysis.
IMF World Economic Outlook Database	https://www.imf.org/en/Publications/WEO	Free	Access IMF macroeconomic forecasts and historical data for GDP, inflation, fiscal balance, and current account for LU4 policy analysis.
UNDP Human Development Index Database	https://hdr.undp.org	Free	Compare Sarawak/Malaysia HDI against regional peers; access composite human development data for LU4 welfare and development analysis.
EPF Malaysia	https://www.kwsp.gov.my	Free (Malaysian)	Understand Malaysia's Employees

Annual Report			Provident Fund (EPF) structure and its macroeconomic role as a forced savings mechanism relevant to LU4 development economics.
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LU5: Trade and International Finance

Tool / Resource	Link	Cost	How to Use for LU5
MATRADE — Malaysia Trade Data	https://www.matrade.gov.my	Free (Malaysian)	Access Malaysia's export and import data by product, country, and sector for LU5 trade analysis; use to identify Sarawak's main export markets.
ASEAN Statistics (ASEAN Stats)	https://data.aseanstats.org	Free (Regional)	Compare Sarawak/Malaysia trade performance with ASEAN peers; access regional FDI and trade integration data for LU5 analysis.

WTO Trade Statistics	https://stats.wto.org	Free	Access global tariff schedules, non-tariff measures, and bilateral trade flow data for LU5 trade policy analysis.
XE Currency Historical Charts	https://www.xe.com/currencycharts	Free	Track historical MYR/USD and other exchange rate movements for LU5 practical analysis; plot the 2023 depreciation and 2024 recovery visually.

LU6: Game Theory in Decision Making

Tool / Resource	Link	Cost	How to Use for LU6
Game Theory Solver — Normal Form	https://www.gametheory.net/Mike/applets/NormalForm/	Free	Input 2×2 payoff matrices and solve for Nash Equilibria interactively — ideal for practising LU6.1 analysis of dominance and Nash Equilibrium

			before the assessment.
Coursera — Game Theory (Stanford/UBC)	https://www.coursera.org/learn/game-theory-1	Freemium	Audit the Yale-Stanford Game Theory course for supplementary video lectures on Nash Equilibrium, extensive-form games, and repeated games (LU6.1–6.5).
Nobel Prize — Game Theory Resources	https://www.nobelprize.org/prizes/economic-sciences/2005/summary	Free	Read Nobel Prize lecture summaries on game theory (Nash 1994; Aumann and Schelling 2005; Shapley 2012) for academic depth and LU6 historical context.
Excel — Payoff Matrix Template	Build manually in Microsoft Excel or Google Sheets	Free	Create 2×2 and 3×3 payoff matrices in

			Excel to model Sarawak business competition scenarios; use conditional formatting to highlight dominant strategies and Nash Equilibria.
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LU7: Digital and Knowledge Economy

Tool / Resource	Link	Cost	How to Use for LU7
SDEC Digital Economy Blueprint 2030	https://www.sdec.com.my/digital-economy-blueprint	Free (Sarawak)	Primary source document for Sarawak's digital economy strategy — essential reading for LU7 contextualisation and CA2/Final assignment Sarawak digital economy analysis.
MDEC — Malaysia	https://mdec.my	Free (Malaysian)	Access current data on Malaysia's digital

Digital Economy Statistics			economy GDP contribution, digital jobs, and tech sector investment for LU7 quantitative analysis.
WEF — Future of Jobs Report 2025	https://www.weforum.org/publications	Free	Benchmark Sarawak/Malaysia workforce digital skills challenges against global IR4.0 and IR5.0 trends; data on automation risk by occupation type.
MIT Technology Review	https://www.technologyreview.com	Freemium (some free articles)	Access frontier analysis of AI, automation, quantum computing, and IR5.0 economic implications; useful for LU7.8 Future Trends section and Final Assignment Q3.

Responsible AI and Data Use — Note for All LUs

Students may use AI tools (e.g., ChatGPT, Perplexity, Microsoft Copilot) to support their learning across all LUs. However, the following principles apply to all AI tool use in this course: (1) Verify AI-generated economic data and statistics against primary sources (BNM, DOSM, IMF, World Bank) before using in assignments — AI tools may generate plausible-sounding but inaccurate figures; (2) Do not submit AI-generated text as your own original analysis; (3) Declare AI tools used in an appendix statement; (4) Do not input assignment

scenarios, case data, or any personal/confidential information into external AI platforms, in compliance with Malaysia's PDPA 2010 and i-CATS academic integrity policies.

APPENDIX B: ECONOMICS FORMULA AND DIAGRAM REFERENCE CARDS

This appendix provides five condensed reference cards covering the core formulas and diagram structures in MBA6664. Use these cards for quick revision and as a companion to assignment preparation. Full derivations and worked examples are provided in the relevant Learning Units.

Card 1: Core Microeconomics Formulas

Formula	Expression	Key Notes
Price Elasticity of Demand	$PED = \% \Delta Q_d / \% \Delta P$	Negative for normal goods; $ PED > 1$ = elastic; $ PED < 1$ = inelastic
Midpoint PED Formula	$PED = \frac{[(Q_2 - Q_1) / ((Q_1 + Q_2) / 2)]}{[(P_2 - P_1) / ((P_1 + P_2) / 2)]}$	Use when calculating PED from two price-quantity observations; symmetric
Price Elasticity of Supply	$PES = \% \Delta Q_s / \% \Delta P$	Always positive; high PES = production can expand quickly in response to price rises
Cross-Price Elasticity	$XED = \% \Delta Q_{d_A} / \% \Delta P_B$	$XED > 0$ = substitutes; $XED < 0$ = complements; $XED = 0$ = independent goods
Income Elasticity	$YED = \% \Delta Q_d / \% \Delta Y$	$YED > 0$ = normal good; $YED < 0$ = inferior good; $YED > 1$ = luxury good
Total Revenue	$TR = P \times Q$	TR rises when P falls if demand is elastic; TR falls when P falls if inelastic
Total Cost	$TC = FC + VC$	In long run, all costs are variable; in short run, FC cannot be avoided

Average Total Cost	$ATC = TC / Q$	U-shaped; minimum point = efficient scale; $ATC = AFC + AVC$
Marginal Cost	$MC = \Delta TC / \Delta Q$	U-shaped; crosses AVC and ATC at their minimum; profit max rule: $P = MC$
Economic Profit	$\pi = TR - TC$ (economic)	Includes implicit costs (opportunity costs); economic profit = 0 in long-run competitive equilibrium
Accounting Profit	AR = Revenue – Explicit Costs	Does not include opportunity costs; typically higher than economic profit
Lerner Index (Markup)	$L = (P - MC) / P = -1/ PED $	Measures market power; $L = 0$ under perfect competition; $L \rightarrow 1$ under monopoly

Card 2: Core Macroeconomics Formulas

Formula	Expression	Key Notes
GDP (Expenditure Approach)	$Y = C + I + G + NX$	$NX = X - M$ (net exports); most widely used GDP measurement approach
GDP (Income Approach)	$Y = W + P + R + I$	Wages + Profits + Rent + Interest; equals expenditure GDP by identity
Real GDP	$\text{Real GDP} = \text{Nominal GDP} / \text{GDP Deflator} \times 100$	Adjusts for inflation; measures actual output growth over time
Consumer Price Index	$\text{CPI} = (\text{Cost of basket}_t / \text{Cost of basket}_{\text{base}}) \times 100$	Measures consumer price level; base year $\text{CPI} = 100$; published monthly by DOSM
Inflation Rate	$\pi = (\text{CPI}_t - \text{CPI}_{t-1}) / \text{CPI}_{t-1} \times 100$	Year-on-year % change in CPI; Malaysia target: 2–3%

Real Interest Rate	$r \approx i - \pi$ (Fisher equation)	Approximate real rate = nominal rate minus inflation; relevant for investment decisions
Simple Money Multiplier	$m = 1 / \text{Reserve Ratio}$	Maximum deposit expansion from RM 1 of base money; actual multiplier is lower due to cash holdings
Government Budget Balance	$BD = G - T$	Positive BD = deficit; negative BD = surplus; Malaysia's fiscal deficit target < 3% of GDP
Current Account Balance	$CA = NX + \text{Net Income} + \text{Net Transfers}$	Malaysia typically runs a CA surplus; positive CA = net foreign asset accumulation
Exchange Rate (Direct Quote)	RM per USD (e.g., 4.35)	MYR appreciation = fewer RM per USD; MYR depreciation = more RM per USD; impacts export/import competitiveness
Unemployment Rate	$U = (\text{Unemployed} / \text{Labour Force}) \times 100$	Malaysia structural unemployment ~3.5%; NAIRU = non-accelerating inflation rate of unemployment

Card 3: Standard Diagram Descriptions

Supply-Demand Diagram:

Axes: P (price, vertical axis), Q (quantity, horizontal axis). Demand curve (D): slopes downward from left to right — higher price leads to lower quantity demanded. Supply curve (S): slopes upward from left to right — higher price leads to higher quantity supplied. Equilibrium: intersection of D and S at (P^*, Q^*) . A price ceiling ($P_{\text{Ceil}} < P^*$) is a horizontal line below equilibrium — creates a shortage ($Q_d > Q_s$). A price floor ($P_{\text{floor}} > P^*$) is a horizontal line above equilibrium — creates a surplus ($Q_s > Q_d$). Demand shift right (e.g., income rises): new equilibrium has higher P and higher Q. Supply shift right (e.g., technology improves): new equilibrium has lower P and higher Q.

AD-AS Diagram:

Axes: P (price level, vertical), Y (real GDP/output, horizontal). AD curve: slopes downward — combines wealth effect, interest rate effect, and exchange rate effect. SRAS: slopes upward (sticky wages in short run). LRAS: vertical line at potential output Y^* (full employment). Positive demand shock (AD shifts right): short-run — higher P and higher Y; long-run — wages adjust, SRAS shifts left, economy returns to Y^* with higher P only. Adverse supply shock (SRAS shifts left): stagflation — higher P and lower Y simultaneously. Malaysia example: 2022 energy/commodity price spike shifted SRAS left, raising CPI while dampening growth.

Cost Curves Diagram:

Axes: Cost in RM (vertical), Q (output, horizontal). AFC (average fixed cost): continuously falling hyperbola. AVC (average variable cost): U-shaped. ATC (average total cost): U-shaped, lies above AVC by the vertical distance AFC. MC (marginal cost): U-shaped, crosses AVC and ATC at their respective minimum points. The minimum of ATC = efficient scale. For a competitive firm, profit maximisation rule: produce where $P = MC$. For a Sarawak palm oil mill, the ATC curve shifts down when crude palm oil input prices fall, lowering the minimum efficient scale and enabling expansion.

Monopoly Diagram:

Same cost curves as above, plus: downward-sloping demand curve ($D = AR$) and MR curve below D with twice the slope (for linear demand). Profit maximisation: set $MR = MC$ to find quantity Q_m . Read the price from the D curve at Q_m to find $P_m > MC$ (monopoly markup). Monopoly profit = $(P_m - ATC) \times Q_m$ = shaded rectangle above ATC. Deadweight loss (DWL) = triangle between competitive output Q_c and monopoly output Q_m , bounded by D and MC. Sarawak Energy Berhad as natural monopoly: regulated tariff setting aims to approximate $P = MC$ to reduce DWL, while ensuring SEB earns sufficient revenue to cover grid investment.

Card 4: Game Theory — 2x2 Payoff Matrix Notation

Standard 2x2 Payoff Matrix Structure:

A 2x2 payoff matrix has Row Player (Player 1) choosing between two strategies on the rows, and Column Player (Player 2) choosing between two strategies on the columns. Each cell shows payoffs as (P1 payoff, P2 payoff). Payoffs represent profit, utility, or any measurable outcome.

P1 / P2	Strategy A	Strategy B
Strategy X	(a, b)	(c, d)
Strategy Y	(e, f)	(g, h)

Key Game Theory Definitions:

- **Dominant Strategy:** A strategy that gives a higher payoff than any other strategy, regardless of what the opponent does. If P1 earns more playing X whether P2 plays A or B (i.e., $a > e$ AND $c > g$), then X dominates Y for P1.
- **Nash Equilibrium:** A combination of strategies where neither player can improve their payoff by unilaterally switching, given the other player's choice. Each player is playing their best response.
- **Prisoner's Dilemma:** Both players have Defect as dominant strategy $\rightarrow$ (Defect, Defect) Nash Equilibrium even though (Cooperate, Cooperate) gives higher joint payoff. Illustrates why rational self-interest can produce collectively suboptimal outcomes.
- **Backward Induction:** Method for solving sequential (extensive-form) games by working from the final decision node back to the first, identifying the optimal action at each stage — produces the Subgame Perfect Equilibrium.
- **Folk Theorem:** In infinitely repeated games with sufficiently patient players, cooperative outcomes can be sustained through trigger strategies (e.g., grim trigger: cooperate until partner defects, then defect forever). Foundation for understanding long-term business relationships.

Blank Matrix Template for Student Practice:

P1 / P2	Strategy A	Strategy B
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Strategy X	(__ , __)	(__ , __)
Strategy Y	(__ , __)	(__ , __)

Card 5: Trade and International Finance Quick Reference

Concept	Formula / Definition	Sarawak / Malaysia Example
Comparative Advantage	Country A has CA in Good X if opportunity cost of X is lower in A than in B	Sarawak: lower OC of LNG and palm oil production → exports these; imports manufactured goods from China
Terms of Trade	ToT = Export Price Index / Import Price Index x 100	Malaysia benefits when palm oil and LNG prices rise relative to import prices; ToT improves when commodity prices surge
Trade Balance	TB = Exports (X) - Imports (M)	Malaysia runs a persistent trade surplus (TB > 0); Sarawak LNG and palm oil are major contributors to export revenue
Balance of Payments Identity	CA + KA + FA = 0 (must balance)	Malaysia CA surplus offset by capital/financial account deficit; BNM accumulates foreign reserves
Exchange Rate (Direct Quote)	Domestic currency units per foreign currency unit	RM 4.75/USD in 2023 (weak MYR): Sarawak palm oil exporters earn more RM per USD of revenue
Real Exchange Rate	$RER = (e \times P^*) / P$	Adjusts nominal rate for inflation differential; measures actual international price competitiveness
Tariff Effect on Welfare	Raises domestic price: producers gain (CS),	EU EUDR non-tariff barrier on Malaysian palm oil: shifts income from

	consumers lose (PS), government gains revenue, DWL created	Sarawak exporters to EU consumers/producers
NPV of Investment	$NPV = \text{Sum}[CF_t / (1+r)^t]$ - Initial Investment	SEB Baleh Dam: NPV positive at $r = 8\%$ discount rate supports investment decision; used to evaluate SCORE corridor FDI

APPENDIX C: OPTIONAL ADVANCED TRACK — QUANTITATIVE ECONOMICS

IMPORTANT NOTE: This appendix is entirely optional. The content here is NOT directly assessed in MBA6664 and is NOT required to achieve a passing grade. It is provided for students with quantitative backgrounds or a strong interest in the mathematical foundations of economics. Students who find this material challenging should focus on the conceptual and applied content in LU1–LU7 and the assessments.

C1: Worked Numerical Examples

Example 1: Price Elasticity of Demand — Sarawak Freshwater Fish Market

Scenario: At the Satok Weekend Market in Kuching, the price of freshwater fish rises from RM 12/kg to RM 15/kg. Quantity demanded falls from 8 kg to 7 kg per stall per hour.

- Step 1 — % Change in Price: $(15 - 12) / 12 \times 100 = +25\%$
- Step 2 — % Change in Quantity Demanded: $(7 - 8) / 8 \times 100 = -12.5\%$
- Step 3 — PED = $\% \Delta Q_d / \% \Delta P = -12.5\% / 25\% = -0.5$

Interpretation: $|PED| = 0.5 < 1 \rightarrow$ demand is INELASTIC. Fish buyers at Satok market have few close substitutes and buy fish regardless of modest price increases.

Total Revenue Check:

- TR before: $RM\ 12 \times 8\ kg = RM\ 96$ per stall per hour
- TR after: $RM\ 15 \times 7\ kg = RM\ 105$ per stall per hour (+9.4%)

Conclusion: When demand is inelastic, a price INCREASE causes total revenue to INCREASE. The fish vendor earns more despite selling fewer kilograms — a counter-intuitive but practically critical insight for small market vendors in Sarawak.

Example 2: The Keynesian Fiscal Multiplier — Sarawak Infrastructure Spending

Scenario: The Sarawak state government announces RM 500 million in new road construction spending on the Kapit-Belaga corridor. Assume Malaysia's Marginal Propensity to Consume (MPC) = 0.75.

- Step 1 — Calculate the Simple Multiplier: $k = 1 / (1 - \text{MPC}) = 1 / (1 - 0.75) = 1 / 0.25 = 4$
- Step 2 — Calculate Total GDP Impact: $\text{GDP Impact} = \Delta G \times k = \text{RM } 500\text{M} \times 4 = \text{RM } 2 \text{ billion}$

Interpretation: The initial RM 500M injection triggers a chain of spending rounds. Construction workers spend 75% of wages on consumption → retailers spend 75% of their earnings → and so on. The total cumulative GDP impact is approximately RM 2 billion — four times the initial spending. This is the Keynesian multiplier at work in Sarawak's regional economy.

Important Caveat: The simple multiplier of 4 overstates real-world effects. In practice, the effective multiplier is smaller because: (a) some consumption leaks to imports (imported materials, Chinese manufactured goods), reducing the domestic multiplier; (b) government borrowing to finance spending may crowd out private investment through higher interest rates; (c) if the economy is near full employment, additional spending generates more inflation than output. The IMF estimates Malaysia's actual fiscal multiplier at approximately 0.8–1.2, reflecting these leakages and structural characteristics.

Example 3: Supply-Demand Equilibrium Shift — Sarawak Palm Oil and the EUDR

Scenario: The global CPO market initially has equilibrium at $P^* = \text{RM } 3,800/\text{tonne}$, $Q^* = 2$ million tonnes/month. The EU Deforestation Regulation (EUDR) 2025 effectively closes the EU market to non-compliant Sarawak palm oil, reducing global effective demand by approximately 200,000 tonnes/month at every price level (a leftward demand shift).

- If CPO demand is inelastic ($|\text{PED}| = 0.3$ in the short run), the % price change required to clear a 10% demand reduction: $\% \Delta P = \% \Delta Q_d / |\text{PED}| = -10\% / 0.3 = -33\%$
- New price approximation (short-run, supply fixed): $P_{\text{new}} = \text{RM } 3,800 \times (1 - 0.33) = \text{approximately RM } 2,546/\text{tonne}$

- In practice, supply also adjusts: producers reduce output as price falls → partially moderating the price decline → new equilibrium at approximately RM 3,200-3,400/tonne with Q around 1.85-1.9M tonnes/month

Managerial Implication: Sarawak palm oil estates — which depend heavily on export revenue priced in USD — would face significantly reduced MYR profitability at RM 3,200/tonne versus RM 3,800/tonne. For an estate producing 100,000 tonnes/year, the revenue loss = $100,000 \times \text{RM } 600 = \text{RM } 60 \text{ million/year}$. This quantification of trade policy risk is precisely the kind of economic analysis that LU5 equips managers to perform.

C2: Simple Excel Workbook Template Structure for Economic Analysis

The following workbook structure can be built in Microsoft Excel or Google Sheets for practical economic analysis tasks throughout MBA6664. This is a pedagogical planning tool, not a professional econometric model.

Sheet Name	Purpose	Key Inputs	Key Outputs
Supply-Demand Calculator	Find market equilibrium given linear supply and demand equations	Demand intercept (a), slope (b), supply intercept (c), slope (d): $Q_d = a - bP$; $Q_s = c + dP$	Equilibrium P^* and Q^* ; consumer surplus; producer surplus; DWL under price controls
Elasticity Calculator	Calculate all four elasticity measures from observed data	P_1, P_2 (prices); Q_1, Q_2 (quantities); Y_1, Y_2 (incomes); P_{B1}, P_{B2} (related good prices)	PED, PES, XED, YED with automatic interpretation (elastic/inelastic, normal/inferior/luxury)
GDP / National Accounts	Apply expenditure and income	C, I, G, X, M values (RM millions)	GDP; net exports (NX); trade balance %; real vs

	approach GDP formulas		nominal GDP comparison
Payoff Matrix Solver	Identify dominant strategies and Nash Equilibria in 2x2 games	4 payoff pairs: (a,b), (c,d), (e,f), (g,h) for P1 rows and P2 columns	Dominant strategy for each player; Nash Equilibrium cell highlighted; Prisoner's Dilemma flag
NPV / Investment Appraisal	Evaluate FDI or infrastructure investment decisions	Initial outlay (I ₀), annual cash flows (CF ₁ to CF ₁₀), discount rate (r%)	NPV; IRR (approximate); payback period; sensitivity table at different r values

Construction Tips: For the payoff matrix sheet, use Excel conditional formatting (Home → Conditional Formatting → Highlight Cell Rules) to automatically highlight the Nash Equilibrium cell in green. For the supply-demand calculator, set up the simultaneous equations system: $Q_d = a - bP$ and $Q_s = c + dP$, then solve algebraically for $P^* = (a - c) / (b + d)$. The elasticity calculator uses the standard arc (midpoint) formula for accuracy. All formulas from Appendix B Cards 1-5 can be directly entered as Excel functions.

C3: Free Online Short Courses and Advanced Resources

The following resources support deeper engagement with the quantitative and theoretical foundations of economics for students who wish to go beyond the MBA6664 curriculum. All are free or freemium.

Resource	Provider and Link	Cost	Best For
Full Economics Course — Micro and Macro	Khan Academy: https://www.khanacademy.org/economics-finance-domain	Free	Self-paced review of all LU1–LU5 foundations

			with interactive exercises and immediate feedback; ideal entry point for quantitative practice
14.01 Principles of Microeconomics	MIT OCW: https://ocw.mit.edu/courses/14-01-principles-of-microeconomics-fall-2018	Free (full notes + exams)	MIT-level microeconomics with complete problem sets, exams, and solutions — equivalent to first-year undergraduate economics depth
14.02 Principles of Macroeconomics	MIT OCW: https://ocw.mit.edu/courses/14-02-principles-of-macroeconomics-spring-2023	Free	Deep coverage of Keynesian models, AD-AS, monetary and fiscal policy — complements LU2, LU4, and LU5 with greater mathematical rigour
Game Theory (Yale/Stanford)	Coursera: https://www.coursera.org/learn/game-theory-1	Freemium (audit free)	Rigorous game theory coverage beyond LU6:

			Nash Equilibrium proofs, extensive-form games, repeated games with mathematical derivations
FRED Economic Data	Federal Reserve: https://fred.stlouisfed.org	Free	World's largest free economics database with 800,000+ data series including Malaysia; build custom charts for Final Assignment macro analysis
Our World in Data — Economics	https://ourworldindata.org/economics	Free	Best free source for long-run development data with excellent visualisations; ideal for LU4 development economics and LU7 digital economy country comparisons

A Note on Data Literacy: Economics is increasingly an empirical discipline. MBA graduates who can download data from DOSM, BNM, IMF, or World Bank, construct charts in Excel, and apply basic elasticity and multiplier calculations have a significant advantage in business and policy analysis roles. Start with Khan Academy and FRED for accessible practice, then progress to MIT OCW materials as your confidence grows. The FRED database in particular is an invaluable free resource for tracking real-time Malaysian and global economic indicators relevant to all seven LUs in MBA6664.

